Simultaneous Fiber Cancellation

Arithmetic convolutions, signed sequences, and the full factorial hierarchy

KEIO*

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Abstract

We develop irrationality and linear-independence criteria for factorial series at every nonzero rational point whose reduced numerator has absolute value one or two. The original all-rank divisor-sum target of Deajim and Siksek is retained, with the factorial-grid architecture attributed to Tokengrinder. The expanded arithmetic theorem admits polynomial dependence on prime valuations, every logarithmic mean order, and closure under pointwise, Dirichlet, and unitary products. It includes all weighted ordered factorizations, higher divisor functions, gcd moments, finite-group and finite-ring counting functions, and rough-prime total-multiplicity weights. All polynomial moments, residue restrictions, and positive and negative derivative orders remain jointly independent after identical normalized profiles are grouped. Rational cyclotomic coordinates cover every root-of-unity order; actual four- and six-direction values are independent over the Gaussian and Eisenstein fields. A complementary elementary criterion characterizes rationality for every integer coefficient sequence of growth $o(n)$, and gives complete two-sided hierarchies for all bounded non-eventually-periodic sequences. Exact normal forms classify nested polynomial integrals and quotients of affine coordinate forms. A separately attributed classical E-function argument treats all inverse exponential moments and arbitrary repeated integer-pole weights at nonzero algebraic arguments. The original proofs are preserved, the new arithmetic proofs are supplied in full, and eight exact auxiliary verifiers are embedded in this single source. The manuscript is unrefereed and not Lean-formalized. Historical priority is not certified; the classical companion is not claimed as a new transcendence result, and no transcendence assertion is made for the nontrivial arithmetic-profile values.

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1 The unchanged historical target

For positive integers k, n, define

$$\sigma_k(n) = \sum_{d|n} d^k, \qquad \alpha_k = \sum_{n=1}^{\infty} \frac{\sigma_k(n)}{n!}. \tag{1}$$

The individual irrationality question is the Erdős–Kac problem, recorded as Erdős Problem 252. Deajim and Siksek [1] studied the stronger, specific target

$$1, \alpha_1, \ldots, \alpha_K \quad \text{linearly independent over } \mathbb{Q}. \tag{2}$$

Their criterion assumes Schinzel's hypothesis and was checked for $K = 50$. The exact target and this qualification are recorded in the primary publisher/Warwick abstract and independently in Pratt's introduction [2]. We use the descriptive phrase "the linear-independence question studied by Deajim and Siksek": we do not invent a numbered conjecture or claim that this phrase was already an established eponym.

Tokengrinder's public development [4] supplies a proof of irrationality of each individual α_k. The later extension directory [5] expressly distinguishes a polynomial perturbation $\sigma_k(n)+P(n)$ from a polynomial in the divisors, $\sum_{d|n} P(d)$, and identifies the latter, equivalently the joint target (2), as further work. Individual irrationality alone does not imply joint independence.

Theorem 1.1. *For every integer $K \geq 1$, the numbers $1, \alpha_1, \ldots, \alpha_K$ are linearly independent over $\mathbb{Q}$. Equivalently, for every nonzero polynomial $P \in \mathbb{Q}[X]$ with $P(0) = 0$,*

$$\sum_{n=1}^{\infty} \frac{\sum_{d|n} P(d)}{n!} \notin \mathbb{Q}. \tag{3}$$

The proof neither changes the denominator $n!$ nor restricts the divisor exponents or their rational coefficients. No prime-pattern conjecture is assumed. The equivalence follows by writing $P(X) = \sum_{k=1}^{K} a_k X^k$, interchanging a finite sum with an absolutely convergent series, and clearing rational coefficients.

Theorem 14.1 is the common transfer mechanism. The enlarged arithmetic input is supplied by Theorems 16.5 and 19.4. The latter permits a rough total-multiplicity factor multiplied by any admissible polynomial valuation profile. The separate sequence criterion in Theorem 20.1 admits arbitrary signed sublinear integer coefficients, and Theorem 20.3 handles every bounded non-eventually-periodic sequence. These statements have different hypotheses: their families are combined only where an explicit joint theorem proves the combination, such as Theorem 20.4. All earlier arguments are retained in full so that the historical target and every new dependency can be checked separately.

The progression from the previous revision is substantive: specifying an exact prime valuation distinguishes profiles invisible to a divisibility-only test; preserving residues gives simultaneous signed and oscillatory values; and negative grid coordinates plus extra global constraints admit integration orders of arbitrary size. The permitted evaluation magnitudes remain $2/b$, with $b \geq 1$.

1.1 How to use the expanded results

For the arithmetic theorem, compute a normalized prime-power profile, group equal profiles, and check the stated positivity, integrality, and growth conditions. Theorem 16.5 and Section 17 supply entire closed arithmetic classes. One may then use any fixed residue class, all two-sided jets, and the rational weights with simple negative-integer poles. The stronger rough-prime

product theorem has its own growth restriction and does not authorize arbitrary products of rough factors.

For signed coefficients of growth $o(n)$, eventual equality of sequences decides all rational relations between their factorial values. Bounded aperiodic sequences automatically have independent full jet hierarchies. For nested polynomial kernels, reduce to the unique normal form of Theorem 23.1; for quotients of affine forms, use the exact rank test of Theorem 23.2. The classical inverse-moment result in Section 24 is a separate, stronger transcendence theorem for the pure exponential channel. It is not a mixed-independence assertion with the other arithmetic channels.

These distinctions prevent double counting. Two integrals with the same rational coefficient multiplier are one coordinate, and two counting functions with the same normalized profile belong to the same arithmetic block. The concrete examples are instances of infinite families, not a claim that each was a previously named open constant.

1.2 What is added to the existing method

The factorial expansion, squared-congruence alignment, integer limit, and progression-mean separation are the architecture of [4]. The new step of this argument is to solve all normalized cancellation equations simultaneously. Multiplicativity introduces the different factors $1/\sigma_k(p)$ for different k; these factors cannot simply be ignored or replaced by a common binomial weight. A sufficiently wide finite grid instead supplies a rational kernel. Terminal-shift injectivity makes every nonzero kernel vector usable.

The supplied corpus was inspected at commit `510a32c9a3fd2aa0b5c78d3f6812ded723b9ac8f` [6]. Its fixed-box relation methods in [013] and mixed congruence kernels in [078] motivated looking for simultaneous relations. No result in an unreviewed corpus manuscript is a hypothesis of the proof below. The source of the factorial-grid architecture is [4], rather than those Mahler-series papers.

2 A finite auxiliary problem and its solution

The finite question is useful independently of the asymptotic argument. For $L, K \geq 1$, put $\mathcal{V} = \{0, \dots, L-1\}^K$. A coordinate-$j$ fiber is the set of vertices obtained by fixing every coordinate except e_j, for $0 \leq j < K$.

Auxiliary problem 2.1. Given finitely many rational coefficient arrays on $\mathcal{V}$, each assigned to one coordinate direction, can one choose nonzero integer vertex weights whose weighted sum for each array vanishes on every fiber in its assigned direction? Can such cancellation be arranged without annihilating the final expression needed for an application?

Lemma 2.2 (Simultaneous fiber cancellation)**.** *Suppose there are at most C assigned rational arrays, and $L > C$. There are integers c_e, not all zero, satisfying every prescribed fiber equation. If a remaining expression has one distinct formal label r_e for each vertex and a nonzero multiplier u_e at that label, then $\sum_e c_e u_e[r_e]$ is nonzero.*

Proof. There are L^K unknowns and at most CL^{K-1} homogeneous equations over $\mathbb{Q}$. Rank-nullity gives a nonzero rational solution; multiplication by a common denominator gives integer weights. If $c_{e_*} \neq 0$, the coefficient of the unique label r_{e_*} is $c_{e_*} u_{e_*} \neq 0$. □

This is elementary finite-dimensional linear algebra. Its role here is specific: it replaces an explicit one-function finite difference by simultaneous equations involving several different multiplicative normalizations. The terminal labels in the application will be actual integer shifts, and their distinctness will be proved rather than assumed. No bound on the size of

the integer weights is required: the entire finite construction is fixed before the asymptotic variable tends to infinity.

3 Factorial tails with a sufficiently small remainder

Let $\tau(m)$ count the positive divisors of m. Complementary-divisor pairing gives $\tau(m) \le 2\sqrt{m}$. Consequently, for $k \ge 1$,

$$\sigma_k(m) \le 2m^{k+1/2}, \qquad \rho_k(m) := \frac{\sigma_k(m)}{m^k} = \sum_{d\mid m} d^{-k} \le 2\sqrt{m}. \tag{4}$$

These deliberately weak bounds are sufficient. In particular all the series (1) converge absolutely.

Write $\left\{ {i \atop j} \right\}$ for the Stirling numbers of the second kind, including $\left\{ {0 \atop 0} \right\} = 1$ and $\left\{ {i \atop 0} \right\} = 0$ for $i > 0$.

Lemma 3.1 (Common-order tail expansion)**.** *Fix $K \ge k \ge 1$. As $n \to \infty$ through positive integers,*

$$\begin{aligned} T_k(n) &:= n! \sum_{m>n} \frac{\sigma_k(m)}{m!} \\ &= \sum_{j=0}^{K} \sum_{i=j}^{K} \left\{ {i \atop j} \right\} \frac{\sigma_k(n+j+1)}{(n+j+1)^{i+1}} + O_K(n^{-3/2}). \end{aligned} \tag{5}$$

In particular the remainder remains $o(1)$ after multiplication by n.

Proof. For fixed $0 \le j \le K$ and $x \to \infty$, the generating-function identity

$$\prod_{r=1}^{j} (1 - rz)^{-1} = \sum_{u=0}^{\infty} \left\{ {u+j \atop j} \right\} z^u$$

gives

$$\frac{1}{x(x-1)\cdots(x-j)} = \sum_{i=j}^{K} \left\{ {i \atop j} \right\} x^{-i-1} + O_K(x^{-K-2}). \tag{6}$$

For $j = 0$ the formula is exact. For $j > 0$, the generating identity follows either by multiplying the geometric series or from the recurrence $\left\{ {i+1 \atop j} \right\} = j \left\{ {i \atop j} \right\} + \left\{ {i \atop j-1} \right\}$. Since the finitely many rational functions are analytic at $z = 0$, their Taylor remainders have the stated bounds.

Apply (6) to the first $K+1$ summands of $T_k(n)$, with $x = n + j + 1$. After multiplication by $\sigma_k(x)$, the remainder is $O_K(n^{k-K-3/2})$, hence $O_K(n^{-3/2})$. For the remaining summands $h \ge K+2$, use the majorants

$$U_h = \frac{2(n+h)^{k+1/2}}{(n+1)\cdots(n+h)}.$$

Their consecutive ratio is

$$\frac{U_{h+1}}{U_h} = \frac{(1 + 1/(n+h))^{k+1/2}}{n+h+1} \le \frac{1}{2}$$

for all sufficiently large n, uniformly in $h \ge K+2$. Also $U_{K+2} = O_K(n^{k-K-3/2})$. The omitted tail is at most $2U_{K+2}$. This proves (5). □

If integers $a_1, \dots, a_K$ satisfy $\alpha = \sum_k a_k \alpha_k \in \mathbb{Q}$, then

$$T(n) := \sum_{k=1}^{K} a_k T_k(n) = n!\alpha - \sum_{m=1}^{n} \frac{n!}{m!} \sum_{k=1}^{K} a_k \sigma_k(m) \in \mathbb{Z} \tag{7}$$

for all sufficiently large n. Indeed the denominator of α divides $n!$ once n is sufficiently large. The coefficients a_k may have either sign; positivity of the combined tail is unnecessary.

4 A coprime grid with distinct terminal shifts

Fix $K \geq 1$ and set

$$\begin{aligned} C &= \frac{K^2(K+1)}{2}, \qquad L = C + 1, \qquad b = K(L-1) + 1, \\ F &= \max(b^K - 1, K + 1), \qquad D = F!, \qquad B = 1 + KDF. \end{aligned} \tag{8}$$

For $e = (e_0, \dots, e_{K-1}) \in \mathcal{V} = \{0, \dots, L-1\}^K$, define

$$\begin{aligned} I_e &= \sum_{a=0}^{K-1} e_a b^a, & J_e &= \sum_{a=0}^{K-1} (a+1) e_a b^a, \\ p_e &= B + DI_e, & t_e &= DJ_e, \\ s_{e,j} &= (j+1)p_e - t_e & & (0 \leq j \leq K). \end{aligned} \tag{9}$$

The notation p_e denotes positive integer multipliers; they need not be primes.

Lemma 4.1 (Grid geometry and arithmetic)**.** *The construction has the following properties.*

1. *The p_e are pairwise coprime, and $\gcd(p_e, h) = 1$ for every $1 \leq h \leq K+1$.*
2. *$0 \leq t_e < B \leq p_e$, so all $s_{e,j}$ are positive.*
3. *For $j < K$, $s_{e,j}$ is independent of e_j along its fiber.*
4. *All terminal shifts $s_{e,K}$ are distinct. Each is at least $(K+1)B$, whereas every $s_{e,j}$ with $j < K$ is smaller than $(K+1)B$.*

Proof. Base-b encoding makes the I_e distinct and gives $0 \leq I_e \leq F$. Also $J_e \leq KI_e$, whence $t_e \leq KDF = B - 1$. Every p_e is congruent to 1 modulo D. If a prime ℓ divided two distinct multipliers, it would divide $D(I_e - I_f)$ but not D. It would therefore divide the nonzero integer $I_e - I_f$, of absolute value at most F. Hence $\ell \leq F$ and $\ell \mid F! = D$, a contradiction. Since $F \geq K + 1$, every $h \leq K + 1$ divides D, giving the other coprimality assertion.

Expanding the shifts gives

$$s_{e,j} = (j+1)B + D \sum_{a=0}^{K-1} (j-a) e_a b^a. \tag{10}$$

For $j < K$, the coefficient of e_j is zero. At $j = K$, the digits $(K-a)e_a$ lie between 0 and $K(L-1) = b - 1$. Uniqueness of base-b digits makes $e \mapsto s_{e,K}$ injective; those digits are nonnegative, so $s_{e,K} \geq (K+1)B$. For $j < K$ we have

$$s_{e,j} \leq Kp_e \leq K(B + DF) = (K+1)B - 1.$$

This also proves separation from every lower shift. □

Lemma 4.2 (Arithmetic cancellation weights)**.** *There are integers c_e, not all zero, such that for $1 \le k \le K$, $0 \le i < K$, $0 \le j \le i$, and each coordinate-j fiber,*

$$\sum_{e_j=0}^{L-1} c_e \frac{p_e^{i+1}}{\sigma_k(p_e)} = 0. \tag{11}$$

Consequently, for every function g on the positive integers,

$$\sum_{e\in\mathcal{V}} c_e \frac{p_e^{i+1}}{\sigma_k(p_e)} g(s_{e,j}) = 0 \tag{12}$$

for the same triples (k, i, j).

Proof. There are $K\sum_{i=0}^{K-1}(i+1) = C$ rational coefficient arrays. Each denominator $\sigma_k(p_e)$ is a positive integer. Apply Lemma 2.2, since $L > C$. On each fiber the shift $s_{e,j}$ is fixed, so multiplication of (11) by $g(s_{e,j})$ and summation over the fibers proves (12). □

5 Progression means distinguish an isolated shift

We first prove the final obstruction independently of the grid. For $k, Q, a \ge 1$, put

$$\mu_k(Q, a) = \sum_{\substack{d\ge 1\\ \gcd(d,Q)\mid a}} \frac{\gcd(d, Q)}{d^{k+1}}. \tag{13}$$

It is finite, depends only on a mod Q, and is positive because its $d = 1$ summand is 1.

Lemma 5.1 (Exact progression average)**.** *For positive integers k, Q, a,*

$$\lim_{M\to\infty} \frac{1}{M} \sum_{v=0}^{M-1} \rho_k(a + Qv) = \mu_k(Q, a). \tag{14}$$

If $\ell \nmid Q$ is prime and $a' \equiv a \pmod{Q}$, then

$$\mu_k(Q\ell, a') = (1 - \ell^{-k-1} + \mathbf{1}_{\ell\mid a'}\ell^{-k})\mu_k(Q, a). \tag{15}$$

Proof. Use $\rho_k(m) = \sum_{d\mid m} d^{-k}$. For each fixed d, the congruence $d \mid a + Qv$ has no solution unless $\gcd(d, Q) \mid a$; otherwise its density is $\gcd(d, Q)/d$. Truncating the divisor sum at H therefore gives the desired limit for the truncated expression. The average of the omitted positive terms is at most

$$Q \sum_{d>H} d^{-k-1} + \frac{1}{M} \sum_{d\le a+QM} d^{-k}.$$

The second expression tends to zero, including when $k = 1$, when it is $O(\log M/M)$. The first tends to zero as $H \to \infty$. This justifies the limit without assuming that ρ_1 is bounded.

To prove (15), write $d = \ell^u d_0$ with $\ell \nmid d_0$. In (13) for modulus Q, the sum over $u \ge 0$ contributes $(1 - \ell^{-k-1})^{-1}$. For modulus $Q\ell$, only $u = 0$ is allowed if $\ell \nmid a'$. If $\ell \mid a'$, the terms $u \ge 1$ instead have the additional factor ℓ from $\gcd(d, Q\ell)$. Thus the additional factor, relative to the original mean, is

$$(1 - \ell^{-k-1}) \sum_{u\ge 1} \ell^{1-u(k+1)} = \ell^{-k}.$$

The condition involving d_0 is unchanged because $a' \equiv a \pmod{Q}$. □

Proposition 5.2 (No constant limit with an isolated shift)**.** *Fix $k, Q \geq 1$, an integer A, finitely many integer shifts r_i, and real coefficients β_i. Suppose one shift r_* occurs exactly once and its coefficient β_* is nonzero. Then*

$$\sum_i \beta_i \rho_k(A + Qv + r_i)$$

cannot converge to any finite constant as $v \to \infty$. Only sufficiently large v, for which all arguments are positive, are relevant.

Proof. Choose a prime ℓ larger than Q, the number of distinct shifts, and all differences $|r_i - r_j|$. Distinct shifts are then distinct modulo ℓ. Choose one residue for N mod ℓ missing all $-r_i$, and another residue equal to $-r_*$. By the Chinese remainder theorem both can be combined with $N \equiv A \pmod{Q}$. Take positive representatives $A_{\text{off}}, A_{\text{on}}$ large enough for all shifted arguments to be positive.

If the asserted finite limit existed, its value would also be the Cesàro mean along both refined progressions. By Lemma 5.1, however, the difference between those two means is exactly

$$\beta_* \ell^{-k} \mu_k(Q, A + r_*) \neq 0. \tag{16}$$

Here a positive representative is used in the mean if needed. Every other shift has the same factor $1 - \ell^{-k-1}$ in both progressions, while only r_* acquires the extra factor ℓ^{-k}. Positivity of the original mean gives the contradiction. □

6 Proof of the joint linear-independence theorem

Suppose a nontrivial rational relation with 1 exists. Clear its denominators and choose the largest index K with a nonzero coefficient. We then have

$$\alpha = \sum_{k=1}^{K} a_k \alpha_k \in \mathbb{Q}, \qquad a_k \in \mathbb{Z}, \quad a_K \neq 0.$$

Construct the grid and integer weights from Lemmas 4.1 and 4.2. Put

$$Q = \prod_{e \in \mathcal{V}} p_e^2.$$

Pairwise coprimality supplies an integer A with

$$A \equiv t_e \pmod{p_e^2} \qquad (e \in \mathcal{V}). \tag{17}$$

Let $N = A + Qv$ tend to infinity and set

$$n_e = \frac{N - t_e}{p_e}.$$

These are eventually positive integers, divisible by p_e. Hence, for $0 \leq j \leq K$,

$$\gcd(p_e, n_e + j + 1) = \gcd(p_e, j + 1) = 1, \qquad p_e(n_e + j + 1) = N + s_{e,j}.$$

The divisor-sum function is multiplicative on coprime arguments: each divisor of a product of coprime integers decomposes uniquely as a product of divisors. It follows exactly that

$$\frac{\sigma_k(n_e + j + 1)}{(n_e + j + 1)^{i+1}} = \frac{p_e^{i+1}}{\sigma_k(p_e)} \frac{\sigma_k(N + s_{e,j})}{(N + s_{e,j})^{i+1}}. \tag{18}$$

By (7), the quantity

$$W(N) = \sum_{e \in \mathcal{V}} c_e T(n_e)$$

is an integer for all sufficiently large N in this progression. Substitute the common-order expansion (5) and the rescaling (18). For each $i < K$, the sum over e is zero by (12), using $g(s) = \sigma_k(N+s)/(N+s)^{i+1}$. We obtain

$$W(N) = \sum_{k=1}^{K} a_k \sum_{e \in \mathcal{V}} \sum_{j=0}^{K} \frac{c_e p_e^{K+1} \left\{ {K \atop j} \right\}}{\sigma_k(p_e)} \frac{\sigma_k(N+s_{e,j})}{(N+s_{e,j})^{K+1}} + O(N^{-3/2}). \tag{19}$$

The implicit constant can depend on the fixed grid, weights, and coefficients. This causes no difficulty: K and all those objects are fixed before N varies. In particular $n_e \asymp N$ for each fixed vertex, so the scaled errors still have order $N^{-3/2}$.

By (4), each main term in (19) is $O(N^{k-K-1/2})$, hence $O(N^{-1/2})$. Thus $W(N) \to 0$. An integer sequence converging to zero is eventually zero. Multiplying (19) by N, the error tends to zero, and the terms with $k < K$ tend to zero because $k - K + 1/2 \le -1/2$. Define

$$\beta_{e,j} = \frac{a_K c_e p_e^{K+1} \left\{ {K \atop j} \right\}}{\sigma_K(p_e)}.$$

The remaining assertion is

$$\sum_{e,j} \beta_{e,j} \frac{N}{N+s_{e,j}} \rho_K(N+s_{e,j}) \longrightarrow 0.$$

For every fixed shift s, replacing $N/(N+s)$ by 1 changes a summand by at most a constant times $|s|\rho_K(N+s)/(N+s) = O(N^{-1/2})$. Consequently

$$\sum_{e \in \mathcal{V}} \sum_{j=0}^{K} \beta_{e,j} \rho_K(N+s_{e,j}) \longrightarrow 0. \tag{20}$$

Choose e_* with $c_{e_*} \neq 0$. Since $\left\{ {K \atop K} \right\} = 1$ and $a_K \neq 0$, the coefficient $\beta_{e_*,K}$ is nonzero. Lemma 4.1 says that its shift $s_{e_*,K}$ occurs nowhere else among the displayed summands. Thus (20) contradicts Proposition 5.2. This proves Theorem 1.1. □

6.1 A further consequence involving the exponential constant

Corollary 6.1. *For every $K \ge 1$, the numbers $1, e, \alpha_1, \dots, \alpha_K$ are linearly independent over $\mathbb{Q}$.*

Proof. Suppose $\lambda e + \sum_{k=1}^{K} a_k \alpha_k \in \mathbb{Q}$ with integral coefficients and with some $a_k \neq 0$; let K be the largest such index. A constant numerator contributes to the combined factorial tail the additional term

$$\lambda U(n), \qquad U(n) = n! \sum_{m>n} \frac{1}{m!} = \frac{1}{n+1} + O(n^{-2}).$$

The combined tails are still eventually integral; the omitted $m = 0$ contribution is rational. Using the same grid and weights, the additional contribution to $W(N)$ tends to zero, while after multiplication by N it tends to $\lambda \sum_e c_e p_e$. The preceding proof therefore forces the expression in (20) to converge to a finite constant instead of zero. Proposition 5.2 excludes that as well. If all $a_k = 0$, a nontrivial relation would make e rational. The elementary factorial-tail proof rules this out: rationality of e would make $U(n)$ an integer eventually, whereas $0 < U(n) < 1$ for all sufficiently large n. □

This corollary is a consequence of the proved mechanism, not a replacement for the original target. It does not assert algebraic independence or transcendence of any particular α_k.

7 Several primes separate distinct local responses

The preceding argument uses one fresh prime to separate a surviving divisor sum. We now give a transfer principle in which several primes separate different arithmetic functions at the same terminal shift. This permits polynomial weights, Jordan and Dedekind functions, and products of these functions. The separation is stronger than linear independence of one-prime responses: distinct responses suffice.

For a function $\rho : \mathbb{N} \to \mathbb{R}$, write

$$M_\rho(Q, a) = \lim_{X\to\infty} \frac{1}{X} \sum_{v=0}^{X-1} \rho(Qv + a), \qquad Q, a \geq 1,$$

when the limit exists. This mean depends only on a modulo Q.

Lemma 7.1 (Separation by several fresh primes)**.** *Let $\rho_1, \dots, \rho_s$ have positive means $M_u(Q, a)$ on every arithmetic progression. Suppose that for every prime $\ell \nmid Q$, and $a' \equiv a \pmod{Q}$,*

$$M_u(Q\ell, a') = M_u(Q, a) \left[b_u(1/\ell) + \mathbf{1}_{\ell|a'} \delta_u(1/\ell) \right]. \tag{21}$$

Here the same functions b_u, δ_u work for every Q, a, and the δ_u are pairwise distinct, nonzero rational functions. Consider a finite sum

$$V(N) = \sum_{u=1}^{s} \sum_{r} \beta_{u,r} \rho_u(N + r)$$

on $N \equiv A \pmod{Q}$. If a shift r_* occurs at a distinguished location, and some $\beta_{u,r_*} \neq 0$, then $V(N)$ cannot have a finite limit. Repeated shifts in the sum are first grouped together.

Proof. Suppose that the limit is L. Call a prime admissible if it exceeds Q, the number of distinct shifts, and every nonzero absolute difference between two shifts, and avoids all poles required below. At an admissible prime choose an off-residue avoiding every $-r$, and an on-residue equal to $-r_*$; the on-residue hits no other shift.

For a nonempty finite set S of admissible primes, CRT gives $2^{|S|}$ refinements of $N \equiv A \pmod{Q}$, one for every choice of on/off states. The mean of V on each refinement equals L. Take their alternating sum, with sign $(-1)^{\text{number of off states}}$. Every other shift has an off-factor at every prime, independently of the state, and cancels. Iterating (21) at the distinguished shift gives

$$\sum_{u=1}^{s} B_u \prod_{\ell\in S} \delta_u(1/\ell) = 0, \qquad B_u = \beta_{u,r_*} M_u(Q, A + r_*). \tag{22}$$

If necessary the starting residue is increased by a multiple of Q to make all arguments positive. Crucially, each subset S refines only its own primes. Thus the coefficients B_u are the same for every subset; no unused off-factor is incorporated into them.

Fix an index t. Choose an admissible base prime ℓ_0 with $\delta_t(1/\ell_0) \neq 0$. For each $u \neq t$, choose a distinct admissible prime ℓ_u, also different from ℓ_0, such that $\delta_t(1/\ell_u) \neq \delta_u(1/\ell_u)$. These choices are possible because a nonzero rational function has only finitely many zeros, while there are infinitely many primes. Expand the product in

$$\sum_{v=1}^{s} B_v \delta_v(1/\ell_0) \prod_{u\neq t} [\delta_v(1/\ell_u) - \delta_u(1/\ell_u)] = 0. \tag{23}$$

Every term in its expansion is a constant multiple of an equation (22) for a subset containing ℓ_0, so the displayed equality follows. For $v \neq t$, the factor indexed by $u = v$ is zero. The multiplier of B_t is nonzero. Hence $B_t = 0$. As t was arbitrary and the means are positive, every $\beta_{u,r_*} = 0$, a contradiction. □

The hypothesis about the distinguished location is automatic for a terminal grid shift: its vertex and shift index are unique, although several arithmetic functions may contribute there. No linear independence assumption on the rational functions δ_u is used. For example,

$$\frac{z}{1+z^2} = \frac{z}{1-z^2} - \frac{2z^3}{1-z^4}$$

is a genuine one-prime linear dependence. It causes no difficulty for Lemma 7.1, since the three functions are distinct.

Theorem 7.2 (Transfer from local responses to factorial values)**.** *Let $\mathcal{F}$ be a collection of positive integer-valued multiplicative functions. Assign each $f \in \mathcal{F}$ an integer $h_f \geq 1$ such that $f(n) = O_f(n^{h_f+1/2})$, and put $\rho_f(n) = f(n)/n^{h_f}$. Suppose that these normalized functions have positive progression means and satisfy (21). Within each fixed degree h, their response functions are nonzero and pairwise distinct rational functions. Then, for every fixed $c \in \mathbb{Z}\setminus\{0\}$, the family*

$$\{1, e^{1/c}\} \ \cup \ \left\{ \sum_{n\geq 1} \frac{f(n)}{c^n n!} : f \in \mathcal{F} \right\} \tag{24}$$

is linearly independent over $\mathbb{Q}$. Independence of an infinite family means independence of every finite subfamily.

Proof. It suffices to consider a finite proposed relation, with integral coefficients a_f and λ, in which $\lambda e^{1/c} + \sum_f a_f \sum_{m\geq 1} f(m)/(c^m m!)$ is rational. If some a_f is nonzero, retain just these functions and put $H = \max h_f$, and let s be their number. Define

$$T_{f,c}(n) = c^n n! \sum_{m>n} \frac{f(m)}{c^m m!}.$$

The same falling-product expansion proved earlier, now with the factor c^{-j-1}, gives

$$T_{f,c}(n) = \sum_{j=0}^{H} \sum_{i=j}^{H} \left\{ {i \atop j} \right\} \frac{f(n+j+1)}{c^{j+1}(n+j+1)^{i+1}} + O(n^{-3/2}). \tag{25}$$

Indeed each finite truncation error is $O(n^{h_f+1/2-H-2}) = O(n^{-3/2})$; the omitted tail is bounded by a geometric series with first term of the same order. The inequality $|c| \geq 1$ preserves this bound even when $c < 0$.

For the exponential term put

$$U_c(n) = c^n n! \sum_{m>n} \frac{1}{c^m m!} = \frac{1}{c(n+1)} + O(n^{-2}).$$

The combined tail $T(n) = \lambda U_c(n) + \sum_f a_f T_{f,c}(n)$ is an integer for all large n. The scaled rational total is an integer eventually, and every scaled partial-sum term is an integer, since $c^{n-m} n!/m! \in \mathbb{Z}$ for $m \leq n$.

Use the grid in (8)–(9) with $K = H$, but choose $L > sH(H+1)/2$. Its geometric proof works for every such L, with $b = H(L-1)+1$ and the same formulas for F, D, B. Impose, for every f, $0 \le i < H$, $0 \le j \le i$, and coordinate-j fiber,

$$\sum_{e_j} w_e \frac{p_e^{i+1}}{f(p_e)} = 0. \tag{26}$$

There are at most $sH(H+1)L^{H-1}/2 < L^H$ equations. Since all denominators are positive integers, Lemma 2.2 gives nonzero integral weights w_e.

Choose $N \equiv A \pmod{Q}$, where $Q = \prod_e p_e^2$ and $A \equiv t_e \pmod{p_e^2}$. Then $n_e = (N - t_e)/p_e$ is an integer, $p_e \mid n_e$, and multiplicativity gives exactly

$$\frac{f(n_e + j + 1)}{(n_e + j + 1)^{i+1}} = \frac{p_e^{i+1}}{f(p_e)} \frac{f(N + s_{e,j})}{(N + s_{e,j})^{i+1}}.$$

The fiber equations remove every $i < H$ term. The eventual integer $W(N) = \sum_e w_e T(n_e)$ is therefore $O(N^{-1/2})$ and is eventually zero. Multiplication by N eliminates all functions with $h_f < H$ and the remainders, and forces

$$\sum_{h_f = H} \sum_e \sum_{j=0}^{H} \frac{a_f w_e p_e^{H+1} \left\{ {H \atop j} \right\}}{c^{j+1} f(p_e)} \rho_f(N + s_{e,j}) \longrightarrow -\frac{\lambda}{c} \sum_e w_e p_e. \tag{27}$$

Replacing $N/(N+s)$ by 1 costs $O(N^{-1/2})$, because $\rho_f(N+s) = O(N^{1/2})$.

Select a vertex e_* with $w_{e_*} \neq 0$. Its terminal shift $s_{e_*,H}$ occurs at no other grid location. For each retained top-degree function its coefficient there is $a_f w_{e_*} p_{e_*}^{H+1}/(c^{H+1} f(p_{e_*})) \neq 0$. Lemma 7.1 contradicts (27). If there was no nonzero a_f, the relation contradicts the irrationality of $e^{1/c}$: its cth power is e, whose elementary irrationality was proved above. This completes the proof. □

8 Polynomial Euler factors and all polynomial moments

Let $\mathcal{P}$ consist of polynomials $P \in \mathbb{Z}[X]$ satisfying

$$P(0) = 1, \qquad P \neq 1, \qquad P(1/p) > 0 \quad \text{for every prime } p. \tag{28}$$

Positivity on $0 < x \le 1/2$ is a convenient sufficient condition. For $P \in \mathcal{P}$ put $h_P = \deg P$ and

$$F_P(n) = n^{h_P} \prod_{p \mid n} P(1/p). \tag{29}$$

Distinct polynomials are used only once. This convention avoids counting two factorizations of the same polynomial as different arithmetic functions.

Theorem 8.1 (All moments of polynomial Euler factors)**.** *For every fixed nonzero integer c, the following entire family is linearly independent over $\mathbb{Q}$:*

$$\{1, e^{1/c}\} \cup \left\{ \sum_{n \ge 1} \frac{n^d \sigma_k(n)}{c^n n!} : k \ge 1,\ d \ge 0 \right\} \cup \left\{ \sum_{n \ge 1} \frac{n^d F_P(n)}{c^n n!} : P \in \mathcal{P},\ d \ge 0 \right\}. \tag{30}$$

The statement is at one fixed c at a time; it does not assert joint independence across different evaluation points $1/c$.

Lemma 8.2 (Growth and means for polynomial Euler factors)**.** *The function F_P is positive integer-valued and multiplicative. For every $\epsilon > 0$, $F_P(n) = O_{P,\epsilon}(n^{h_P+\epsilon})$. Its normalized phase $\rho_P(n) = F_P(n)/n^{h_P}$ has the positive progression mean*

$$M_P(Q,a) = \prod_{p\nmid Q}\left(1+\frac{P(1/p)-1}{p}\right)\prod_{p\mid \gcd(Q,a)} P(1/p). \tag{31}$$

Its off-factor and response are

$$b_P(z) = \frac{1}{1+z(P(z)-1)}, \qquad \delta_P(z) = \frac{P(z)-1}{1+z(P(z)-1)}. \tag{32}$$

Proof. For $a \geq 1$, $F_P(p^a) = p^{h_P(a-1)}[p^{h_P}P(1/p)]$ is a positive integer. This proves integrality and multiplicativity. Since $P(1/p) = 1 + O_P(p^{-1})$, each sufficiently large prime factor contributes at most p^ϵ; the finitely many smaller primes give a fixed constant. This proves the growth bound.

Define $a_P(1) = 1$, set $a_P(d) = 0$ for nonsquarefree d, and on squarefree d put $a_P(d) = \prod_{p\mid d}(P(1/p)-1)$. Then

$$\rho_P(n) = \sum_{d\mid n} a_P(d), \qquad |a_P(d)| \ll_{P,\epsilon} d^{-1+\epsilon}.$$

For the bound, use $|P(1/p)-1| \leq C_P/p$ and $C_P^{\omega(d)} \ll_{P,\epsilon} d^\epsilon$ on squarefree integers; the latter follows by separating the finitely many primes with $p^\epsilon < C_P$.

Counting divisibility in an arithmetic progression now yields

$$M_P(Q,a) = \sum_{\gcd(d,Q)\mid a} a_P(d)\frac{\gcd(d,Q)}{d}.$$

To justify passage to the limit, for a cutoff Y bound the remaining average in absolute value by

$$Q\sum_{d>Y}\frac{|a_P(d)|}{d} + \frac{1}{X}\sum_{d\leq QX+a}|a_P(d)|.$$

Choose $0 < \epsilon < 1$. The second term is $O(X^{-1+\epsilon})$ and the first tends to zero with Y. The absolutely convergent Euler product is exactly (31). Every factor is positive: an unrestricted factor is $(1-1/p) + P(1/p)/p$. Since it is $1 + O_P(p^{-2})$, the infinite product is strictly positive.

At a new prime, replace the unrestricted local factor $1 + (P(1/p)-1)/p$ by 1 in the off case and by $P(1/p)$ in the on case. Their ratios and difference give (32). □

Proof of Theorem 8.1. Assign degree $d+k$ to $n^d\sigma_k(n)$ and degree $d+h_P$ to $n^dF_P(n)$. These functions are positive integer-valued and multiplicative, with the growth required by Theorem 7.2. Their normalized phases are independent of d. The divisor phase has the previously proved off-factor and response

$$b_{\sigma,k}(z) = 1 - z^{k+1}, \qquad \delta_{\sigma,k}(z) = z^k.$$

The polynomial Euler phases satisfy Lemma 8.2.

Their response functions are pairwise distinct and nonzero. Indeed the transformation $H \mapsto H/(1+zH)$ has inverse $\delta \mapsto \delta/(1-z\delta)$. For δ_P this inverse is the nonzero polynomial $P-1$, so distinct P give distinct responses. For $\delta_{\sigma,k} = z^k$ the inverse is $z^k/(1-z^{k+1})$, which is not a polynomial. Thus a divisor response cannot equal a polynomial Euler response, and different divisor indices give different monomials. At any fixed total degree, a given P or k determines its weight d uniquely, so no duplicate response occurs in that degree. Apply Theorem 7.2. □

8.1 Named arithmetic functions included in the theorem

The statement is not limited to newly defined examples. The standard Jordan and generalized Dedekind functions are

$$J_k(n) = n^k \prod_{p|n}(1-p^{-k}), \qquad \Psi_k(n) = n^k \prod_{p|n}(1+p^{-k}).$$

Here $J_1 = \varphi$ and $\Psi_1 = \psi$, the ordinary Dedekind psi function. They correspond to $P(X) = 1 - X^k$ and $1 + X^k$. More generally, for nonnegative integers u_k, v_k of finite support, not all zero,

$$P(X) = \prod_k (1-X^k)^{u_k}(1+X^k)^{v_k} \quad \Longrightarrow \quad F_P(n) = \prod_k J_k(n)^{u_k}\Psi_k(n)^{v_k}. \tag{33}$$

The required positivity holds at every $1/p$. Different descriptions in (33) can represent the same P, so equality of polynomials must be checked first. For instance $J_k(n)\Psi_k(n) = J_{2k}(n)$; these are one function, not two independent coordinates.

The theorem therefore covers all polynomial moments of all positive powers $\varphi(n)^m$ and $\psi(n)^m$, and of every distinct product in (33), jointly with all divisor-sum moments. For example it proves the irrationality of

$$\sum_{n\geq 1} \frac{(n^2-3n+1)\sigma_2(n) + 2\varphi(n)^3 - 5\psi(n)^2}{n!}. \tag{34}$$

The three nonpolynomial arithmetic coordinates here have distinct base functions, and their displayed coefficients do not all vanish.

Erdős and Straus [3], Theorem 3.7, already imply the joint independence of $1, e, \sum \sigma(n)/n!$, and $\sum \varphi(n)/n!$. Thus the degree-one specialization is not claimed as new. The all-moment and full polynomial-Euler-factor statements above are additional results of this manuscript, whose historical priority remains subject to literature review.

9 All derivative values and an exact rationality criterion

For $k \geq 1$ and $P \in \mathcal{P}$ define the entire functions

$$A_k(z) = \sum_{n\geq 1} \frac{\sigma_k(n)}{n!} z^n, \qquad E_P(z) = \sum_{n\geq 1} \frac{F_P(n)}{n!} z^n.$$

Their entireness and termwise differentiability follow from the polynomial growth bounds. Pratt [2] explicitly discussed the divisor-sum moment functions and their nonclosing derivative hierarchy. The next corollary addresses rational linear relations among all these derivative values at reciprocal integers. It does not establish Pratt's broader expectation of transcendence at every nonzero algebraic argument.

Corollary 9.1 (Joint independence of all derivative values)**.** *For each fixed $c \in \mathbb{Z} \setminus \{0\}$, the family*

$$\{1, e^{1/c}\} \ \cup \ \{A_k^{(j)}(1/c) : k \geq 1,\ j \geq 0\} \ \cup \ \{E_P^{(j)}(1/c) : P \in \mathcal{P},\ j \geq 0\}$$

is linearly independent over $\mathbb{Q}$.

Proof. For either numerator function f, termwise differentiation gives

$$\left(\sum_{n\geq 1} \frac{f(n)z^n}{n!}\right)^{(j)}_{z=1/c} = c^j \sum_{n\geq 1} \frac{n^{\underline{j}} f(n)}{c^n n!}.$$

The falling factorial $X^{\underline{j}}$ is a monic polynomial of degree j. The change from powers to falling factorials is triangular over $\mathbb{Q}$ with diagonal entries 1; the additional factors c^j are nonzero rational numbers. Thus every finite proposed derivative relation becomes a nontrivial relation in Theorem 8.1. □

In particular $A_1'(1)/A_1(1)$ is irrational: the denominator is positive, and a rational ratio would be a forbidden relation between two derivative values. The inclusion of $j = 0$ is essential; these coordinates are the original factorial constants.

For the exact criterion below, write

$$T_d(x) = \sum_{j=0}^{d} \left\{ {d \atop j} \right\} x^j, \qquad \mathcal{B}_c(R) = \sum_d r_d T_d(1/c) \quad \text{if } R(X) = \sum_d r_d X^d.$$

These are the Touchard polynomials; at $x = 1$ they give Bell numbers. For rational R and integer $c \neq 0$, $\mathcal{B}_c(R)$ is rational and is computable by finite exact arithmetic.

Corollary 9.2 (Complete rationality test for this class)**.** *Let $R, U_k, V_P \in \mathbb{Q}[X]$, only finitely many nonzero, with equal polynomial Euler factors grouped together. Set*

$$S = \sum_{n \geq 1} \frac{R(n) + \sum_k U_k(n)\sigma_k(n) + \sum_P V_P(n)F_P(n)}{c^n n!}.$$

Then $S \in \mathbb{Q}$ if and only if every U_k, V_P is the zero polynomial and $\mathcal{B}_c(R) = 0$. In that case $S = -R(0)$.

Proof. Expanding n^d in falling factorials and summing gives

$$\sum_{n \geq 1} \frac{R(n)}{c^n n!} = e^{1/c}\mathcal{B}_c(R) - R(0).$$

Expand the remaining polynomial coefficients into powers of n, and apply the joint independence in Theorem 8.1. □

The qualification about the polynomial part is necessary: $\sum_{n \geq 1}(n-1)/n! = 1$. Thus a nonzero polynomial numerator alone need not produce an irrational value. By contrast, any nonzero arithmetic polynomial coefficient in the canonical form of Corollary 9.2 forces irrationality.

This is a complete classification inside the displayed class, not a decision procedure for arbitrary convergent series. It makes no claim about Euler's constant γ, Robin's inequality, or the transcendence of an individual factorial constant.

10 A concrete value at two and a denominator bound

We now prove the concrete irrationality statement

$$\boxed{C_* := \sum_{n=1}^{\infty} \frac{2^n \sigma_5(n)}{n!} \notin \mathbb{Q}.} \tag{35}$$

This is $A_5(2)$ in the notation of Section 9. It is not an instance of the earlier reciprocal-integer evaluation theorem: 2 cannot be written $1/c$ with c a nonzero integer. An additional denominator estimate, proved below, supplies the extension.

Historical scope. Pratt [2] studies the functions $A_k(z)$ and discusses their expected transcendence at nonzero algebraic arguments. The stated geometric theorem in [5] treats $z = 1/c$ with integral $c \neq 0$. The sources examined did not supply an earlier proof of (35). However, we did not locate a primary source explicitly designating this particular value as a previously open, named irrationality problem. Consequently we claim a concrete irrationality theorem and an extension of the stated evaluation range, not a certified resolution of a separately named historical conjecture. In particular this is not a solution of the full algebraic-point transcendence expectation, nor of Erdős Problems 68, 249, or 1049.

Let $s_2(n)$ be the number of ones in the binary expansion of the nonnegative integer n. We will use

$$v_2(n!) = n - s_2(n), \qquad 2^{s_2(n)} \leq n + 1. \tag{36}$$

For the first identity, sum $\lfloor n/2^j \rfloor$ and expand n in binary. For the second, the least nonnegative integer with s binary ones is $2^s - 1$.

Lemma 10.1 (A denominator bound for a normalized tail)**.** *Fix $x = \varepsilon 2/b$, where $b \geq 1$ is an integer and $\varepsilon \in \{1, -1\}$. Suppose that integers a_m of polynomial growth satisfy $\sum_{m\geq 0} a_m x^m/m! = u/q \in \mathbb{Q}$, with $q \geq 1$. Put*

$$T_x(n) = n! x^{-n} \sum_{m>n} \frac{a_m x^m}{m!}.$$

Then $q2^{s_2(n)} T_x(n) \in \mathbb{Z}$ for every $n \geq 0$.

Proof. Subtract the finite partial sum from u/q and multiply by $n!x^{-n}$. The total term becomes integral after multiplication by $q2^{s_2(n)}$, by (36). For $0 \leq m \leq n$, the scaled partial-sum coefficient is

$$\frac{n!}{m!} x^{m-n} = \varepsilon^{m-n} \frac{n!}{m!} \frac{b^{n-m}}{2^{n-m}}.$$

There are no odd denominator factors, and its 2-adic valuation is at least

$$v_2(n!) - v_2(m!) - (n - m) = s_2(m) - s_2(n) \geq -s_2(n).$$

Multiplication by $2^{s_2(n)}$ therefore clears every partial term. The proof also permits even b; those cases include the old reciprocal-integer arguments. □

Lemma 10.2. *For every $x = \varepsilon 2/b$ as above, e^x is irrational.*

Proof. If $e^x = u/q$ were rational, apply Lemma 10.1 to $a_m = 1$ and write the resulting tail as $U_x(n)$. Its first terms give

$$U_x(n) = \frac{x}{n} + \frac{x^2 - x}{n^2} + O(n^{-3}), \qquad U_x(n) - U_x(n+1) = \frac{x}{n^2} + O(n^{-3}).$$

The difference is nonzero eventually. On the other hand, it lies in the lattice with denominator $q2^{\max(s_2(n), s_2(n+1))} \leq q(n + 2)$. Multiplication by that denominator yields a nonzero integer tending to zero, a contradiction. The tail expansion follows by keeping the first two factorial terms and bounding the remaining terms by $O(n^{-3})$; it is valid for either sign of x. □

Theorem 10.3 (Evaluation at rational points of numerator at most two)**.** *Let $\mathcal{F}_*$ denote the numerator functions*

$$\{n \mapsto n^d \sigma_k(n) : d \geq 0, k \geq 1\} \ \cup \ \{n \mapsto n^d F_P(n) : d \geq 0, P \in \mathcal{P}\}.$$

For every fixed $x = \varepsilon 2/b$, $b \geq 1$ and $\varepsilon \in \{1, -1\}$, the family

$$\{1, e^x\} \ \cup \ \left\{ \sum_{n \geq 1} \frac{f(n)x^n}{n!} : f \in \mathcal{F}_* \right\}$$

is linearly independent over $\mathbb{Q}$. Equivalently, this covers every nonzero rational argument whose reduced numerator has absolute value one or two, at one fixed argument at a time.

Proof. Consider a nontrivial integral-coefficient relation $\lambda e^x + \sum_f a_f \sum_{m \geq 1} f(m)x^m/m! \in \mathbb{Q}$. If all $a_f = 0$, use Lemma 10.2. Otherwise retain the finitely many functions with $a_f \neq 0$, and let $K \geq 1$ be their largest growth degree. Write the rational total as u/q. The combined coefficients, including the $m = 0$ exponential term, are integers. Thus their combined normalized tail $T_x(n)$ obeys Lemma 10.1.

Set $H = K + 1$. In addition to the retained arithmetic functions, include $g_0(n) = 1$ in the fiber constraints. Choose a grid of dimension H and side length $L > (s+1)H(H+1)/2$, where s is the number of retained arithmetic functions. Lemmas 4.1 and 2.2 supply nonzero integral weights satisfying

$$\sum_{e_j} w_e \frac{p_e^{i+1}}{g(p_e)} = 0 \quad (0 \leq i < H, \ 0 \leq j \leq i)$$

on each coordinate-j fiber, for every retained $g = f$ and for g_0. The squared CRT alignment and the shifts are exactly as in Theorem 7.2.

For every such g, the centered expansion through order H is

$$T_{g,x}(n) = \sum_{j=0}^{H} \sum_{i=j}^{H} x^{j+1} \left\{ {i \atop j} \right\} \frac{g(n+j+1)}{(n+j+1)^{i+1}} + O(n^{-5/2}). \tag{37}$$

The error follows from $g(n) = O(n^{K+1/2})$ and $K + 1/2 - H - 2 = -5/2$. The same geometric tail estimate applies for fixed $|x| \leq 2$.

Along the aligned progression, put $n_e = (N - t_e)/p_e$ and $W(N) = \sum_e w_e T_x(n_e)$. All $i < H$ terms vanish, including those of the exponential. Each remaining arithmetic term has size $O(N^{K+1/2-H-1}) = O(N^{-3/2})$, so

$$W(N) = O(N^{-3/2}).$$

Unlike the earlier proof, $W(N)$ need not itself be an integer. Its common denominator divides

$$D_N = q 2^{\max_e s_2(n_e)} \leq q(1 + \max_e n_e) = O(N).$$

All defects are powers of the same prime, so a maximum suffices; one does not multiply the individual denominators. Therefore $D_N W(N) \in \mathbb{Z}$ and $D_N W(N) = O(N^{-1/2}) \to 0$. It follows that $W(N) = 0$ eventually.

Multiply the remaining identity by N^2. The errors are $O(N^{-1/2})$, all degrees below K disappear, and the top degrees give

$$\sum_{h_f = K} \sum_{e} \sum_{j=0}^{H} \frac{a_f w_e p_e^{H+1} x^{j+1} \left\{ {H \atop j} \right\}}{f(p_e)} \rho_f(N + s_{e,j}) \longrightarrow 0. \tag{38}$$

Indeed $H = K + 1$, so the retained normalized factor before taking the limit is $N^2/(N+s)^2$; replacing it by 1 costs $O(N^{-1/2})$. The g_0 contribution is $O(N^{-K})$ and vanishes. Choose e_* with $w_{e_*} \neq 0$. Its terminal shift is unique, and a top-degree coefficient there is $a_f w_{e_*} p_{e_*}^{H+1} x^{H+1}/f(p_{e_*}) \neq 0$. The distinct response functions proved in Theorem 8.1 and Lemma 7.1 contradict (38). □

Taking $x = 2$, $d = 0$, and $k = 5$ proves (35).

For identification, exact rational interval arithmetic gives

$$C_* = 3412.81774524922338054544082302233096428831099819146187\ldots.$$

The first 50 decimal places are certified by the bounds

$$S_{80} < C_* < S_{80} + \frac{4 \cdot 2^{81} 81^5}{81!}, \qquad S_{80} = \sum_{n=1}^{80} \frac{2^n \sigma_5(n)}{n!}.$$

Indeed $\sigma_5(n) < 2n^5$, and the consecutive-term ratio of the majorant $2^{n+1}n^5/n!$ is $2(n+1)^4/n^5$, decreasing and below $1/2$ for $n \geq 81$. This is an identification certificate, not the irrationality proof.

The derivative and rationality-classification corollaries also extend to each permitted x: replace c^j by x^{-j} in the derivative conversion and evaluate the Touchard polynomials at x.

The new argument has a precise limitation. If the reduced numerator of x contains a prime power p^a with $a > 1/(p-1)$, then the corresponding defect in $n!x^{-n}$ has valuation

$$v_p(n!) - an = -\left(a - \frac{1}{p-1}\right) n + O(\log n).$$

Its denominator grows exponentially, so the fixed extra cancellation order used here does not clear it. No extension to numerator three, to all rational arguments, or to all algebraic arguments is asserted.

11 Prime-power profiles: products, radicals, and counting functions

Divisibility by a fresh prime observes only one average of the prime-power values of a multiplicative function. A refinement that specifies the exact valuation observes an individual prime-power value. This extra freedom gives a substantially larger class than polynomial Euler factors or single divisor sums.

11.1 An admissible profile and its progression means

Write $R \in \mathbb{Q}(z)[w]$ for a polynomial in w whose coefficients are rational functions of z. We call (R, h) admissible if $h \geq 1$ is an integer, the coefficients of R have no poles at $z = 1/p$ for primes p, and the following conditions hold:

1. There is a constant C_R such that
$$0 < R(1/p, p^{-a}) \leq C_R \qquad (p \text{ prime},\ a \geq 1).$$
2. $R(z, z) = 1 + O(z)$ as $z \to 0$ through positive real values.
3. $p^{ha} R(1/p, p^{-a})$ is an integer for every prime p and every $a \geq 1$.

Define the positive integer-valued multiplicative function $f_{R,h}$ by

$$f_{R,h}(1) = 1, \qquad f_{R,h}(p^a) = p^{ha} R(1/p, p^{-a}), \qquad \rho_R(n) = \frac{f_{R,h}(n)}{n^h}. \tag{39}$$

The normalized function ρ_R is independent of h. For each $\epsilon > 0$, boundedness of the local factors gives $\rho_R(n) = O_{R,\epsilon}(n^\epsilon)$: separate the finitely many primes for which $C_R > p^\epsilon$. Thus the factorial series below converge absolutely and define entire functions.

Lemma 11.1 (Means from a bounded prime-power profile)**.** *For every admissible profile, the normalized function ρ_R has a finite, strictly positive mean $M_R(Q,a)$ on every arithmetic progression. Put $\rho_R(p^0)=1$, and write $R(z,w)=\sum_{j=0}^{d} r_j(z)w^j$. The unrestricted local mean is the rational function*

$$U_R(z)=(1-z)\left(1+\sum_{j=0}^{d} r_j(z)\frac{z^{j+1}}{1-z^{j+1}}\right). \tag{40}$$

At a fresh prime $\ell\nmid Q$, the ratios of refined to original means are $1/U_R(1/\ell)$ when the new residue is a unit modulo ℓ, and $R(1/\ell,\ell^{-t})/U_R(1/\ell)$ when its valuation is exactly $t\geq 1$.

Proof. Let $g=\rho_R*\mu$, where $*$ denotes Dirichlet convolution. Then g is multiplicative and

$$g(p)=O_R(p^{-1}),\qquad g(p^a)=\rho_R(p^a)-\rho_R(p^{a-1})=O_R(1)\quad (a\geq 2).$$

Consequently

$$\sum_{a\geq 1}\frac{|g(p^a)|}{p^a}=O_R(p^{-2}),\qquad \sum_{d\geq 1}\frac{|g(d)|}{d}<\infty.$$

The second assertion follows by multiplying the absolutely convergent nonnegative Euler factors; the finitely many small primes also have finite factors. Since $\rho_R(n)=\sum_{d\mid n} g(d)$, counting divisibility along a progression gives

$$M_R(Q,a)=\sum_{\gcd(d,Q)\mid a} g(d)\frac{\gcd(d,Q)}{d}. \tag{41}$$

Here is a justification that requires no pointwise bound on $g(d)$. The contribution beyond a fixed cutoff Y to a progression average of length X is bounded by

$$Q\sum_{d>Y}\frac{|g(d)|}{d}+\frac{1}{X}\sum_{d\leq QX+|a|}|g(d)|.$$

The second term tends to zero. Indeed split its sum at Y; the fixed initial part is $o(X)$, while the rest divided by X is at most $(Q+o(1))\sum_{d>Y}|g(d)|/d$. Let first X, then Y, tend to infinity. This also justifies the absolutely convergent series in (41).

Its Euler factors are particularly transparent as local conditional expectations. At $p\nmid Q$ the factor is

$$(1-p^{-1})\sum_{v\geq 0}\rho_R(p^v)p^{-v}=U_R(1/p).$$

At $p^s\|Q$, if $v_p(a)=t<s$, the factor is $\rho_R(p^t)$. If $p^s\mid a$, it is $(1-p^{-1})\sum_{v\geq s}\rho_R(p^v)p^{-(v-s)}$. Every factor is positive. The unrestricted factors are $1+O_R(p^{-2})$, so their product is strictly positive. Expanding R in powers of w and summing geometric series gives (40). In particular $U_R(z)=1+O_R(z^2)$ near $z=0$: this holds along $z=1/p$, and the rational function $U_R(z)-1$ therefore has order at least two at zero. Fixing the exact valuation at a fresh prime replaces its unrestricted factor by the corresponding single value $\rho_R(\ell^t)$; making it a unit replaces that factor by 1. □

11.2 A vector of valuation responses separates every profile

For $t\geq 1$ define the rational function

$$\delta_{R,t}(z)=\frac{R(z,z^t)-1}{U_R(z)}. \tag{42}$$

It is the relative difference between an exact-valuation-t mean and a unit mean. Individual responses may vanish or coincide. The entire vector of responses has neither defect once identical profiles have been grouped.

Lemma 11.2 (Injectivity of valuation responses)**.** *If $\delta_{R,t} = \delta_{S,t}$ as rational functions for every $t \geq 1$, then $R = S$ in $\mathbb{Q}(z)[w]$. The response vector of R vanishes identically if and only if $R = 1$.*

Proof. For sufficiently small real $0 < z < 1$, the geometric series converge and the denominators are nonzero. From (40),

$$\sum_{t \geq 1} \delta_{R,t}(z) z^t = \frac{U_R(z) - 1}{(1 - z) U_R(z)}.$$

Equality of response vectors therefore implies $U_R = U_S$ and then $R(z, z^t) = S(z, z^t)$ for every t. For each such fixed z, the polynomial $R(z, w) - S(z, w)$ has infinitely many distinct roots $w = z^t$, hence is zero. Its rational coefficient functions vanish on an interval, so they are identically zero. The zero-vector assertion is the special case $S = 1$, for which $U_S = 1$. □

Lemma 11.3 (Separation by fresh prime powers)**.** *Let $R_1, \ldots, R_s$ be pairwise distinct admissible profiles, none equal to 1. After identical shifts have been grouped, any finite sum of their shifted normalized functions with at least one nonzero coefficient cannot tend to a finite constant on a progression.*

Proof. Group repeated shifts and choose a shift r_* with a nonzero coefficient vector. Suppose the sum has limit L on $N \equiv A \pmod{Q}$. Write its coefficients at r_* as β_j, and put $B_j = \beta_j M_{R_j}(Q, A + r_*)$. Choose all fresh primes larger than the absolute differences between distinct shifts and larger than the number of shifts. For a chosen prime ℓ and depth $t \geq 1$, the on state is

$$N + r_* \equiv \ell^t \pmod{\ell^{t+1}}.$$

It has exactly the desired valuation and makes every other shift a unit modulo ℓ. Choose an off residue modulo ℓ missing all shifts, and lift it to a residue modulo ℓ^{t+1}. It makes all shifts units. The Chinese remainder theorem allows these two states independently at distinct primes while preserving the base progression.

Take progression means on the resulting refinements. For any nonempty finite set S of selected primes, the alternating sum over its $2^{|S|}$ states cancels the constant limit and every shift other than r_*. Lemma 11.1 gives

$$\sum_{j=1}^{s} B_j \prod_{\ell \in S} \delta_{R_j, t_\ell}(1/\ell) = 0. \tag{43}$$

For each subset only its own primes refine the original progression; thus the same B_j occur in all these identities.

Fix a target index j_0. By Lemma 11.2, choose a depth t_0 for which $\delta_{R_{j_0}, t_0}$ is nonzero. For each $j \neq j_0$, choose a depth t_j for which $\delta_{R_{j_0}, t_j} - \delta_{R_j, t_j}$ is nonzero. Choose distinct fresh primes ℓ_0, ℓ_j avoiding the finitely many zeros and poles of these rational functions. A linear combination of (43), over subsets containing ℓ_0, yields

$$\sum_{v} B_v \delta_{R_v, t_0}(1/\ell_0) \prod_{j \neq j_0} \left(\delta_{R_v, t_j}(1/\ell_j) - \delta_{R_j, t_j}(1/\ell_j) \right) = 0.$$

Every term except $v = j_0$ vanishes, and the remaining multiplier is nonzero. Thus $B_{j_0} = 0$. All progression means are positive, so all coefficients at the distinguished shift vanish, a contradiction. □

11.3 The larger irrationality theorem

Theorem 11.4 (All moments of distinct prime-power profiles)**.** *Let $\mathcal{R}$ be any collection of pairwise distinct profiles $R \neq 1$, and choose for each R an integer $h_R \geq 1$ such that (R, h_R) is admissible. For each fixed $x = \varepsilon 2/b$, where $b \geq 1$ and $\varepsilon \in \{1, -1\}$, the family*

$$\{1, e^x\} \ \cup \ \left\{ \sum_{n \geq 1} \frac{n^d f_{R,h_R}(n) x^n}{n!} : R \in \mathcal{R}, \ d \geq 0 \right\} \tag{44}$$

is linearly independent over $\mathbb{Q}$. The assertion is at one fixed evaluation point at a time.

Proof. Only finitely many coordinates occur in a proposed relation. Each numerator $f(n) = n^d f_{R,h_R}(n)$ is positive integer-valued and multiplicative, has degree $h_R + d \geq 1$, and satisfies $f(n) = O(n^{h_R+d+1/2})$. Its normalized phase is precisely ρ_R. These are all the arithmetic and growth inputs used in the proof of Theorem 10.3 before its final separation step.

For completeness, let K be the largest retained degree and use the same grid with $H = K + 1$, including 1 in the cancellation constraints. The normalized-tail expansion has error $O(n^{-5/2})$. All lower fiber terms cancel, giving $W(N) = O(N^{-3/2})$. Its denominator divides $q 2^{\max_e s_2(n_e)} = O(N)$, so $W(N) = 0$ eventually. Multiplication by N^2 gives exactly the limiting identity (38), now with phases ρ_R. At degree K, each profile occurs at most once: its weight is uniquely $d = K - h_R$. The terminal shift has a nonzero coefficient because the grid weight and $f(p_e)$ are nonzero. Apply Lemma 11.3 to contradict that limiting identity. If no arithmetic coordinate was present, use Lemma 10.2. □

The existing derivative and Touchard-polynomial arguments now apply to this entire class. In particular, if $F_R(z) = \sum_{n \geq 1} f_{R,h_R}(n) z^n/n!$, all numbers $F_R^{(j)}(x)$, together with $1, e^x$, are jointly independent. For a finite expression

$$S = \sum_{n \geq 1} \left[V(n) + \sum_R V_R(n) f_{R,h_R}(n) \right] \frac{x^n}{n!}, \qquad V, V_R \in \mathbb{Q}[X],$$

the exact rationality criterion is: every V_R vanishes, and $\sum_d v_d T_d(x) = 0$ for $V(X) = \sum_d v_d X^d$. Then $S = -V(0)$. Profiles must first be grouped by equality in $\mathbb{Q}(z)[w]$; this is an exact finite algebraic test after clearing the z-denominators. If the same profile is written with different base degrees, first factor out the corresponding power of n.

11.4 Named families covered by the theorem

The following table gives the growth degree and the normalized profile. All parameters are positive integers unless otherwise specified. The notation $p^a \| n$ specifies the exact prime power dividing n.

Numerator function	h	$R(z, w)$
$\sigma_k(n)$	k	$(1 - z^k w^k)/(1 - z^k)$
$J_k(n)$	k	$1 - z^k$
$\Psi_k(n)$	k	$1 + z^k$
$\sigma_k^*(n) = \prod_{p^a \| n}(1 + p^{ak})$	k	$1 + w^k$
$J_k^*(n) = \prod_{p^a \| n}(p^{ak} - 1)$	k	$1 - w^k$
$\mathrm{rad}(n) = \prod_{p \mid n} p$	1	w/z
$a_r(n)$, index-n subgroups of $\mathbb{Z}^r$, $r \geq 2$	$r - 1$	$\displaystyle\prod_{j=1}^{r-1} \frac{1 - z^j w}{1 - z^j}$
$g(n) = \sum_{d \mid n} d\varphi(d)$	2	$(1 + zw^2)/(1 + z)$

The original polynomial Euler factors are the profiles $R(z,w)=P(z)$. Every finite pointwise product of entries in the table and of such factors is again admissible: multiply the profiles and add the degrees. This includes arbitrary powers and mixed products of divisor sums, ordinary and unitary Jordan or Dedekind factors, radicals, and subgroup counts. Positivity and integrality follow from their definitions. Each displayed profile is uniformly bounded at $(z,w)=(1/p,p^{-a})$ and equals $1+O(z)$ when $w=z$. For example, $w/z=p^{1-a}\le 1$, so the pole of this coefficient at $z=0$ causes no difficulty. All assertions survive multiplication of any fixed finite number of factors.

These are products of arithmetic functions *inside* a factorial series. The theorem does not assert algebraic independence of the resulting constants or irrationality of their arbitrary products. Nor may algebraically identical profiles be counted as separate coordinates. For example,

$$J_k\Psi_k=J_{2k},\qquad J_k^*\sigma_k^*=J_{2k}^*,\qquad a_2=\sigma_1.$$

Here is a useful nonredundant subfamily. Let P range over the polynomials of Section 8, now permitting $P=1$, and let $\mathbf{m}=(m_1,m_2,\dots)$ be a finitely supported sequence of nonnegative integers. Exclude $(P,\mathbf{m})=(1,\mathbf{0})$ and put

$$f_{P,\mathbf{m}}(n)=F_P(n)\prod_{k\ge1}\sigma_k(n)^{m_k},\quad F_1(n)=1,\quad h=\deg P+\sum_k km_k.$$

Its profile is

$$R_{P,\mathbf{m}}(z,w)=P(z)\prod_k\left(\frac{1-z^kw^k}{1-z^k}\right)^{m_k}. \tag{45}$$

All these profiles are distinct. Indeed equality of two such profiles, first at $w=0$ and then at arbitrary w, would give $\prod_k(1-(zw)^k)^{m_k-n_k}=1$. The largest index with a nonzero exponent has a cyclotomic factor not supplied by smaller indices, a contradiction. Thus $\mathbf{m}=\mathbf{n}$, and then $P=Q$. Consequently this whole family, with every polynomial moment, is covered simultaneously without further duplicate tests.

11.5 Lattice counts, least common multiples, and explicit constants

For clarity the classical subgroup formula [8] in the table can be obtained without using its interpretation as a black box. Hermite normal form gives

$$a_r=1*\mathrm{id}*\mathrm{id}^2*\cdots*\mathrm{id}^{r-1},\qquad \sum_{a\ge0}a_r(p^a)T^a=\prod_{j=0}^{r-1}(1-p^jT)^{-1}.$$

The coefficient is the Gaussian binomial

$$a_r(p^a)=p^{(r-1)a}\prod_{j=1}^{r-1}\frac{1-p^{-(a+j)}}{1-p^{-j}},$$

which proves the displayed profile and its integrality. Distinct $r\ge2$ give distinct profiles by their different degrees in w. In particular

$$\sum_{n\ge1}\frac{a_3(n)2^n}{n!}\notin\mathbb{Q},$$

and all its derivative values join the same independent family. This is a factorial-weighted count of finite-index sublattices of the ordinary integer lattice $\mathbb{Z}^3$.

Define the standard least-common-multiple sum [10] $L(n) = \sum_{j=1}^{n} \operatorname{lcm}(j,n)$. Grouping terms by $\gcd(j,n)$, and using the pairing of reduced residues modulo m, gives the exact identity

$$L(n) = \frac{n}{2}\left(1 + \sum_{d\mid n} d\varphi(d)\right) = \frac{n}{2}(1+g(n)). \tag{46}$$

For $m > 1$ the reduced residues sum to $m\varphi(m)/2$; the term $m = 1$ contributes 1, which accounts for the extra 1 above. The multiplicative function g has

$$g(p^a) = 1 + \sum_{j=1}^{a} p^j(p^j - p^{j-1}) = \frac{p^{2a+1}+1}{p+1},$$

so its profile is $(1+zw^2)/(1+z)$, as claimed. Therefore

$$\sum_{n\ge 1} \frac{x^n}{n!} \sum_{j=1}^{n} \operatorname{lcm}(j,n) = \frac{xe^x}{2} + \frac{1}{2}\sum_{n\ge 1} \frac{ng(n)x^n}{n!} \notin \mathbb{Q} \tag{47}$$

at every permitted x. Its mean connection with zeta values is also exact:

$$U_g(z) = \frac{1-z^2}{1-z^3}, \qquad \lim_{X\to\infty} \frac{1}{X} \sum_{n\le X} \frac{g(n)}{n^2} = \frac{\zeta(3)}{\zeta(2)}.$$

The appearance of this ratio in the mean does not establish its irrationality; the theorem concerns the factorial series.

For additional concrete instances, each of

$$\sum_{n\ge 1} \frac{2^n \operatorname{rad}(n)}{n!}, \qquad \sum_{n\ge 1} \frac{2^n \sigma_1(n)^2}{n!}, \qquad \sum_{n\ge 1} \frac{2^n \varphi(n)^3 \sigma_5(n)^2}{n!}$$

is irrational, and their polynomially weighted versions are covered by the same theorem. The corresponding series with unitary totients J_1^* or unitary divisor sums σ_1^* are also covered. These examples illustrate the breadth of the proved implication; their individual historical open status and priority have not been established by a literature search.

12 Every residue class, signed twists, and four real transforms

A periodic restriction is not absorbed by changing the evaluation point in the preceding theorem. The grid can nevertheless preserve an entire residue class, giving independence of the separate pieces of every series. For an integer $q \ge 1$, a residue $0 \le r < q$, and $x \ne 0$, put

$$E_{q,r}(x) = \sum_{\substack{n\ge 0 \\ n\equiv r \ (q)}} \frac{x^n}{n!}, \qquad B_{f,q,r}(x) = \sum_{\substack{n\ge 1 \\ n\equiv r \ (q)}} \frac{f(n)x^n}{n!}.$$

The $n = 0$ convention is intentional: exponential terms include it, whereas arithmetic functions are defined on positive integers.

Lemma 12.1 (Periodic exponential independence)**.** *For every fixed $q \ge 1$ and $x = \varepsilon 2/b$, $b \ge 1$, $\varepsilon \in \{1,-1\}$, the numbers $1, E_{q,0}(x), \dots, E_{q,q-1}(x)$ are linearly independent over $\mathbb{Q}$.*

Proof. A nontrivial relation would give a nonzero integer-valued q-periodic sequence $a(n)$ such that $\sum_{n\ge 0} a(n)x^n/n! = u/v \in \mathbb{Q}$. Define $U(n) = n!x^{-n}\sum_{m>n} a(m)x^m/m!$. Lemma 10.1 gives

$v2^{s_2(n)}U(n) \in \mathbb{Z}$. Since a is bounded, expansion of its first two tail terms, followed by a geometric bound for the rest, gives

$$U(n) = \frac{xa(n+1)}{n} + \frac{x^2a(n+2) - xa(n+1)}{n^2} + O(n^{-3}).$$

Choose a residue of n on which $a(n+1) \neq 0$. Periodicity makes both displayed coefficients identical for n and $n+q$, whence on this infinite progression

$$U(n) - U(n+q) = \frac{qxa(n+1)}{n^2} + O(n^{-3}).$$

This difference is eventually nonzero. Multiplying it by $v2^{\max(s_2(n),s_2(n+q))} \leq v(n+q+1)$ gives a nonzero integer tending to zero. The contradiction proves the claim. The term $a(0)$ is already included in the denominator lemma and creates no exception. □

Theorem 12.2 (Periodic transfer principle)**.** *Let $\mathcal{F}$ consist of positive integer-valued multiplicative functions with assigned integers $h_f \geq 0$ and $f(n) = O_f(n^{h_f+1/2})$. Suppose that, within each fixed degree, the normalized phases $\rho_f(n) = f(n)/n^{h_f}$ have the isolated-shift separation property: no finite shifted linear combination with a nonzero grouped coefficient at some shift has a finite limit on an arithmetic progression. This is the conclusion of Lemma 7.1; a proved stronger local-response separation criterion may be used in its place. Then, for every fixed $q \geq 1$ and $x = \varepsilon 2/b$ as above,*

$$\{1\} \ \cup \ \{E_{q,r}(x) : 0 \leq r < q\} \ \cup \ \{B_{f,q,r}(x) : f \in \mathcal{F}, \ 0 \leq r < q\} \tag{48}$$

is linearly independent over $\mathbb{Q}$. In particular this applies to the arithmetic family of Theorem 10.3, including every polynomial moment. The modulus and the positive magnitude of the evaluation point are fixed; different moduli are not listed as independent coordinates.

Proof. Clear denominators in a proposed finite relation. It says that

$$\sum_{m\geq 0} a_0(m)\frac{x^m}{m!} + \sum_f \sum_{m\geq 1} a_f(m)f(m)\frac{x^m}{m!} \in \mathbb{Q},$$

where every a_f and a_0 is an integer-valued q-periodic sequence. If all arithmetic sequences vanish, apply Lemma 12.1. Otherwise discard the zero sequences, put $K = \max h_f$, and set $H = K + 1$. Include $g_0(n) = 1$ in the cancellation constraints, just as in Theorem 10.3.

Use the grid construction with this H, a sufficiently large L, and the same F, but replace $D = F!$ by $D = qF!$ and retain $B = 1 + HDF$. Every proof in Lemma 4.1 is unchanged: $F! \mid D$, and distinct multipliers remain pairwise coprime. Additionally,

$$p_e \equiv 1 \pmod q, \qquad t_e \equiv 0 \pmod q.$$

The nonzero integer weights obey

$$\sum_{e_j} w_e \frac{p_e^{i+1}}{g(p_e)} = 0 \quad (0 \leq j \leq i < H)$$

on every coordinate-j fiber, for every retained arithmetic function and for g_0. Pick a top-degree function f_* and a residue A such that $a_{f_*}(A + H + 1) \neq 0$. Since $\gcd(q, p_e) = 1$, CRT simultaneously imposes

$$N \equiv t_e \pmod{p_e^2} \quad \text{for every } e, \qquad N \equiv A \pmod q.$$

Writing $n_e = (N - t_e)/p_e$, we have $n_e \equiv A \pmod q$. Consequently the periodic multiplier in every centered tail term is

$$a_f(n_e + j + 1) = a_f(A + j + 1),$$

independent of the vertex. Thus every lower fiber cancellation remains exact despite the periodic coefficients. Multiplicativity still gives $f(n_e + j + 1) = f(N + s_{e,j})/f(p_e)$.

The common expansion through order H has remainder $O(n^{-5/2})$, since the periodic coefficients are bounded. After cancellation, $W(N) = \sum_e w_e T(n_e) = O(N^{-3/2})$. Its denominator divides $v2^{\max_e s_2(n_e)} = O(N)$, where v is the denominator of the rational total. Hence $W(N) = 0$ eventually. Multiplication by N^2 removes all degrees below K and all errors, giving a finite limit for

$$\sum_{h_f=K} \sum_e \sum_{j=0}^{H} \frac{w_e p_e^{H+1} x^{j+1} \left\{ {H \atop j} \right\}}{f(p_e)} a_f(A + j + 1)\rho_f(N + s_{e,j}). \tag{49}$$

For $K \geq 1$ the limit is zero. For $K = 0$, the auxiliary g_0 terms converge to a constant instead; this is harmless because the separation hypothesis excludes every finite limit. As before, replacing $N^2/(N + s)^2$ by 1 costs $O(N^{-1/2})$.

Choose e_* with $w_{e_*} \neq 0$. The terminal shift $s_{e_*,H}$ is unique. Its f_* coefficient in (49) is

$$\frac{w_{e_*} p_{e_*}^{H+1} x^{H+1}}{f_*(p_{e_*})} a_{f_*}(A + H + 1) \neq 0.$$

This contradicts the assumed isolated-shift separation and proves the theorem, including the possible degree-zero case. □

12.1 Simultaneous positive, negative, and oscillatory values

Write $B_f(z) = \sum_{n\geq 1} f(n)z^n/n!$. For a positive permitted $x = 2/b$, the modulus-two theorem and the invertible rational change of coordinates

$$B_f(x) = B_{f,2,0}(x) + B_{f,2,1}(x), \qquad B_f(-x) = B_{f,2,0}(x) - B_{f,2,1}(x)$$

prove joint independence of

$$\{1, e^x, e^{-x}\} \ \cup \ \{B_f(x), B_f(-x) : f \in \mathcal{F}\}.$$

There is an equally explicit four-direction statement.

Corollary 12.3 (Four real transforms at once)**.** *For every positive $x = 2/b$, the real numbers in*

$$\{1, e^x, e^{-x}, \cos x, \sin x\} \ \cup \ \{B_f(x), B_f(-x), \operatorname{Re} B_f(ix), \operatorname{Im} B_f(ix) : f \in \mathcal{F}\}$$

are jointly linearly independent over $\mathbb{Q}$.

Proof. Apply Theorem 12.2 with $q = 4$. Relative to the four residue coordinates, the displayed transforms have matrix

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix},$$

whose determinant is -8. The same matrix takes the four exponential residue sums to $e^x, e^{-x}, \cos x, \sin x$. Apply this invertible rational transformation separately to every function block. □

For example, with $f(n) = \sigma_5(n)$, this proves that both oscillatory series

$$\sum_{m\geq 1} \frac{(-1)^m \sigma_5(2m)\, 2^{2m}}{(2m)!}, \qquad \sum_{m\geq 0} \frac{(-1)^m \sigma_5(2m+1)\, 2^{2m+1}}{(2m+1)!}$$

are irrational, jointly with $B_f(2)$ and $B_f(-2)$ and the listed classical exponential and trigonometric values. These are statements about real rational linear relations. They do not assert linear independence over algebraic numbers, algebraic independence, or transcendence of the arithmetic values.

12.2 Periodic signs and conditional Poisson expectations

Any nonzero rational-valued periodic weight $w(n)$ now gives

$$\sum_{n\geq 1} \frac{w(n) f(n) x^n}{n!} \notin \mathbb{Q}.$$

This includes real Dirichlet characters and the integer Ramanujan weights $c_d(n) = \sum_{1\leq a\leq d,\ (a,d)=1} e^{2\pi i a n/d}$. More generally, a finite collection of periodic weights which is linearly independent over $\mathbb{Q}$ as functions on $\mathbb{Z}/q\mathbb{Z}$ gives independent weighted values within each arithmetic block.

A probabilistic interpretation supplies further explicit constants. Let N have the Poisson distribution of positive mean $x = 2/b$, and extend f by $f(0) = 0$. Then

$$\mathbb{E}[f(N) \mid N \equiv r \pmod q] = \frac{B_{f,q,r}(x)}{E_{q,r}(x)} \notin \mathbb{Q}. \tag{50}$$

The denominator is strictly positive, and a rational quotient would contradict independence of those two coordinates. Thus the expected totient, divisor-power sum, or any covered moment, even after conditioning on an arbitrary residue class, is irrational. This is an exact consequence of independence, not of a numerical approximation.

12.3 A complete test for periodic polynomial numerators

For each residue r, let $R_r, U_{k,r}, V_{P,r} \in \mathbb{Q}[X]$, with only finitely many nonzero arithmetic coefficients and canonical grouping of equal polynomial Euler factors. Consider

$$S = \sum_{n\geq 1} \left[R_{n \bmod q}(n) + \sum_k U_{k,n \bmod q}(n)\sigma_k(n) + \sum_P V_{P,n \bmod q}(n) F_P(n) \right] \frac{x^n}{n!}.$$

Expand the ordinary polynomial part in falling factorials, $R_r(X) = \sum_j c_{r,j} (X)_{\underline{j}}$, and set

$$C_s(x) = \sum_{j\geq 0} c_{s+j \bmod q,j}\, x^j \qquad (0 \leq s < q).$$

Then S is rational if and only if every $U_{k,r}$ and $V_{P,r}$ is zero and every $C_s(x)$ is zero. In that case $S = -R_0(0)$. Indeed the ordinary polynomial contribution is exactly

$$\sum_{s=0}^{q-1} C_s(x) E_{q,s}(x) - R_0(0),$$

because $(n)_{\underline{j}}/n! = 1/(n-j)!$ for $n \geq j$. Apply (48) to all remaining moment coordinates. The same test applies to any additional base arithmetic family whose complete moment family satisfies the transfer hypothesis.

12.4 All derivatives and rational differential relations

Let $\mathcal{B}$ be a base arithmetic family whose full moment family $\{n \mapsto n^d f(n) : f \in \mathcal{B}, d \geq 0\}$ satisfies Theorem 12.2. For fixed q and permitted x,

$$\{1, E_{q,r}(x) : 0 \leq r < q\} \ \cup \ \{B^{(j)}_{f,q,r}(x) : f \in \mathcal{B},\ 0 \leq r < q,\ j \geq 0\}$$

is linearly independent over $\mathbb{Q}$. In fact

$$B^{(j)}_{f,q,r}(x) = x^{-j} \sum_{\substack{n \geq 1 \\ n \equiv r \ (q)}} (n)_{\underline{j}} f(n) \frac{x^n}{n!},$$

and the falling factorials give an invertible triangular change from ordinary moments, separately for each base function and residue.

Consequently the analytic functions 1, all $E_{q,r}(z)$, and all $B^{(j)}_{f,q,r}(z)$ are linearly independent over $\mathbb{Q}(z)$. To prove this, evaluate a putative finite rational-function relation at $x = 2/b$, avoiding the finitely many poles and the finitely many zeros of one nonzero coefficient. The value relation contradicts the preceding independence. In particular no covered base residue function satisfies a nontrivial linear differential equation over $\mathbb{Q}(z)$, even with an inhomogeneous term in the $\mathbb{Q}(z)$-span of the exponential residue sums and 1. This differential consequence does not exclude nonlinear differential equations, and no separate historical priority claim is made for the special non-D-finiteness consequences.

13 A logarithmic channel: the divisor-counting function

The divisor-counting function $\tau(n) = \sum_{d|n} 1$ has a logarithmically divergent mean. It therefore does not satisfy the finite-mean hypothesis of Theorem 7.2. Nevertheless its entire polynomial-moment family can be added, by separating logarithmic growth before the finite local means.

Lemma 13.1 (Logarithmic progression mean)**.** *For positive integers Q, a, put*

$$L_\tau(Q, a) = \frac{1}{Q} \sum_{r=1}^{Q} \gcd(r, Q)\, \mathbf{1}_{\gcd(r,Q)|a} > 0.$$

Then

$$\frac{1}{X} \sum_{v=0}^{X-1} \tau(Qv + a) = L_\tau(Q, a) \log X + O_{Q,a}(1).$$

For every prime $\ell \nmid Q$ and $a' \equiv a \pmod{Q}$,

$$L_\tau(Q\ell, a') = L_\tau(Q, a)(1 - \ell^{-1} + \mathbf{1}_{\ell|a'}). \tag{51}$$

Proof. The bounded Q-periodic function $w(d) = \gcd(d, Q)\mathbf{1}_{\gcd(d,Q)|a}$ has mean $L_\tau(Q, a)$. Partial summation of its bounded partial-sum discrepancy gives

$$\sum_{d \leq Z} \frac{w(d)}{d} = L_\tau(Q, a) \log Z + O_Q(1).$$

Divisor pairing, with a correction of at most $\sqrt{Y}$ for squares, gives

$$\sum_{\substack{n \leq Y \\ n \equiv a \ (Q)}} \tau(n) = 2 \sum_{d \leq \sqrt{Y}} \#\{m \geq d : dm \leq Y,\ dm \equiv a \pmod{Q}\} + O(\sqrt{Y}).$$

Writing $g = \gcd(d, Q)$, the inner count is zero unless $g \mid a$; otherwise it equals $(Y/d - d)g/Q + O(1)$. Thus the last display equals

$$\frac{2Y}{Q} \sum_{d \le \sqrt{Y}} \frac{w(d)}{d} + O_Q(Y) = \frac{Y}{Q} L_\tau(Q, a) \log Y + O_Q(Y).$$

Taking $Y = Q(X-1) + a$ proves the claimed average, up to a bounded initial segment. The terms with $\gcd(r, Q) = 1$ prove positivity. Finally CRT makes residues modulo Q and ℓ independent in the defining finite average: the new prime contributes weight 1 on its $\ell - 1$ nonzero residues and weight $\ell \mathbf{1}_{\ell \mid a'}$ on its zero residue. Their average is precisely the multiplier in (51). □

Lemma 13.2 (Separation of a logarithmic phase)**.** *Let $\rho_1, \dots, \rho_s$ have finite means on every arithmetic progression. A finite expression*

$$V(N) = \sum_r b_r \tau(N + r) + \sum_{u,r} c_{u,r} \rho_u(N + r)$$

cannot have a finite limit on $N \equiv A \pmod{Q}$ if, after grouping repeated shifts, at least one b_r is nonzero.

Proof. Choose r_* with $b_{r_*} \neq 0$ and a fresh prime larger than the number of shifts and all their nonzero absolute differences. Compare an off-residue avoiding every $-r$ with an on-residue equal to $-r_*$. On either refined progression the mean of V, divided by $\log X$, tends to zero if V converges. The finite-mean terms disappear. By Lemma 13.1, subtracting the two resulting equalities gives $b_{r_*} L_\tau(Q, A + r_*) = 0$, contrary to positivity. Positive representatives of the starting residues may be used throughout. □

Theorem 13.3 (Adjoining all divisor-counting moments)**.** *At each fixed $x = \pm 2/b$, $b \ge 1$, the family in Theorem 10.3 remains linearly independent over $\mathbb{Q}$ after adjoining all the values*

$$T_d(x) = \sum_{n \ge 1} \frac{n^d \tau(n) x^n}{n!}, \qquad d = 0, 1, 2, \dots.$$

More generally, let $\mathcal{F}$ be a family satisfying the hypotheses of Theorem 12.2, and suppose all its normalized phases have finite progression means. For each common period q, its independent family remains independent after adjoining all residue-class values

$$T_{d,r}^{[q]}(x) = \sum_{\substack{n \ge 1 \\ n \equiv r \ (q)}} \frac{n^d \tau(n) x^n}{n!}, \qquad d \ge 0, \quad 0 \le r < q.$$

The periodic base values are the $E_{q,r}(x)$ of that theorem; polynomial weights on those base values are reduced by the stated finite identities.

Proof. In a hypothetical finite relation assign degree d to $n^d \tau(n)$, and retain the assigned degrees of the other arithmetic coordinates. Let $K \ge 0$ be the largest degree and use $H = K + 1$. Include $g_0 = 1$ in the fiber equations. The bound $\tau(n) \le 2\sqrt{n}$, positivity, integer values, and multiplicativity supply every algebraic and growth hypothesis used in the cancellation and dyadic-height steps. Thus $W(N) = O(N^{-3/2})$, its denominator is $O(N)$, and $W(N) = 0$ eventually. After multiplication by N^2, the top-degree normalized shifted phases have a finite limit; every lower-degree phase vanishes. If $K = 0$, the constant phase contributes a finite constant rather than vanishing, which is permitted here.

If there is no top-degree divisor-counting coordinate, use the finite-mean separation argument. Otherwise it is the unique coordinate $n^K\tau(n)$. At a vertex e_* with nonzero grid weight, its isolated terminal-shift coefficient is

$$\frac{a_\tau w_{e_*} p_{e_*}^{H+1} x^{H+1}}{p_{e_*}^K \tau(p_{e_*})} \neq 0.$$

All other top-degree phases have finite progression means. Lemma 13.2 supplies the contradiction.

For the general periodic statement, adjoin $n^d\tau(n)$ with assigned degree d. At any fixed degree there is exactly one additional normalized phase, namely τ. If a grouped shifted combination contains a nonzero τ coefficient, Lemma 13.2 excludes a finite limit; otherwise the assumed separation for $\mathcal{F}$ applies. Thus the enlarged family satisfies every hypothesis of Theorem 12.2, proving the assertion. Equivalently, in its grid proof one chooses the progression residue A so that the nonzero top-degree periodic coefficient is nonzero at $A + H + 1$; its isolated terminal coefficient survives exactly as above. □

Let $T(z) = \sum_{n\geq 1} \tau(n) z^n/n!$. Every derivative of T at a permitted point joins the independent derivative family, by the same falling-factorial change of basis. The exact rationality criterion also extends: a polynomial coefficient multiplying $\tau(n)$ must vanish, as must every nonconstant arithmetic coordinate already present. The corresponding statements hold residue by residue in the periodic setting. These are joint results; no transcendence assertion is inferred. Individual divisor-counting factorial irrationality has classical precedents, so its specialization alone is not claimed as new.

14 The full derivative–antiderivative hierarchy

The preceding polynomial-moment results extend to genuine integration constants. The new ingredient is a grid whose coordinates include negative indices, together with different truncation orders for different factorial shifts. A finite set of additional equations removes the exponential tail to sufficiently high order. These modifications are necessary: a shifted factorial denominator is not a polynomial weight in the earlier theorem.

Theorem 14.1 (Two-sided factorial hierarchy)**.** *Let $\mathcal{B}$ be a family of positive integer-valued multiplicative functions. For each $f \in \mathcal{B}$ assign an integer $h_f \geq 0$ such that $f(n) = O_f(n^{h_f+1/2})$. Suppose that the normalized phases $\rho_f(n) = f(n)/n^{h_f}$, taken across the whole canonical base family, have the isolated-shift separation property of Theorem 12.2. In particular, positive progression means and pairwise distinct nonzero scalar responses as in Lemma 7.1 suffice. A proved profile-vector separation criterion, such as Lemma 11.3, also suffices. For every fixed $q \geq 1$ and $x = \varepsilon 2/b$, $b \geq 1$ an integer and $\varepsilon \in \{-1, 1\}$, the family*

$$\begin{aligned} &\{1\} \cup \{E_{q,a}(x) : 0 \leq a < q\} \\ &\cup \left\{ \sum_{\substack{n\geq 1 \\ n\equiv a \ (q)}} \frac{n^d f(n) x^n}{n!} : f \in \mathcal{B},\ d \geq 0,\ 0 \leq a < q \right\} \\ &\cup \left\{ \sum_{\substack{n\geq 1 \\ n\equiv a \ (q)}} \frac{f(n) x^n}{(n+r)!} : f \in \mathcal{B},\ r \geq 1,\ 0 \leq a < q \right\} \end{aligned} \tag{52}$$

is linearly independent over $\mathbb{Q}$.

The divisor functions σ_k and the distinct polynomial Euler factors F_P satisfy these hypotheses by the preceding sections. More generally, the result applies to any additional family for which these hypotheses have actually been verified. In particular, choose one arithmetic base for each distinct nontrivial admissible profile and adjoin τ with degree zero. If a shifted combination contains a nonzero τ coefficient, Lemma 13.2 excludes a finite limit; otherwise Lemma 11.3 does. Thus this entire enlarged base family satisfies the theorem, across all effective degrees. The assertion is at one fixed x; arbitrary unrelated evaluation points are not asserted to be independent.

Proof. We give the details because both negative shifts and arbitrarily high integration orders matter. Consider a finite proposed rational relation. Its arithmetic channels have one of the canonical forms

$$(g, r) = (n^d f, 0), \quad h = h_f + d, \qquad \text{or} \qquad (g, r) = (f, r), \quad h = h_f, \quad r \geq 1.$$

Define the effective degree of a channel to be $w = h - r$. Group residue-class coefficients into periodic functions before assigning channels. At a fixed effective degree each base function occurs at most once, so the top normalized phases belong to the separated base family. If every arithmetic coefficient vanishes, apply Lemma 12.1. Henceforth assume an arithmetic channel remains. Retain the channels with nonzero coefficients, put

$$R = \max r, \qquad w_* = \max(h - r), \qquad J = \max(0, w_* + 1), \qquad M = J + R.$$

If there is an arithmetic channel then $M \geq 1$. Reindexing gives

$$\sum_{n\geq 1} \frac{g(n) x^n}{(n+r)!} = x^{-r} \sum_{m\geq r+1} \frac{g(m-r) x^m}{m!}.$$

Clear all rational coefficients, including the factors x^{-r}. Thus the putative rational relation is an integer-coefficient factorial series with a q-periodic exponential coefficient $a_0(m)$ and nonzero q-periodic arithmetic coefficient functions $a_{g,r}(m-r)$. These functions are evaluated at the original arithmetic argument. Write the rational value as u/v. For its normalized tail $T(n)$, Lemma 10.1 gives $v2^{s_2(n)}T(n) \in \mathbb{Z}$. Choose a top effective-degree channel and a residue A such that $a_{g,r}(A+J+1) \neq 0$. Fix this residue before solving for the weights.

The offset grid. Index the coordinates by $\ell = -R, \dots, J-1$ and let $e_\ell \in \{0, \dots, L-1\}$, where L will be chosen below. Put

$$b_0 = M(L-1)+1, \quad F = \max(b_0^M - 1, M+2), \quad D = qF!, \quad C = \max(R,J)+1,$$
$$I_e = \sum_{\ell=-R}^{J-1} e_\ell b_0^{\ell+R}, \qquad T_e = \sum_{\ell=-R}^{J-1} (\ell+1) e_\ell b_0^{\ell+R},$$
$$B = 1 + (J+2C+M+2)DF, \qquad p_e = B + DI_e, \qquad t_e = DT_e,$$
$$s_{e,\ell} = (\ell+1)p_e - t_e \qquad (-R \le \ell \le J).$$

The integers t_e and some lower shifts may be negative; all arguments below are positive for sufficiently large N. Exactly as in Lemma 4.1, the p_e are pairwise coprime, and each is coprime to every integer at most $M+2$. A lower shift $s_{e,\ell}$ is constant on its coordinate-ℓ fiber. Moreover,

$$s_{e,J} = (J+1)B + D \sum_{\ell=-R}^{J-1} (J-\ell) e_\ell b_0^{\ell+R}.$$

Its displayed digits lie between 0 and $M(L-1) = b_0 - 1$, so the terminal shifts are distinct. They are at least $(J+1)B$, whereas every lower shift is at most $JB + (J+C)DF < (J+1)B$. Here we used $0 \le I_e \le F$ and $|T_e| \le CF$.

We also have $p_e \equiv 1 \pmod q$ and $t_e \equiv 0 \pmod q$. Take the CRT progression specified by

$$N \equiv t_e + (R+1)p_e \pmod{p_e^2} \quad \text{for every } e, \qquad N \equiv A \pmod q.$$

Put $n_e = (N - t_e)/p_e$. Then $n_e \equiv R+1 \pmod{p_e}$ and

$$\gcd(p_e, n_e + \ell + 1) = \gcd(p_e, R + \ell + 2) = 1, \qquad -R \le \ell \le J,$$

since $2 \le R + \ell + 2 \le M + 2$. In particular the otherwise problematic shift $\ell = -1$ is covered. Multiplicativity therefore gives the exact rescaling identity

$$\frac{g(n_e + \ell + 1)}{(n_e + \ell + 1)^{i+1}} = \frac{p_e^{i+1}}{g(p_e)} \frac{g(N + s_{e,\ell})}{(N + s_{e,\ell})^{i+1}}.$$

Moreover $n_e \equiv A \pmod q$, so the periodic coefficient of the term at index ℓ is $a_{g,r}(A+\ell+1)$, independent of the vertex. Every fiber cancellation below therefore remains exact.

Different truncation orders. For a channel (g,r) expand its reindexed tail through order $J+r$. If $y = n+j+1-r$, the relevant denominator is

$$\frac{n!}{(n+j+1)!} = \frac{1}{(y+r)(y+r-1)\cdots(y+r-j)}.$$

Its expansion in y^{-1} has integral coefficients $c_{r,j,i}$, with $c_{r,j,j} = 1$. The same geometric remainder argument as in Lemma 3.1 gives

$$T_{g,r}(n) := n! x^{-n} \sum_{m>n} \frac{g(m-r)x^m}{m!}$$

$$= \sum_{j=0}^{J+r} \sum_{i=j}^{J+r} x^{j+1} c_{r,j,i} \frac{g(n+j+1-r)}{(n+j+1-r)^{i+1}} + O(n^{h-J-r-3/2}). \tag{53}$$

For example, $c_{r,j,i}$ is $(-1)^{i-j}$ times the complete homogeneous polynomial of degree $i-j$ in $r, r-1, \dots, r-j$. The omitted tail has the same bound. The error in (53) is $O(n^{w-J-3/2})$.

For every term with $\ell = j - r < J$, impose on every coordinate-ℓ fiber the rational equation

$$\sum_{e_\ell} v_e \frac{p_e^{i+1}}{g(p_e)} = 0.$$

There are a fixed number C_0 of coefficient arrays, hence at most $C_0 L^{M-1}$ equations. These equations cancel every displayed term except $j = i = J+r$, whose shift is the common terminal index $\ell = J$.

The exponential at negative effective degrees. One cannot simply discard the exponential when $w_* < 0$. For any fixed integer $T \geq 2$, the periodic exponential tail, restricted to $n \equiv A \pmod q$, has a rational expansion

$$U_x(n) = n! x^{-n} \sum_{m>n} \frac{a_0(m) x^m}{m!} = \sum_{a=1}^{T-1} u_a n^{-a} + O(n^{-T}), \qquad u_a \in \mathbb{Q}.$$

Substituting $n_e = (N - t_e)/p_e$ yields

$$U_x(n_e) = \sum_{v=1}^{T-1} K_v(p_e, t_e) N^{-v} + O(N^{-T}), \quad K_v(p,t) = \sum_{a=1}^{v} u_a \binom{v-1}{v-a} p^a t^{v-a}.$$

Set $T = \max(2, J + 2 - w_*)$ and add the $T-1$ global equations

$$\sum_e v_e K_v(p_e, t_e) = 0 \qquad (1 \leq v < T).$$

Choose $L > C_0 + T$. The total number of equations is less than L^M, so rank–nullity supplies a nonzero integer vector (v_e) satisfying all the fiber and global constraints. All its data are fixed before N tends to infinity.

The integer limit. The weighted combined tail $W(N) = \sum_e v_e T(n_e)$ now satisfies

$$W(N) = \sum_{g,r} \sum_e \frac{a_{g,r}(A+J+1) x^{J+r+1} v_e p_e^{J+r+1}}{g(p_e)} \frac{g(N+s_{e,J})}{(N+s_{e,J})^{J+r+1}} + O(N^{w_*-J-3/2}) + O(N^{-T}).$$

The terminal terms are $O(N^{w_*-J-1/2})$, so $W(N) = O(N^{-3/2})$. Its denominator divides

$$v 2^{\max_e s_2(n_e)} = O(N).$$

Multiplication by this integer gives an integer tending to zero. Consequently $W(N) = 0$ eventually.

Multiply by N^{J+1-w_*}. The errors tend to zero, every smaller effective degree disappears, and replacing the fixed shifted power ratio by 1 costs $O(N^{-1/2})$. We obtain

$$\sum_{h-r=w_*} \sum_e \frac{a_{g,r}(A+J+1) x^{J+r+1} v_e p_e^{J+r+1}}{g(p_e)} \rho_g(N+s_{e,J}) \longrightarrow 0.$$

Choose a vertex with $v_e \neq 0$. Its terminal shift is isolated, and the top channel used to choose A has a nonzero coefficient there. The separation property of the canonical bases is a contradiction. If the original relation had no arithmetic channel, use Lemma 12.1. This proves the theorem. □

Corollary 14.2 (All derivatives and all normalized antiderivatives)**.** *First take $q = 1$. For $f \in \mathcal{B}$, put $A_f(z) = \sum_{n\ge1} f(n)z^n/n!$. For $r \ge 1$, let*

$$A_f^{[-r]}(x) = \frac{1}{(r-1)!}\int_0^x (x-t)^{r-1}A_f(t)\,dt = \sum_{n\ge1}\frac{f(n)x^{n+r}}{(n+r)!}.$$

For nonnegative j use $A_f^{[j]} = A_f^{(j)}$. At every fixed permitted x, the family

$$\{1, e^x\} \ \cup\ \{A_f^{[j]}(x) : f \in \mathcal{B},\ j \in \mathbb{Z}\}$$

is linearly independent over $\mathbb{Q}$. For general fixed q, the same assertion holds with all residue-restricted functions

$$A_{f,q,a}(z) = \sum_{\substack{n\ge1\\ n\equiv a\ (q)}} \frac{f(n)z^n}{n!} \qquad (0 \le a < q)$$

in place of A_f, and with $E_{q,0}(x), \dots, E_{q,q-1}(x)$ in place of e^x.

Proof. Negative orders multiply the second family in (52) by the nonzero rational number x^r. For nonnegative orders, the falling-factorial change of basis from Corollary 9.1 is triangular and invertible over $\mathbb{Q}$. Apply Theorem 14.1 to the finite set of coordinates in any proposed relation. The change of basis acts separately on each residue-restricted function, so the same proof establishes the last assertion. □

Corollary 14.3 (The two-sided hierarchy in four real directions)**.** *For a positive permitted $x = 2/b$, the following real numbers are jointly linearly independent over $\mathbb{Q}$:*

$$1,\ e^x,\ e^{-x},\ \cos x,\ \sin x,$$
$$A_f^{[j]}(x),\quad A_f^{[j]}(-x),\quad \operatorname{Re} A_f^{[j]}(ix),\quad \operatorname{Im} A_f^{[j]}(ix) \qquad (f \in \mathcal{B},\ j \in \mathbb{Z}).$$

For negative orders, the antiderivative is normalized to vanish at zero together with its lower derivatives; its entire series defines it also at complex arguments.

Proof. Use the residue-restricted statement of Corollary 14.2 with modulus four. For $\zeta \in \{1, -1, i\}$, the power series gives

$$A_f^{[j]}(\zeta x) = \zeta^{-j}\sum_{a=0}^{3}\zeta^a A_{f,4,a}^{[j]}(x), \qquad j \in \mathbb{Z}.$$

This identity is valid for derivatives and normalized antiderivatives alike. For each fixed j, its real transformation is the invertible rational matrix in Corollary 12.3, followed by sign changes and possibly interchanging the real and imaginary rows. The exponential block uses the same matrix. Independence is therefore preserved block by block in every finite proposed relation. □

Corollary 14.4 (Rational weights with simple negative-integer poles)**.** *Let finitely many $R_f \in \mathbb{Q}(X)$ have arbitrary polynomial parts and at most simple poles, all among $-1, -2, \dots$. If at least one R_f is nonzero, then, for every $\lambda, q \in \mathbb{Q}$,*

$$q + \lambda e^x + \sum_f \sum_{n\ge1} \frac{R_f(n)f(n)x^n}{n!} \notin \mathbb{Q}.$$

In particular the quantities

$$M_{f,a}(x) := \int_0^1 t^{a-1}A_f(xt)\,dt = \sum_{n\ge1}\frac{f(n)x^n}{n!(n+a)}, \qquad a \ge 1,$$

are jointly independent together with $1, e^x$ *and every polynomial moment in the first family of* (52).

Proof. For each $R \geq 1$, the rational functions

$$\frac{1}{(X+1)(X+2)\cdots(X+r)}, \qquad 1 \leq r \leq R,$$

form a basis of the rational functions vanishing at infinity with at most simple poles in $\{-1, \ldots, -R\}$. Indeed the residue at $-R$ first determines the last coefficient, and downward induction determines the others; the dimension is R by partial fractions. Combining this basis with ordinary polynomials converts the claim to Theorem 14.1. Termwise integration is justified by absolute uniform convergence on the compact integration interval. For the last assertion, the functions $(X+a)^{-1}$ are themselves linearly independent, again by their residues. □

The rational-weight statement also holds separately and jointly on all residue classes of any fixed modulus. Apply the same basis conversion to each $R_{f,a}$, and then use the residue-restricted family (52). Consequently the class permits, at once, periodic signs, residue restrictions, polynomial weights, simple negative-integer poles, and canonical arithmetic bases with a proved separation property.

For example, the theorem includes simultaneously

$$\int_0^1 t^6 A_5(2t)\, dt, \qquad \int_0^2 (2-t)^9 A_5(t)\, dt, \qquad A_5^{(17)}(2),$$

and their counterparts for every admissible arithmetic base, after the canonical basis conversion just proved. The displayed integral with $(2-t)^9$ is $9!A_5^{[-10]}(2)$. Independence must always be checked after this conversion: listing two different integral representations of the same linear functional does not create two independent coordinates.

This extension does not include a pole at zero, such as the weight $1/n$, or repeated poles such as $(n+1)^{-2}$. It also does not infer irrationality of an arbitrary integral from pointwise irrationality of its integrand. Here the precise factorial coefficients and the new tail argument prove the integral statements directly. Their historical priority, including classical special cases of low degree, requires a separate literature review.

15 Consequences across arithmetic, lattices, and finite groups

The preceding theorems control several operations together: products of arithmetic functions, residue restrictions, differentiation, and the specific integral transforms proved in Corollary 14.4. The following examples make the scope concrete. The underlying counting functions are classical; their factorial evaluations are the objects studied here.

15.1 Cyclic subgroups of finite Abelian groups

For $r \geq 2$ let

$$c_r(n) = \#\{\text{cyclic subgroups of } (\mathbb{Z}/n\mathbb{Z})^r\}.$$

The identity

$$c_r(n) = \sum_{d \mid n} \frac{J_r(d)}{\varphi(d)} \tag{54}$$

is a specialization of the subgroup formulas in [9]. It also has a short direct proof. For $d \mid n$, the number of vectors whose order divides d is d^r. Möbius inversion shows that exactly $J_r(d)$

vectors have order d. Each cyclic subgroup of that order has $\varphi(d)$ generators, proving (54), including the trivial subgroup at $d = 1$.

The summand J_r/φ is multiplicative, so c_r is multiplicative and integer-valued. Put $h = r - 1$ and

$$D_r(z) = \frac{1 - z^r}{(1 - z)(1 - z^h)}.$$

At a prime power, summing the geometric progression gives

$$c_r(p^a) = 1 + \frac{p^r - 1}{p - 1} \sum_{j=0}^{a-1} p^{hj}, \qquad \frac{c_r(p^a)}{p^{ha}} = D_r(z) + (1 - D_r(z))w^h, \quad z = p^{-1}, \quad w = p^{-a}.$$

For $0 < z \le 1/2$, $D_r(z)$ is positive and bounded. The last expression is a convex combination of $D_r(z)$ and 1, so it is positive and uniformly bounded for all $a \ge 1$; at $w = z$ it is $1 + O(z)$. It is therefore an admissible, nonconstant profile. Different r give different degrees in w. Consequently Theorem 11.4 covers all c_r and all their polynomial moments jointly with $1, e^x$; any further profiles may be included after the stated equality test. Their fixed finite pointwise products are also admissible.

15.2 Explicit constants and one simultaneous statement

To keep the formulas readable, write

$$\mathcal{E}[f] = \sum_{n \ge 1} \frac{2^n f(n)}{n!}, \qquad A_5(z) = \sum_{n \ge 1} \frac{\sigma_5(n) z^n}{n!}.$$

Every entry in the left column below is irrational. The second column identifies the coefficients or the proved operation.

Constant	Arithmetic or geometric meaning
$\mathcal{E}[\mathrm{rad}]$	Product of the distinct prime factors of each index
$\mathcal{E}[\sigma_1^2]$	Square of the divisor sum at each index
$\mathcal{E}[\varphi^3 \sigma_5^2]$	A mixed product of totients and divisor sums
$\mathcal{E}[a_3]$	Index-n sublattices of $\mathbb{Z}^3$ [8]
$\mathcal{E}[c_3]$	Cyclic subgroups of $(\mathbb{Z}/n\mathbb{Z})^3$ [9]
$\mathcal{E}[L]$, $L(n) = \sum_{j=1}^{n} \mathrm{lcm}(j, n)$	The classical least-common-multiple sum [10]
$\displaystyle\sum_{\substack{n \ge 1 \\ n \equiv 1 \,(\mathrm{mod}\ 3)}} \frac{2^n \sigma_5(n)}{n!}$	One arithmetic progression of indices
$\displaystyle\int_0^1 t^6 A_5(2t)\, dt = \sum_{n \ge 1} \frac{2^n \sigma_5(n)}{(n+7)n!}$	A rational-weight integral covered by the new tail argument

The first six follow from Theorem 11.4 and (47); the last two follow from Theorem 12.2 and Corollary 14.4, respectively. None of these statements requires numerical approximation to certify irrationality.

A stronger assertion connects four of these arithmetic settings with the integral transform. The following seven numbers are linearly independent over $\mathbb{Q}$:

$$\boxed{1,\ e^2,\ A_5(2),\ \int_0^1 t^6 A_5(2t)\, dt,\ \mathcal{E}[a_3],\ \mathcal{E}[\mathrm{rad}],\ \mathcal{E}[L].} \tag{55}$$

Indeed the profiles of σ_5, a_3, rad, and $g(n) = \sum_{d|n} d\varphi(d)$ are distinct. Their degrees in w are $5, 2, 1, 2$; the a_3 profile has a nonzero coefficient of w, whereas the g profile has none. Within the σ_5 coordinate, the rational weights 1 and $(n+7)^{-1}$ are independent. Finally, (46) gives

$$\mathcal{E}[L] = e^2 + \tfrac{1}{2}\mathcal{E}[ng],$$

an invertible rational change of coordinates in a family supplied by Corollary 14.4. This proves (55); in particular every rational linear combination having a nonzero coefficient on one of the six nonconstant coordinates is irrational.

The square in $\mathcal{E}[\sigma_1^2]$ acts on each coefficient: it does not denote $\mathcal{E}[\sigma_1]^2$. Products of the resulting constants, algebraic independence, and transcendence require different arguments. Likewise, the exact appearance of $\zeta(3)/\zeta(2)$ in the normalized mean of g, established above, does not transfer this conclusion to that zeta ratio. Classical low-growth factorial results [7] remain part of the prior literature. The catalog illustrates the range of the proved theorems; it does not certify that each displayed constant was previously a named open problem.

16 Polynomial valuation profiles and every logarithmic order

The finite-mean profiles and the separate divisor-counting channel fit into one larger class. The enlargement admits polynomial dependence on a prime exponent, arbitrarily high powers of the divisor-counting function, all ordered-factorization functions, and products of these with the earlier arithmetic bases. We prove the necessary progression averages here; no unproved distribution hypothesis or general Selberg–Delange theorem is needed.

Definition 16.1 (An admissible valuation profile)**.** Let $\mathcal{R}(z, w, u) \in \mathbb{Q}(z)[w, u]$. Its coefficient functions have no poles at $z = 1/p$ for primes p. We call $\mathcal{R}$ admissible if:

1. there are constants C, D such that, for every prime p and integer $a \geq 1$,

$$0 < \mathcal{R}(p^{-1}, p^{-a}, a) \leq C(1+a)^D;$$

2. for an integer $\kappa_{\mathcal{R}} \geq 1$,

$$\mathcal{R}(z, z, 1) = \kappa_{\mathcal{R}} + O(z) \qquad (z \longrightarrow 0);$$

3. for some integer $h \geq 0$, every $p^{ha}\mathcal{R}(p^{-1}, p^{-a}, a)$ is a positive integer.

Set $\rho_{\mathcal{R}}(1) = 1$ and extend

$$\rho_{\mathcal{R}}(p^a) = \mathcal{R}(p^{-1}, p^{-a}, a)$$

multiplicatively. An integral realization is $f_{\mathcal{R},h}(n) = n^h \rho_{\mathcal{R}}(n)$. The value at exponent zero is defined to be 1; it need not equal $\mathcal{R}(z, 1, 0)$. Distinct profiles always mean distinct elements of $\mathbb{Q}(z)[w, u]$ after simplification.

The integer $\kappa_{\mathcal{R}} - 1$ is the order of logarithmic growth of the progression mean. The former admissible profiles are the u-independent case with $\kappa = 1$. The profile $u + 1$ gives τ and has $\kappa = 2$. The constant profile $c \geq 2$ gives $c^{\omega(n)}$, where $\omega(n)$ counts distinct prime divisors: this example explains the exponent-zero convention.

Lemma 16.2 (Growth and an absolutely convergent convolution)**.** *For an admissible profile and every $\epsilon > 0$, $\rho_{\mathcal{R}}(n) = O_{\mathcal{R},\epsilon}(n^{\epsilon})$. Writing $\kappa = \kappa_{\mathcal{R}}$, let*

$$g = \rho_{\mathcal{R}} * \mu^{*\kappa}, \qquad \rho_{\mathcal{R}} = d_\kappa * g,$$

where $*$ denotes Dirichlet convolution, $\mu^{*\kappa}$ its κ-fold power, and $d_\kappa = 1^{*\kappa}$. For every real $\sigma > 1/2$,

$$\sum_{n\ge 1} \frac{|g(n)|}{n^\sigma} < \infty.$$

Moreover, with

$$\mathcal{L}_{\mathcal{R}}(z) = 1 + \sum_{a\ge 1} \mathcal{R}(z, z^a, a) z^a, \qquad U_{\mathcal{R}}(z) = (1-z)\mathcal{L}_{\mathcal{R}}(z),$$

these are rational functions of z, and

$$G_{\mathcal{R}} = \sum_{n\ge 1} \frac{g(n)}{n} = \prod_p (1-p^{-1})^\kappa \mathcal{L}_{\mathcal{R}}(p^{-1}) > 0.$$

Proof. The polynomial bound in a implies that $B = \sup_{a\ge 1}[C(1+a)^D]^{1/a}$ is finite. If $p \ge B^{1/\epsilon}$, then $\rho(p^a) \le p^{\epsilon a}$. For each of the finitely many remaining primes, the supremum of $\rho(p^a)p^{-\epsilon a}$ over $a \ge 0$ is finite. Multiplying those finitely many bounds proves the growth assertion.

At a prime, $g(p) = \rho(p) - \kappa = O(p^{-1})$. At higher powers,

$$g(p^a) = \sum_{j=0}^{\min(a,\kappa)} (-1)^j \binom{\kappa}{j} \rho(p^{a-j}) = O_{\mathcal{R}}((1+a)^D).$$

Thus, for fixed $\sigma > 1/2$,

$$\sum_p \sum_{a\ge 1} |g(p^a)| p^{-a\sigma} < \infty.$$

The absolutely convergent Euler product gives the asserted Dirichlet series. Each monomial $r(z)w^j u^m$ contributes $r(z)\sum_{a\ge 1} a^m z^{(j+1)a}$ to $\mathcal{L} - 1$. The series $\sum_{a\ge 1} a^m v^a$ is rational, as follows by repeatedly applying $v\,d/dv$ to $v/(1-v)$. This proves rationality. Finally every Euler factor defining G is positive and equals $1 + O(p^{-2})$. The convergent product is therefore strictly positive. □

Lemma 16.3 (All progression means and their local factors)**.** *For every fixed positive Q, a,*

$$\frac{1}{X}\sum_{v=0}^{X-1} \rho_{\mathcal{R}}(Qv + a) = L_{\mathcal{R}}(Q, a)(\log X)^{\kappa-1} + \begin{cases} o(1), & \kappa = 1, \\ O_{\mathcal{R},Q,a}((\log X)^{\kappa-2}), & \kappa \ge 2, \end{cases}$$

where $L_{\mathcal{R}}(Q, a) > 0$. More explicitly, put $R_p(t) = \rho_{\mathcal{R}}(p^t)$, including $R_p(0) = 1$, and $z = 1/p$. For $p^v \| Q$ define

$$A_{\mathcal{R},p^v}(a) = \begin{cases} \dfrac{R_p(t)}{U_{\mathcal{R}}(z)}, & t = v_p(a) < v, \\ \dfrac{\sum_{j\ge 0} R_p(v+j) z^j}{\mathcal{L}_{\mathcal{R}}(z)}, & p^v \mid a. \end{cases}$$

Then

$$L_{\mathcal{R}}(Q, a) = \frac{G_{\mathcal{R}}}{(\kappa-1)!} \prod_{p^v \| Q} A_{\mathcal{R},p^v}(a).$$

In particular, at a fresh prime $\ell \nmid Q$, refinement to a unit class modulo ℓ^{t+1} multiplies L by $1/U_{\mathcal{R}}(1/\ell)$, whereas refinement to exact valuation $t \ge 1$ multiplies it by $\mathcal{R}(1/\ell, \ell^{-t}, t)/U_{\mathcal{R}}(1/\ell)$.

Proof. We supply an elementary progression argument to include nonunit classes. First,

$$\sum_{n\le Y} d_\kappa(n) = \frac{Y(\log Y)^{\kappa-1}}{(\kappa-1)!} + O_\kappa(Y(1+\log Y)^{\kappa-2}) \quad (\kappa \ge 2),$$

while the sum for $\kappa = 1$ is $Y + O(1)$. For $\kappa = 2$ this follows from the harmonic sum. Induction uses $d_\kappa = 1 * d_{\kappa-1}$ and

$$\sum_{m\le Y} \frac{(\log(Y/m))^j}{m} = \frac{(\log Y)^{j+1}}{j+1} + O_j((1+\log Y)^j).$$

The displayed estimate follows by comparison with the integral.

For a nonprincipal Dirichlet character χ modulo a fixed integer, its partial sums are bounded. The κ-fold hyperbola argument gives

$$\sum_{n\le Y} \chi(n) d_\kappa(n) = O_{\chi,\kappa}(Y^{1-1/\kappa}) \quad (\kappa \ge 1),$$

where the exponent is zero when $\kappa = 1$. For completeness, in the sum over κ factors with product at most Y, at least one factor is at most $Y^{1/\kappa}$. Use inclusion–exclusion on these κ conditions. For a term with $j < \kappa$ restricted factors, induction on $\ell = \kappa - j$ bounds the remaining convolution by $O((Y/(m_1 \cdots m_j))^{1-1/\ell})$. Summing absolute values over $m_i \le Y^{1/\kappa}$ gives $O(Y^{1-1/\ell+j/(\kappa\ell)}) = O(Y^{1-1/\kappa})$. The term with all factors restricted is a product of bounded character sums. This proves the estimate.

For the principal character modulo Q, removing the primes dividing Q from d_κ is convolution with the finite Euler polynomial $\prod_{p\mid Q}(1-p^{-s})^\kappa$. Its leading coefficient is therefore $\prod_{p\mid Q}(1-p^{-1})^\kappa$. Now convolve with $g(n)\chi(n)$ from Lemma 16.2. For a nonprincipal character choose β with $\max(1/2, 1-1/\kappa) < \beta < 1$; the character sum is $O(Y^\beta)$. For the principal character, absolute convergence of $\sum |g(n)| n^{-\beta}$ also gives all the logarithmic moments $\sum |g(n)|(\log n)^j/n < \infty$ needed to expand the leading term. The convolution consequently preserves the stated main term and error. Character orthogonality now proves, when $(a, Q) = 1$, the asserted mean with coefficient

$$\frac{G_{\mathcal{R}}}{(\kappa-1)!} \prod_{p\mid Q} \frac{1}{U_{\mathcal{R}}(1/p)}.$$

For $\kappa = 1$ the same proof gives a power-saving error before division by X, which in particular is $o(1)$ afterwards.

For a general class put $c = (a, Q)$, $Q' = Q/c$, and $a' = a/c$. Then $(a', Q') = 1$ and the function

$$\rho_c(m) = \frac{\rho_{\mathcal{R}}(cm)}{\rho_{\mathcal{R}}(c)}$$

is multiplicative. Its local factors differ from those of ρ only at primes dividing c, where its values are $R_p(v_p(c)+j)/R_p(v_p(c))$. Those finitely many modified factors still have polynomial growth in j; the convolution proof just given applies unchanged to them. Use the unit-class result on $Q'v + a'$ and multiply by $\rho(c)$. If $v_p(a) < v_p(Q)$, the modified prime is removed from the unit-class Euler product, leaving $R_p(v_p(a))/U_{\mathcal{R}}(1/p)$. If $p^{v_p(Q)} \mid a$, it remains unrestricted and leaves the second factor in the statement. These are exactly the asserted formulas, and every factor is positive. The fresh-prime assertions follow at once. □

Lemma 16.4 (Injectivity and separation for logarithmic profiles)**.** *For every nontrivial admissible profile $\mathcal{R} \ne 1$, set*

$$\delta_{\mathcal{R},t}(z) = \frac{\mathcal{R}(z, z^t, t) - 1}{U_{\mathcal{R}}(z)}, \qquad t \ge 1.$$

The vector $(\delta_{\mathcal{R},t})_{t\ge1}$ is nonzero, and distinct profiles have distinct vectors. Consequently, a nontrivial finite linear combination of shifted phases belonging to distinct nontrivial admissible profiles cannot have a finite limit on an arithmetic progression, once identical shifts for the same profile have been grouped.

Proof. Direct summation gives

$$\sum_{t\ge1}\delta_{\mathcal{R},t}(z)z^t=\frac{U_{\mathcal{R}}(z)-1}{(1-z)U_{\mathcal{R}}(z)}.$$

Hence equality of two response vectors implies equality of their U's, and then equality of $\mathcal{R}(z,z^t,t)$ for every positive integer t. If a polynomial $H(z,w,u)\in\mathbb{Q}(z)[w,u]$ vanishes at $(w,u)=(z^t,t)$ for every such t, then $H=0$. Indeed, specialize z to a sufficiently small positive number avoiding all coefficient poles and retaining at least one nonzero coefficient. Writing $H(z,w,u)=\sum_j P_j(u)w^j$, choose the least j with $P_j\neq0$. Division by z^{jt} would force the nonzero polynomial $P_j(t)$ to equal an exponentially decaying polynomial-exponential sum as $t\to\infty$, which is impossible. This also proves that the zero response vector corresponds only to $\mathcal{R}=1$.

Suppose now that the shifted combination has a finite limit. Among its nonzero grouped coefficients choose the largest value κ_*, and a shift r_* carrying a profile of that order. On any fixed refinement of the starting progression, divide the progression average by $(\log X)^{\kappa_*-1}$ and use Lemma 16.3. Terms of smaller order disappear. When $\kappa_*>1$ the resulting limit is zero; when $\kappa_*=1$ it is the same finite constant on every refinement.

Choose fresh primes exceeding the number of shifts and all their nonzero absolute differences. At each such prime an off-class avoids all the shifts modulo the prime, while an on-class gives exact valuation $t\ge1$ to $N+r_*$. All the other shifted arguments are units in both classes. For each nonempty finite set of selected primes take the alternating sum over their on/off choices. The constant limit and all other shifts cancel. If $B_{\mathcal{R}}$ denotes the grouped coefficient at r_* multiplied by $L_{\mathcal{R}}(Q,A+r_*)$, the resulting equations are

$$\sum_{\kappa_{\mathcal{R}}=\kappa_*}B_{\mathcal{R}}\prod_{\ell\in S}\delta_{\mathcal{R},t_\ell}(1/\ell)=0.$$

The $B_{\mathcal{R}}$ do not depend on the chosen finite subset S. Use one prime and valuation giving a nonzero response for a selected profile. For each competing profile use a new prime and valuation that distinguishes its response from the selected one. Such choices exist by vector injectivity and the fact that a nonzero rational function has only finitely many zeros. The product-interpolation argument of Lemma 7.1, now allowing a chosen valuation at each prime, forces the selected $B_{\mathcal{R}}$ to vanish. Positivity of its progression mean contradicts its choice. This proves the claim. □

Theorem 16.5 (A unified arithmetic hierarchy)**.** *Choose one positive integral realization $f_{\mathcal{R},h}$ for each of any collection of distinct nontrivial admissible valuation profiles. For every fixed $q\ge1$ and $x=\pm2/b$, $b\ge1$, this entire canonical base collection satisfies Theorem 14.1. Thus all its residue-class polynomial moments and shifted-factorial values in (52) are jointly linearly independent over $\mathbb{Q}$ together with 1 and the exponential residue values. All derivative orders, all normalized antiderivative orders, the four real directions, and the simple-negative-integer-pole weights in the succeeding corollaries apply simultaneously to this enlarged family.*

Proof. Integral realizations are positive integer-valued and multiplicative. Lemma 16.2 gives $f_{\mathcal{R},h}(n)=O(n^{h+1/2})$, and Lemma 16.4 proves the required separation across the whole canonical base collection, including different logarithmic orders. These are exactly the hypotheses of Theorem 14.1. □

Choosing the least possible integral realization degree is convenient but not necessary. If two constructions have the same profile and different degrees, they differ by an integral power of n and must first be combined in the polynomial-weight coordinates. The profile 1 is also reduced to the exponential-polynomial block, rather than counted as a new base. No list below is intended to override this canonical reduction.

16.1 Ordered factorizations, divisor powers, and multiplicative mixtures

The number $d_r(n)$ of ordered r-tuples of positive integers with product n satisfies

$$d_r(p^a) = \binom{a+r-1}{r-1}.$$

This follows by distributing the exponent a among the r ordered factors. Thus every d_r, $r \geq 2$, is an admissible base of degree zero, with profile $\binom{u+r-1}{r-1}$ and logarithmic parameter $\kappa = r$. The collection may be enlarged at once by

$$\tau(n)^m \quad (m \geq 1), \qquad c^{\omega(n)} \quad (c \geq 2), \qquad \prod_{j=1}^{s} d_{r_j}(n)^{m_j},$$

whose respective profiles and parameters are

$$(u+1)^m, \quad \kappa = 2^m; \qquad c, \quad \kappa = c; \qquad \prod_{j=1}^{s} \binom{u+r_j-1}{r_j-1}^{m_j}, \quad \kappa = \prod_{j=1}^{s} r_j^{m_j}.$$

The products are pointwise products, whereas d_r itself was defined by Dirichlet convolution. Their arithmetic meanings and their different operations should not be conflated. For instance, d_4 and τ^2 both have $\kappa = 4$, but their profiles differ; the response-vector proof establishes their joint independence rather than relying only on different logarithmic growth orders.

Any finite pointwise product of admissible integral realizations is again admissible: multiply the profiles, add the realization degrees, and multiply the integers κ. Positivity, integrality, and the polynomial bound in u are preserved. Consequently the above families can be multiplied by arbitrary finite products of the earlier divisor, Jordan, Dedekind, unitary, radical, lattice-counting, and cyclic-subgroup bases. After identical profiles are grouped, their full two-sided hierarchies remain jointly independent.

Here is one concrete block at $x = 2$. Write $\mathscr{A}_f(z) = \sum_{n \geq 1} f(n) z^n/n!$. The four bases

$$f_1 = \tau^2, \qquad f_2 = d_4, \qquad f_3 = 4^{\omega}, \qquad f_4 = \sigma_3 \tau^2$$

have profiles

$$(u+1)^2, \qquad \binom{u+3}{3}, \qquad 4, \qquad (u+1)^2 \frac{1 - z^3 w^3}{1 - z^3},$$

which are distinct, even though all have $\kappa = 4$. Therefore the family containing

$$1, e^2, e^{-2}, \cos 2, \sin 2,$$
$$\mathscr{A}_{f_j}^{[v]}(2), \quad \mathscr{A}_{f_j}^{[v]}(-2), \quad \operatorname{Re} \mathscr{A}_{f_j}^{[v]}(2i), \quad \operatorname{Im} \mathscr{A}_{f_j}^{[v]}(2i) \qquad (1 \leq j \leq 4,\ v \in \mathbb{Z})$$

is jointly linearly independent over $\mathbb{Q}$. Every finite choice is covered by a single instance of the proved theorem. These are unconditional consequences of the manuscript's mechanism; no separate assertion of historical priority is made for individual classical specializations.

17 Closure under divisor convolutions and a larger arithmetic catalogue

The valuation-polynomial theorem supplies a structural extension, not just another finite list of numerators. The class is closed under the three standard ways of combining multiplicative counting functions: pointwise multiplication, Dirichlet convolution, and unitary convolution. Repeated local poles, which produce polynomials in the prime-power exponent, are precisely why the additional valuation variable is needed.

17.1 Three closure operations

In this section an admissible valuation profile has the meaning of Theorem 16.5. We write its data as (R, h, κ), where

$$f(p^a) = p^{ha} R(1/p, p^{-a}, a), \qquad R(z, z, 1) = \kappa + O(z), \qquad \kappa \geq 1.$$

Here $h \geq 0$ is an integer, $R \in \mathbb{Q}(z)[w, u]$, and the sampled local values are positive, polynomially bounded in a, and give integers after multiplication by p^{ha}. The normalization is always $f(1) = 1$. Identical normalized profiles are grouped, regardless of the degree used to realize them.

For multiplicative functions normalized at 1, define

$$(f * g)(n) = \sum_{d|n} f(d)g(n/d), \qquad (f \star_u g)(n) = \sum_{\substack{d|n \\ (d,n/d)=1}} f(d)g(n/d).$$

The second operation is unitary convolution. It assigns a whole prime power to one of its two factors.

Proposition 17.1 (Arithmetic closure)**.** *Let $f_1, \dots, f_s$ have admissible valuation profiles (R_i, h_i, κ_i).*

1. *The pointwise product $\prod_i f_i$ has profile $\prod_i R_i$, degree $\sum_i h_i$, and parameter $\prod_i \kappa_i$.*
2. *Their Dirichlet convolution $f_1 * \cdots * f_s$ has an admissible valuation profile of degree $H = \max_i h_i$ and parameter*
$$\kappa = \sum_{i:h_i=H} \kappa_i.$$
3. *Their iterated unitary convolution has degree H and the same parameter as in part 2. Its profile is explicitly*
$$R(z, w, u) = \sum_{i=1}^{s} w^{H-h_i} R_i(z, w, u).$$

These operations therefore preserve every factorial-series conclusion of Theorem 16.5, after the usual grouping and removal of the constant profile 1*.*

Proof. Multiplicativity, positive integer values, and the asserted first-prime parameters are immediate for all three operations. For part 2 put

$$G_i(z, T) = 1 + \sum_{a \geq 1} R_i(z, z^a, a) T^a.$$

Expanding R_i into monomials $w^j u^m$ expresses G_i as a rational function of z, T whose possible T-poles are $T = z^{-j}$, $j \geq 0$. Indeed $\sum_{a \geq 1} a^m (z^j T)^a$ has only the pole $1 - z^j T = 0$. Moreover

G_i is bounded at $T=\infty$, so its polynomial part in T is at most constant. The normalized local generating function of the convolution is exactly

$$G(z,T)=\prod_{i=1}^{s} G_i(z,z^{H-h_i}T). \tag{56}$$

Its poles have the form $T=z^{-d}$ with $d\ge 0$. Partial fractions in T show that, for $a\ge 1$, its coefficient is a finite sum

$$\sum_{d,m} c_{d,m}(z)\binom{a+m-1}{m-1}z^{da}.$$

This is $R(z,z^a,a)$ for a polynomial $R\in\mathbb{Q}(z)[w,u]$. The additional denominators introduced by partial fractions are powers of z and differences z^d-z^e with $d\neq e$, so none introduces a pole at $z=1/p$.

For a uniform bound, suppose $R_i(1/p,p^{-a},a)\le C_i(1+a)^{D_i}$, also allowing $a=0$ by increasing C_i. A coefficient of (56) is a sum over $a_1+\cdots+a_s=a$. The additional factors $p^{-(H-h_i)a_i}$ are at most one. Thus it is bounded by

$$\left(\prod_i C_i\right)(1+a)^{D_1+\cdots+D_s+s-1}$$

up to a constant depending only on s. The prime coefficient is

$$\sum_i z^{H-h_i}R_i(z,z,1)=\sum_{i:h_i=H}\kappa_i+O(z).$$

This proves part 2. The displayed profile in part 3 follows by assigning p^a to exactly one factor; its bounds follow at once. Part 1 follows by multiplying the local data. □

17.2 All weighted ordered factorizations

Let $\mathbf{m}=(m_0,m_1,\ldots)$ have finite support, with $m_0\ge 1$ and $m_j\ge 0$. Put $H=\max\{j:m_j>0\}$ and define

$$D_{\mathbf{m}}=\mathop{*}_{j=0}^{H}(\mathrm{id}^{H-j})^{*m_j},\qquad \mathrm{id}^b(n)=n^b. \tag{57}$$

A zeroth convolution power is omitted. Thus $D_{\mathbf{m}}(n)$ is a weighted count of ordered factorizations of n; the list of factor weights contains m_j copies of $H-j$. Replacing every weight by that weight plus $c\ge 0$ multiplies the answer by n^c, which is already covered by the polynomial-moment coordinates.

The degree is H, the parameter is m_0, and its profile is uniquely determined by

$$1+\sum_{a\ge 1}R_{\mathbf{m}}(z,z^a,a)T^a=\prod_{j\ge 0}(1-z^jT)^{-m_j}. \tag{58}$$

This formula is also an exact algorithm for computing the profile. It includes arbitrary repeated weights, rather than just the distinct weights occurring in the subgroup-counting functions.

Define $U_{\mathbf{m}}$ by replacing every Dirichlet convolution in (57) with unitary convolution. It counts only ordered factorizations whose factors are pairwise coprime. Its normalized profile is particularly simple:

$$R^{u}_{\mathbf{m}}(z,w,u)=\sum_{j\ge 0}m_jw^j. \tag{59}$$

In particular, weights all equal to zero give

$$D_{(r)}(n)=d_r(n),\qquad U_{(r)}(n)=r^{\omega(n)}.$$

Here d_r counts ordered factorizations into r positive factors, and $\omega(n)$ is the number of distinct prime divisors. Both are normalized to take the value 1 at $n=1$.

Corollary 17.2 (A nonredundant factorization hierarchy)**.** *Exclude the single vector* $(1,0,0,\ldots)$*. The profiles of all* $D_{\mathbf{m}}$ *and all* $U_{\mathbf{m}}$ *are pairwise distinct, both within and between the two families. Consequently the family*

$$\{1, e^x\} \ \cup \ \left\{ \sum_{n\geq 1} \frac{n^d D_{\mathbf{m}}(n) x^n}{n!}, \ \sum_{n\geq 1} \frac{n^d U_{\mathbf{m}}(n) x^n}{n!} : d \geq 0, \ \mathbf{m} \neq (1,0,\ldots) \right\}$$

is linearly independent over $\mathbb{Q}$ *at every permitted* x*. Every residue-class, two-sided jet, and simple-negative-pole extension in Theorem 16.5 applies to the same canonical list.*

Proof. At the first prime power, both profiles give $\sum_j m_j z^j$. Thus equality of profiles forces equality of the vectors $\mathbf{m}$. For a fixed vector, write the positive numbers z^j, repeated m_j times, as $b_1, \ldots, b_\ell$. At the second prime power the ordinary and unitary values differ by

$$\sum_{1\leq i<j\leq \ell} b_i b_j > 0$$

when $\ell \geq 2$. The excluded vector is the only case with $\ell = 1$ under the convention $m_0 \geq 1$ and no redundant overall power of n. Thus all retained profiles are distinct and none is 1. The conclusion follows from the valuation-profile theorem. □

For example, if $h, m \geq 1$, let

$$B_{h,m}(n) = \sum_{d\mid n} d^h d_m(n/d).$$

This is the subfamily with $m_0 = 1$, $m_h = m$, and all other entries zero. The case $m = 1$ is σ_h, whereas $m \geq 2$ produces genuine valuation dependence. For instance

$$R_{B_{h,2}}(z,w,u) = \frac{1 - z^h w^h\big((u+2) - (u+1)z^h\big)}{(1-z^h)^2}. \tag{60}$$

Its coefficient of uw^h is nonzero. Polynomial-exponential independence therefore shows that it cannot be represented by a profile in $\mathbb{Q}(z)[w]$ alone. The new theorem really enlarges the earlier class.

The mean constants are explicit without becoming new irrationality claims about zeta values. For the ordinary hierarchy, the normalized unrestricted local mean is

$$U_{R_{\mathbf{m}}}(z) = (1-z)^{1-m_0} \prod_{j\geq 1} (1 - z^{j+1})^{-m_j}.$$

Hence Lemma 16.3 gives

$$\frac{1}{X} \sum_{n\leq X} \frac{D_{\mathbf{m}}(n)}{n^H} \sim \frac{\prod_{j\geq 1} \zeta(j+1)^{m_j}}{(m_0-1)!} (\log X)^{m_0-1}. \tag{61}$$

In particular $B_{h,m}(n)/n^h$ has mean $\zeta(h+1)^m$. Equation (61) is a mean-value formula; the irrational numbers asserted here remain the factorial series.

17.3 All positive gcd moments, including the diagonal case

For $r, k \geq 1$ put

$$G_{r,k}(n) = \sum_{1\leq a_1,\ldots,a_r\leq n} \gcd(a_1,\ldots,a_r,n)^k. \tag{62}$$

Grouping by the common gcd gives

$$G_{r,k} = \mathrm{id}^k * J_r = \mu * \mathrm{id}^k * \mathrm{id}^r . \tag{63}$$

The number of tuples with common gcd $d \mid n$ is $J_r(n/d)$: divide by d and exclude, for each prime dividing n/d, the tuples all of whose coordinates are divisible by that prime. This proves (63) directly. In particular $G_{r,k} = G_{k,r}$, so only one order of the pair is retained.

Set $H = \max(r, k)$ and $\Delta = |r - k|$. The normalized local generating function is

$$1 + \sum_{a \geq 1} \frac{G_{r,k}(p^a)}{p^{Ha}} T^a = \frac{1 - z^H T}{(1 - T)(1 - z^\Delta T)}, \qquad z = 1/p. \tag{64}$$

For $\Delta > 0$ this gives the bounded profile

$$R_{r,k}(z, w, u) = 1 + (1 - z^{\min(r,k)}) \frac{z^\Delta (1 - w^\Delta)}{1 - z^\Delta}.$$

For $r = k$ it gives instead

$$R_{r,r}(z, w, u) = 1 + u(1 - z^r), \qquad \kappa = 2. \tag{65}$$

This diagonal was absent from the bounded-profile theorem. Its first case is the classical gcd-sum, or Pillai, function $G_{1,1}(n) = \sum_{a=1}^{n} \gcd(a, n)$.

The profiles for $1 \leq k \leq r$ are all distinct: their first-prime values are $1 + z^{r-k} - z^r$ when $k < r$, and $2 - z^r$ when $k = r$. They are also distinct from every ordinary and unitary factorization profile above, whose first-prime polynomials have nonnegative coefficients. Therefore all the $G_{r,k}$, $k \leq r$, can be adjoined *simultaneously* to Corollary 17.2, with all the same moments, residue restrictions, and two-sided jets.

For reference, their normalized means are

$$\frac{1}{X} \sum_{n \leq X} \frac{G_{r,k}(n)}{n^H} \sim \begin{cases} \zeta(\Delta + 1)/\zeta(H + 1), & r \neq k, \\ (\log X)/\zeta(r + 1), & r = k. \end{cases}$$

They follow either from (64) or from the general mean theorem, with no transfer of irrationality to these zeta ratios.

17.4 Gcd moments on the unit group

A second natural family sums over units rather than over all residue classes. For $r, k \geq 1$ define

$$M_{r,k}(n) = \sum_{a_1, \ldots, a_r \in (\mathbb{Z}/n\mathbb{Z})^\times} \gcd(a_1 - 1, \ldots, a_r - 1, n)^k. \tag{66}$$

Use the convention that the unit group modulo 1 has one element, so $M_{r,k}(1) = 1$. The basic identity $g^k = \sum_{d \mid g} J_k(d)$ and the surjectivity of reduction of units modulo n onto units modulo $d \mid n$ give

$$M_{r,k}(n) = \varphi(n)^r \sum_{d \mid n} \frac{J_k(d)}{\varphi(d)^r}. \tag{67}$$

Indeed, for each $d \mid n$ there are $(\varphi(n)/\varphi(d))^r$ unit tuples with every coordinate congruent to 1 modulo d. Reduction is surjective prime power by prime power, and then by the Chinese remainder theorem. This proves the identity, its multiplicativity, and its positive integrality. These are classical Menon-type counting identities; see [11]. At $r = k = 1$ one recovers $M_{1,1} = \varphi\tau$.

The prime-power computation from (67) is explicit. With degree $H = \max(r, k)$, the normalized profiles are

$$R^M_{r,k}(z,w,u) = \begin{cases} (1-z)^r w^{k-r} + (1-z^k)\dfrac{1-w^{k-r}}{1-z^{k-r}}, & k > r, \\ (1-z)^r + u(1-z^r), & k = r, \\ (1-z)^r + (1-z^k)\dfrac{z^{r-k}(1-w^{r-k})}{1-z^{r-k}}, & k < r. \end{cases} \tag{68}$$

They are bounded in the off-diagonal cases and polynomially bounded in the diagonal case. Positivity follows directly from the count. The parameter is 1 off the diagonal and 2 on the diagonal. None is the constant profile 1: in the off-diagonal cases the coefficient of the displayed nonconstant power of w is nonzero, and on the diagonal the coefficient of u is nonzero. Consequently every factorial series formed from $M_{r,k}$ is irrational, and every finite product or convolution of these functions with the previous catalogue is covered by Proposition 17.1 whenever its resulting profile is not 1. Joint statements use the same exact profile-grouping rule; they do not presume that every written counting formula is a different coordinate.

17.5 All and cyclic subgroups of rank-two finite groups

Let $C_m = \mathbb{Z}/m\mathbb{Z}$. For $1 \le r \le s$, let $S_{r,s}(n)$ be the number of all subgroups of $C_{n^r} \times C_{n^s}$, and let $C_{r,s}(n)$ be the number of its cyclic subgroups. These familiar counting functions have explicit formulas; see [12]. The following direct argument supplies all arithmetic inputs needed here.

Proposition 17.3. *Both functions are multiplicative and have admissible valuation profiles of degree r and parameter $\kappa = s - r + 1$. Writing $b = s - r$, the profiles are*

$$R^S_{r,s}(z,w,u) = (bu+1)\frac{1-zw^r}{1-z} + \frac{2z\big(1-(ru+1)w^r + ruzw^r\big)}{(1-z)^2}, \tag{69}$$

$$R^C_{r,s}(z,w,u) = bu + 1 + \frac{2z(1-w^r)}{1-z}. \tag{70}$$

All profiles in the combined S and C list are pairwise distinct and nonconstant. Thus all their factorial constants and polynomial moments are jointly linearly independent with $1, e^x$, and the full residue-class and two-sided hierarchy applies.

Proof. For groups of coprime orders every subgroup splits into its primary components, and cyclicity does also. This proves multiplicativity. To count subgroups of $C_{p^A} \times C_{p^B}$, choose the projection onto the first factor, of order p^i, and the intersection with the second factor. If the quotient of the second factor by that intersection has order p^j, the remaining choice is a homomorphism $C_{p^i} \longrightarrow C_{p^j}$. There are $p^{\min(i,j)}$ such homomorphisms. Consequently, for $A = ra$, $B = sa$,

$$S_{r,s}(p^a) = \sum_{i=0}^{ra}\sum_{j=0}^{sa} p^{\min(i,j)} = p^{ra}\sum_{d=0}^{ra}\big((s-r)a + 2d + 1\big)p^{-d}.$$

At $d = 0$ there are $(s-r)a+1$ pairs having minimum ra; at the other depths successive rectangular counts give the coefficient $(s-r)a + 2d + 1$. Summing the finite geometric series and its derivative gives (69).

For cyclic subgroups, count elements of each exact order and divide by the number of generators of a cyclic group of that order. For orders p^j with $1 \le j \le ra$ this gives $(p+1)p^{j-1}$;

for $ra < j \le sa$ it gives p^{ra}. Including the trivial subgroup yields

$$C_{r,s}(p^a) = \big((s-r)a+1\big)p^{ra} + 2\sum_{j=0}^{ra-1} p^j,$$

which proves (70). Positivity and integrality follow from the counts; the two displayed profiles are uniformly $O(1+a)$ at the sampled arguments.

At the first prime power their normalized values are, respectively,

$$\kappa + \sum_{d=1}^{r}(\kappa+2d)z^d, \qquad \kappa + 2\sum_{d=1}^{r} z^d.$$

The polynomial degree and constant term identify r, s within either list. Between the lists the coefficient of z differs, since $\kappa + 2 \ge 3$ in the first list and it is 2 in the second. Every profile is nonconstant. Apply Theorem 16.5. □

In particular the series counting all subgroups of $C_n \times C_n$, $C_n \times C_{n^2}$, and $C_{n^2} \times C_{n^5}$, and the corresponding cyclic-subgroup series, are covered at once. Their arbitrary fixed pointwise products and divisor convolutions remain within Proposition 17.1; cross-family lists are again grouped by their exact profiles before independence is asserted.

17.6 A compact list of concrete consequences

Write $\mathcal{E}_f(x) = \sum_{n\ge1} f(n)x^n/n!$. At every permitted x, each of the following is irrational:

$$\mathcal{E}_{d_r}(x)\ (r\ge2), \qquad \mathcal{E}_{r^\omega}(x)\ (r\ge2), \qquad \mathcal{E}_{\mathrm{id}^h * d_m}(x)\ (h,m\ge1),$$

$$\mathcal{E}_{G_{r,k}}(x)\ (r,k\ge1), \qquad \mathcal{E}_{M_{r,k}}(x)\ (r,k\ge1).$$

For a particularly concrete simultaneous statement, the numbers

$$1,\ e^2,\ \mathcal{E}_{d_3}(2),\ \mathcal{E}_{3^\omega}(2),\ \mathcal{E}_{\mathrm{id}*d_2}(2),\ \mathcal{E}_{G_{1,1}}(2),\ \mathcal{E}_{G_{2,1}}(2),\ \mathcal{E}_{\varphi\tau}(2)$$

are linearly independent over $\mathbb{Q}$. The first-prime profile values are respectively $3, 3, 1+2z, 2-z, 1+z-z^2, 2-2z$; the sole collision, between d_3 and 3^ω, is separated at the second prime power by $6 \ne 3$. All degree-zero, residue-class, primitive-integral, and higher-derivative coordinates permitted by the general theorem can be added without changing this reasoning, after canonical grouping.

The countably infinite statements in this section mean that every finite subfamily is independent. They concern values of factorial series and their specified linear transforms. They neither assert algebraic independence of these values nor identify new historically named open problems for each illustrative constant.

18 Finite-ring matrices, Grassmannians, and flag counts

The profile class also includes counting functions from finite linear algebra. The counting identities below are elementary and classical; the application here is their insertion into the common factorial-value theorem. Put $R_n = \mathbb{Z}/n\mathbb{Z}$, with the usual convention that each counting function has value one at $n = 1$.

Proposition 18.1 (Three finite-geometric counting profiles)**.** *For $m \ge r \ge 1$, let $S_{m,r}(n)$ count surjective R_n-linear maps $R_n^m \longrightarrow R_n^r$. For $m > r \ge 1$, let $G_{m,r}(n)$ count free direct summands of rank r in R_n^m. If $\mathbf{b} = (b_1, \dots, b_s)$ has positive integer entries, $s \ge 2$, and $m = \sum b_i$, let $F_{\mathbf{b}}(n)$*

count flags of free direct summands whose successive quotient ranks are $b_1, \dots, b_s$. All three functions are positive integral and multiplicative. Their degree h and normalized prime-power profile are

function	h	$R(z)$
$S_{m,r}$	mr	$\displaystyle\prod_{j=m-r+1}^{m}(1-z^j)$
$G_{m,r}$	$r(m-r)$	$\dfrac{\prod_{j=m-r+1}^{m}(1-z^j)}{\prod_{j=1}^{r}(1-z^j)}$
$F_{\mathbf{b}}$	$\displaystyle\sum_{i<j} b_i b_j$	$\dfrac{\prod_{j=1}^{m}(1-z^j)}{\prod_{i=1}^{s}\prod_{j=1}^{b_i}(1-z^j)}$.

Here $f(p^a) = p^{ha}R(1/p)$ for every prime p and $a \geq 1$.

Proof. Over $\mathbb{Z}/p^a\mathbb{Z}$, a matrix defines a surjection if and only if its reduction modulo p has full row rank. There are $\prod_{j=0}^{r-1}(p^m - p^j)$ such matrices over $\mathbb{F}_p$, and each has $p^{(a-1)mr}$ lifts. This proves the first formula. The criterion follows directly from a unit $r \times r$ minor; conversely a surjection remains surjective after reduction.

A full-column-rank $m \times r$ matrix over this local ring is an ordered basis of a free direct summand. Its column basis can be extended to a basis of the ambient module, again using a unit minor. Each summand has exactly $|\mathrm{GL}_r(\mathbb{Z}/p^a\mathbb{Z})| = S_{r,r}(p^a)$ ordered bases. Thus $G_{m,r}(p^a) = S_{m,r}(p^a)/S_{r,r}(p^a)$, giving the second formula.

The general linear group acts transitively on the stated flags, since bases of the successive free quotients can be lifted and concatenated. The stabilizer of a standard flag consists of block upper triangular matrices with invertible diagonal blocks of sizes b_i. If $D = \sum_{i<j} b_i b_j$, its order is $p^{aD}\prod_i S_{b_i,b_i}(p^a)$. Dividing $S_{m,m}(p^a)$ by this order gives the last formula, since $m^2 - \sum b_i^2 - D = D$. The Chinese remainder decomposition of R_n proves multiplicativity and the asserted product formulas for arbitrary n. □

Each displayed rational function has no pole on $0 < z \leq 1/2$, is positive there, is bounded there, and has value $1 + O(z)$ at zero. It is consequently an admissible profile from Section 11. It is nonconstant for the specified nontrivial ranks. For $S_{m,r}$ this follows from the product formula. For $G_{m,r}$ the profile is the Gaussian binomial polynomial $\left[\begin{smallmatrix} m \\ r \end{smallmatrix}\right]_z$, of positive degree $r(m-r)$, with constant and leading coefficients one. The flag profile is a Gaussian multinomial polynomial of positive degree D. These polynomial descriptions also follow inductively by decomposing subspaces over a finite field according to their intersection with a fixed hyperplane; the rational product formulas above are enough for admissibility.

Two especially familiar cases are

$$|\mathrm{GL}_m(R_n)| = n^{m^2}\prod_{p\mid n}\prod_{j=1}^{m}(1-p^{-j}), \qquad |\mathrm{SL}_m(R_n)| = n^{m^2-1}\prod_{p\mid n}\prod_{j=2}^{m}(1-p^{-j}) \quad (m \geq 2).$$

The second identity uses the surjective determinant map onto $R_n^\times$. For complete flags, $\mathbf{b} = (1, \dots, 1)$, the profile is $\prod_{j=1}^{m}(1 + z + \cdots + z^{j-1})$.

Corollary 18.2 (Factorial constants from finite linear algebra)**.** *For every nontrivial counting function f above, every permitted $x = \pm 2/b$, and every integer j, the number $A_f^{[j]}(x)$ is irrational. The same holds after any fixed residue restriction and with every weight n^d ($d \geq 0$) or $1/(n+a)$ ($a \geq 1$). All these values obey the joint-independence and exact rationality rules of the canonical profile theorem. In particular a single fixed f supplies the entire independent two-sided hierarchy.*

Proof. The preceding proposition verifies all the arithmetic hypotheses. Choose f itself as the integral realization of its normalized profile and apply Theorem 14.1 and Corollary 14.4. For a collection of counting functions, choose one integral realization per distinct profile. Any other realization of the same profile differs by an integral power of n; after choosing the least degree in a finite collection, this is a nonnegative polynomial weight. Shifted-factorial values reduce to polynomial weights and simple negative-integer poles by partial fractions. Their coefficient vectors, rather than their different counting names, therefore determine all joint relations. □

For example, the following are irrational at $x = 2$:

$$\sum_{n\geq 1} \frac{2^n |\mathrm{GL}_3(R_n)|}{n!}, \qquad \sum_{n\geq 1} \frac{2^n G_{6,2}(n)}{n!(n+7)}, \qquad \int_0^1 t^4(1-t)^3 A_{F_{(2,1,3)}}(2t)\, dt.$$

There are genuine equalities among these counting functions, which the profile criterion preserves. For example $G_{2,1}(n) = \Psi_1(n)$ and $|\mathrm{SL}_2(R_n)| = nJ_2(n)$. Also $G_{m,r} = G_{m,m-r}$, and permutations of the block sizes have the same flag count. These are identical arithmetic coefficients, not extra independent constants.

One further matrix family comes for free. Let $M_{u,v,r}(n)$ count $u \times v$ matrices whose image is a free rank-r direct summand, where $1 \leq r \leq \min(u, v)$ and $G_{r,r} := 1$. Choosing the image and then a surjection onto it gives

$$M_{u,v,r}(n) = G_{u,r}(n) S_{v,r}(n).$$

Hence its degree is $r(u + v - r)$ and its profile is the product of the two profiles above. It is nontrivial, positive, and admissible, so all the preceding irrationality statements apply. Finally, the closure theorems of Section 17 permit any finite pointwise, ordinary Dirichlet, or unitary convolution of these counts with the other admitted arithmetic families, subject to the same explicit grouping of equal normalized profiles.

19 Total prime-factor multiplicity on rough residue classes

Polynomial dependence on a prime exponent includes $c^{\omega(n)}$, but not $c^{\Omega(n)}$, where Ω counts prime factors with multiplicity. An additional growth observation admits the latter on every residue class avoiding the finitely many primes at most c. This extension uses an exponential, rather than polynomial, local dependence on the valuation.

Lemma 19.1 (Strictly sublinear normalized growth suffices)**.** *In Theorem 14.1, its growth hypothesis can be replaced by the following: for each base f there is a real number $0 \leq \theta_f < 1$ such that*

$$f(n) = O_f(n^{h_f + \theta_f}).$$

The other hypotheses, in particular multiplicativity, positive integer values, and isolated-shift separation, are unchanged. All its residue, derivative, antiderivative, and rational-weight consequences remain valid. There is no requirement that one common $\theta < 1$ work for an infinite base family.

Proof. In a finite proposed relation choose $\theta = \max_f \theta_f < 1$. Keep the same notation r, w, J, T and the same grid as in the proof of Theorem 14.1. The tail expansion (53) now has remainder

$$O(n^{h+\theta-J-r-2}) = O(n^{w+\theta-J-2}).$$

Indeed the first omitted factorial term has denominator order $J + r + 2$, and the same bound holds for the finite denominator expansions. The surviving terminal terms are $O(N^{w_*+\theta-J-1})$.

Since $J \geq w_* + 1$, their total is $O(N^{\theta-2})$, and the separately cancelled exponential remainder is $O(N^{-T})$. The common dyadic denominator is still $O(N)$. Multiplication by it therefore gives an integer of size $O(N^{\theta-1}) + O(N^{1-T})$, tending to zero.

After the weighted tail becomes identically zero, multiply by N^{J+1-w_*} as before. The expansion error is $O(N^{\theta-1})$; each smaller effective degree contributes at most $O(N^{\theta-1})$; and replacement of a fixed shifted power ratio by 1 has the same bound. The exponential remainder tends to zero by the original choice $T = \max(2, J + 2 - w_*)$. The identical isolated-shift contradiction follows. All subsequent changes of basis are exact and hence remain applicable. □

For an integer $c \geq 2$ define

$$M_c = \prod_{p \leq c} p, \qquad n_{>c} = \prod_{\substack{p^a \| n \\ p > c}} p^a, \qquad W_c(n) = c^{\Omega(n_{>c})}.$$

Thus W_c is a positive integer-valued completely multiplicative function, with prime value 1 at $p \leq c$ and c at $p > c$. On $(n, M_c) = 1$ it agrees with the ordinary function $c^{\Omega(n)}$. Let P_c be the least prime exceeding c and put

$$\theta_c = \frac{\log c}{\log P_c} < 1.$$

Prime by prime, $c \leq p^{\theta_c}$ for $p > c$, so

$$1 \leq W_c(n) \leq n^{\theta_c}. \tag{71}$$

Lemma 19.2 (Logarithmic means for the rough multiplicity function)**.** *The function W_c, normalized with $h = 0$, has positive progression means of logarithmic order $c - 1$ as in Lemma 16.3. Its unrestricted leading coefficient is $G_c/(c-1)!$, where*

$$G_c = \prod_{p \leq c} (1 - p^{-1})^{c-1} \prod_{p > c} \frac{(1 - p^{-1})^c}{1 - c/p} > 0.$$

At a fresh prime $p > c$, the unit and exact-valuation-t refinement factors are, respectively,

$$\frac{1}{U_c(1/p)}, \qquad \frac{c^t}{U_c(1/p)}, \qquad U_c(z) = \frac{1 - z}{1 - cz}.$$

In particular the response vector is

$$\delta_{c,t}(z) = (c^t - 1)\frac{1 - cz}{1 - z}, \qquad t \geq 1. \tag{72}$$

Proof. Set $g_c = W_c * \mu^{*c}$. Its prime-power generating series is

$$\sum_{a \geq 0} g_c(p^a) T^a = \begin{cases} (1 - T)^{c-1}, & p \leq c, \\ (1 - T)^c/(1 - cT), & p > c. \end{cases}$$

At a good prime $p > c$ the linear coefficient vanishes, and the higher coefficients are $O_c(c^a)$. Choose any

$$\max(1/2, \theta_c) < \sigma < 1.$$

Then $cP_c^{-\sigma} < 1$ and

$$\sum_{p > c} \sum_{a \geq 2} |g_c(p^a)| p^{-a\sigma} \ll_{c,\sigma} \sum_{p > c} p^{-2\sigma} < \infty.$$

The finitely many remaining Euler factors are polynomials. Therefore $\sum_n |g_c(n)| n^{-\sigma} < \infty$.

This is the exact convolution input used in the proof of Lemma 16.3, with d_c in place of d_κ. For the nonprincipal character estimate one chooses the convolution exponent also larger than $1 - 1/c$; this is possible because all three lower bounds are strictly less than 1. The principal-character calculation gives G_c above, and the nonprincipal sums have a power-saving error. The good Euler factors of G_c are positive and $1 + O_c(p^{-2})$, so the product is positive.

For nonunit progressions there is no new local-growth issue: $W_c(am) = W_c(a)W_c(m)$ holds for all a, m, without a coprimality restriction. Dividing a progression by (a, Q) therefore reduces it to the unit-class calculation with the same function, multiplied by a positive constant. The local conditional factors are those in Lemma 16.3. At $p > c$ its local series is $1/(1 - cz)$, which gives U_c and (72). All errors are $O_{c,Q,a}((\log X)^{c-2})$ after division by X. □

Theorem 19.3 (Adjoining every rough total-multiplicity channel)**.** *Adjoin W_c, with growth degree $h = 0$, for every integer $c \geq 2$ to the canonical family of nontrivial admissible valuation profiles in Theorem 16.5. The enlarged family satisfies the two-sided hierarchy and all its residue-class conclusions. It also satisfies the cyclotomic-coordinate and quadratic-field conclusions of Section 21.*

Proof. The strict growth condition is (71), and the progression means were just proved. It remains to verify joint isolated-shift separation; individual irrationality alone would not suffice.

Use the largest logarithmic order in a finite proposed limit, exactly as in Lemma 16.4. Two different W_c have different orders. At a fixed order $c - 1$, compare W_c with a polynomial valuation profile $\mathcal{R}$ of $\kappa_{\mathcal{R}} = c$. Their response vectors cannot agree. Otherwise, for every $t \geq 1$,

$$\mathcal{R}(z, z^t, t) = 1 + \frac{U_{\mathcal{R}}(z)}{U_c(z)}(c^t - 1).$$

Fix a sufficiently small real $z \in (0, 1/c)$ avoiding all poles and zeros of the nonzero rational prefactor. The left side has at most polynomial growth in t, because $\mathcal{R}$ is polynomial in its last two variables and $z^t \to 0$. The right side has nonzero exponential growth. This is a contradiction.

Thus the response vectors within every logarithmic order are distinct and nonzero. All their fixed-t components are rational functions of z. The same finite fresh-prime interpolation as in Lemma 16.4 separates the surviving coefficients. Primes at most the largest retained c are simply excluded from the finite set of allowed fresh primes. This proves isolated-shift separation for the enlarged family. Apply Lemma 19.1 and then the exact changes of basis already established. □

19.1 Products with every polynomial valuation profile

The rough multiplicity construction can be combined with all the earlier arithmetic profiles, not only placed beside them. Set $W_1(n) = 1$ and $M_1 = 1$. For $c \geq 1$ and an admissible valuation profile $\mathcal{R}$, choose one integral realization $f_{\mathcal{R},h} = n^h \rho_{\mathcal{R}}$ and put

$$F_{c,\mathcal{R}}(n) = W_c(n) f_{\mathcal{R},h}(n), \qquad \rho_{c,\mathcal{R}}(n) = W_c(n)\rho_{\mathcal{R}}(n).$$

Permit $\mathcal{R} = 1$ here with $h = 0$, but exclude the single pair $(c, \mathcal{R}) = (1, 1)$. Profiles equal as elements of $\mathbb{Q}(z)[w, u]$ are grouped before selecting the realization.

Theorem 19.4 (Rough exponential times polynomial profiles)**.** *The entire canonical family $F_{c,\mathcal{R}}$, over all the pairs just specified, satisfies the two-sided hierarchy, its residue-class and rational-weight consequences, and the cyclotomic and quadratic-field consequences. Distinct pairs give separated normalized phases.*

Proof. At a good prime $p > c$ its normalized local values are $c^a\mathcal{R}(p^{-1}, p^{-a}, a)$; at $p \le c$ the factor c^a is omitted. The integral realization remains a positive integer and is multiplicative. The previous bounds give $\rho_{c,\mathcal{R}}(n) = O(n^{\theta_c+\epsilon})$ for every $\epsilon > 0$ (take $\theta_1 = 0$). Choose $0 < \epsilon < 1 - \theta_c$ and use Lemma 19.1.

The logarithmic order is $\kappa - 1$, where $\kappa = c\kappa_{\mathcal{R}}$. Indeed $g = \rho_{c,\mathcal{R}} * \mu^{*\kappa}$ has $g(p) = O(p^{-1})$ at good primes and $g(p^a) = O(c^a(1+a)^D)$ at higher powers. Its Dirichlet series converges absolutely at every $\sigma > \max(1/2, \theta_c)$; the finitely many bad local factors have only polynomial coefficient growth. The character-convolution proof of Lemma 16.3 therefore applies. In the nonunit reduction, a modification at a fixed good prime has values bounded by a constant times $c^a(1+a)^D$, so its local series still converges at the same $\sigma < 1$; at a bad prime it has polynomial growth. Thus that part of the proof also applies.

At good primes the rational local functions are

$$L_{c,\mathcal{R}}(z) = 1 + \sum_{a\ge1} c^a\mathcal{R}(z, z^a, a)z^a, \qquad U_{c,\mathcal{R}}(z) = (1-z)L_{c,\mathcal{R}}(z).$$

Rationality follows term by term from $\sum_{a\ge1} a^m(cz^{j+1})^a$. The leading Euler product is strictly positive: its local factors are $(1-p^{-1})^\kappa L_{c,\mathcal{R}}(1/p)$ at good primes, and $(1-p^{-1})^\kappa\mathcal{L}_{\mathcal{R}}(1/p)$ at bad primes. Every factor is positive and the good factors are $1 + O(p^{-2})$. The fresh-prime response components are

$$\delta_{c,\mathcal{R},t}(z) = \frac{c^t\mathcal{R}(z, z^t, t) - 1}{U_{c,\mathcal{R}}(z)}.$$

These vectors are injective in the pair $(c, \mathcal{R})$. The identity

$$\sum_{t\ge1} \delta_{c,\mathcal{R},t}(z)z^t = \frac{U_{c,\mathcal{R}}(z) - 1}{(1-z)U_{c,\mathcal{R}}(z)}$$

first recovers U. Equality of two vectors consequently gives

$$c^t\mathcal{R}(z, z^t, t) = c'^t\mathcal{R}'(z, z^t, t) \qquad (t \ge 1).$$

Write $\mathcal{R}(z, w, u) = \sum_j r_j(z, u)w^j$ and likewise for $\mathcal{R}'$. Specialize z to a sufficiently small positive real number avoiding the finitely many coefficient poles and zeros needed to retain a nonzero coefficient. If $c \ne c'$, also avoid the finitely many equations $cz^j = c'z^{j'}$. The displayed equality is then an identity of polynomial-exponential sequences with distinct positive bases cz^j and $c'z^{j'}$. Such sequences are independent: divide by the largest base having a nonzero polynomial coefficient; the coefficient would tend to zero exponentially, impossible for a nonzero polynomial. Removing that term inductively proves the assertion. Thus $c = c'$, and the same argument applied to $\mathcal{R} - \mathcal{R}'$ gives $\mathcal{R} = \mathcal{R}'$. Comparison with $(1, 1)$ also shows that only the excluded pair has zero response vector.

At the largest retained logarithmic order the fresh-prime interpolation of Lemma 16.4 now applies to these distinct rational response vectors. All primes at most the largest retained c are avoided. This proves the required isolated-shift separation and completes the proof. □

In particular, on the unit classes modulo M_c the numerator is the unmodified product $c^{\Omega(n)}f_{\mathcal{R},h}(n)$. Thus one may simultaneously include such products with powers of τ, ordered-factorization functions, divisor sums, Jordan functions, and the other proved polynomial valuation profiles. Products that give the same polynomial profile must first be grouped. The construction does not license arbitrary products of the W_c without rechecking growth: for example, the local value of W_2W_3 at 5^a is 6^a, which exceeds the strictly sublinear regime at that prime.

Corollary 19.5 (Unmodified total-multiplicity sums on rough classes)**.** *Define the entire function*

$$B_c(z) = \sum_{\substack{n\ge 1\\(n,M_c)=1}} \frac{c^{\Omega(n)}z^n}{n!}, \qquad c \ge 2.$$

At every permitted positive x, all values $B_c^{[j]}(x)$, $c \ge 2$, $j \in \mathbb{Z}$, are jointly linearly independent with $1, e^x$ and all the arithmetic two-sided jets from the polynomial valuation families. The same holds for individual unit residue classes modulo M_c, after retaining independent residue coordinates. Every one of these values is irrational.

Proof. On the indicated residue classes $W_c(n) = c^{\Omega(n)}$ exactly. Thus B_c is a nonempty sum of residue-restricted A_{W_c} functions. For any finite proposed relation take a common multiple of the finitely many M_c. Each coefficient of $B_c^{[j]}$ is then copied to all its unit residue coordinates. The supports for different base-and-order pairs are disjoint, so a nonzero coefficient cannot vanish in this conversion. The preceding theorem gives the contradiction. One must not list both a sum of residue coordinates and all of its summands as an independent family; the assertion uses either independent individual coordinates or the displayed sums. □

For example, each of

$$\sum_{\substack{n\ge 1\\ n \text{ odd}}} \frac{2^{\Omega(n)+n}}{n!}, \qquad \sum_{\substack{n\ge 1\\(n,6)=1}} \frac{3^{\Omega(n)}2^n}{n!}, \qquad \sum_{\substack{n\ge 1\\(n,6)=1}} \frac{4^{\Omega(n)}2^n}{n!}$$

is irrational, and the three are jointly independent with $1, e^2$ and all the polynomial valuation profile jets at 2. Polynomial moments, arbitrary normalized integration orders, and the permitted rational weights may be added in their canonical bases. The removal of the small primes is essential to the proved growth bound: this section does not assert the analogous unrestricted theorem for $\sum c^{\Omega(n)}x^n/n!$.

20 Every sublinear integer coefficient sequence

There is a complementary principle which requires no multiplicativity, positivity, or progression means. It supplies an exact rationality test for every integer coefficient sequence of sublinear growth. Its proof uses the same dyadic denominator lemma, but only at the particularly favourable indices $n = 2^k$. The principle and its consequences are presented as elementary extensions of the current toolkit, without a claim of historical priority.

Theorem 20.1 (An exact sublinear-coefficient criterion)**.** *Let $a_n \in \mathbb{Z}$ for $n \ge 0$, and suppose $a_n = o(n)$ as $n \to \infty$. For every $x = \varepsilon 2/b$, $b \ge 1$ an integer and $\varepsilon \in \{-1, 1\}$,*

$$S_a(x) = \sum_{n\ge 0} \frac{a_n x^n}{n!} \in \mathbb{Q} \quad \Longleftrightarrow \quad a_n = 0 \text{ for all sufficiently large } n.$$

The same assertion holds for rational coefficients with a common fixed integer denominator. In particular every non-eventually-zero integer sequence satisfying $a_n = O(n^\beta)$ for some $\beta < 1$ gives an irrational value.

Proof. Finite support plainly gives a rational value. Conversely suppose $S_a(x) = u/v \in \mathbb{Q}$, $v \ge 1$, and set

$$T(n) = n!x^{-n} \sum_{m>n} \frac{a_m x^m}{m!}.$$

The growth assumption implies polynomial growth, so Lemma 10.1 applies. At $n = 2^k$ it gives $2vT(2^k) \in \mathbb{Z}$.

Write $y = |x|$. For every $\eta > 0$, the inequality $|a_m| \le \eta m$ holds beyond a fixed threshold. For all sufficiently large n,

$$\begin{aligned} |T(n)| &\le \eta \sum_{j\ge 1} (n+j) y^j \frac{n!}{(n+j)!} \\ &= \eta y \sum_{k\ge 0} y^k \frac{n!}{(n+k)!} \le \frac{\eta y}{1 - y/(n+1)}. \end{aligned}$$

Since η is arbitrary, $T(n) \to 0$. Thus $T(2^k) = 0$ for all sufficiently large k.

It remains to exclude cancellation of an infinite signed tail. Choose $\eta > 0$ so small that $2\eta y < 1$, and take a large index n with $T(n) = 0$. If there were a first $m > n$ with $a_m \neq 0$, then, after factoring out $x^m/m!$, its tail would equal

$$a_m + \sum_{j\ge 1} a_{m+j} x^j \frac{m!}{(m+j)!}.$$

The second term has absolute value at most $\eta y/(1 - y/(m+1)) < 1$ for large m, whereas the nonzero integer a_m has absolute value at least one. The tail could not vanish. Therefore no such m exists, proving eventual zero. Clearing the fixed denominator proves the rational-coefficient version. □

The growth threshold cannot simply be replaced by $O(n)$. For the same $x = \varepsilon 2/b$, the non-eventually-zero integer sequence $a_n = bn - 2\varepsilon$ satisfies

$$\sum_{n\ge 0} \frac{(bn - 2\varepsilon)x^n}{n!} = (bx - 2\varepsilon)e^x = 0.$$

This example identifies a genuine boundary of the universal statement; the arithmetic-profile theorems overcome it only through additional structure of their coefficients.

Corollary 20.2 (Sequence relations classify all value relations)**.** *Let $a_n^{(i)}$ be any collection of integer sequences, each $o(n)$, and put $S_i(x) = \sum_{n\ge 0} a_n^{(i)} x^n/n!$. For finitely supported rational coefficients c_i,*

$$\sum_i c_i S_i(x) \in \mathbb{Q} \quad \Longleftrightarrow \quad \sum_i c_i a_n^{(i)} = 0 \text{ for every sufficiently large } n.$$

Consequently $\{1\} \cup \{S_i(x)\}$ is linearly independent over $\mathbb{Q}$ exactly when the coefficient sequences are linearly independent over $\mathbb{Q}$ modulo eventual equality. When the displayed sum is rational, its value is the corresponding finite initial factorial sum.

Proof. Every finite rational combination has a common denominator and remains $o(n)$. Apply Theorem 20.1. □

The criterion applies to derivatives and normalized antiderivatives without multiplying the coefficients by powers of their indices. Extend a sequence a_n, $n \ge 0$, by $a_n = 0$ for $n < 0$, and set $A(z) = \sum_{n\ge 0} a_n z^n/n!$. For every integer j,

$$A^{[j]}(x) = \sum_{n\ge 0} \frac{a_{n+j} x^n}{n!},$$

where negative orders have the zero-origin normalization used earlier. Each shifted sequence is still $o(n)$. Thus a rational relation among any finite number of these two-sided values is equivalent to an eventual constant-coefficient relation among the corresponding shifts of a_n. This is an exact criterion; it does not assume that all shifts of every sequence are independent.

Theorem 20.3 (The full hierarchy of every bounded aperiodic sequence)**.** *Let $(a_n)_{n\ge 0}$ be a bounded integer sequence which is not eventually periodic, and put $A(z)=\sum_{n\ge 0} a_n z^n/n!$. For every fixed $q\ge 1$ and every permitted $x=\pm 2/b$, the family*

$$\{1\}\cup\{E_{q,r}(x):0\le r<q\}\cup\{A^{[j]}(x):j\in\mathbb{Z}\}$$

is linearly independent over $\mathbb{Q}$. In particular all derivatives and all normalized antiderivatives of A, together with $1, e^x$, are jointly independent.

Proof. By Corollary 20.2, a rational relation would give, for every sufficiently large n,

$$w(n)+\sum_{j\in J} c_j a_{n+j}=0,$$

where w is rational-valued and q-periodic, and J is finite. If every c_j is zero then w is eventually zero, hence identically zero, so all its residue coefficients vanish.

Otherwise translate the index and retain the largest shift to obtain an affine recurrence

$$a_{n+d}=-\frac{w'(n)+\sum_{i=0}^{d-1} c'_i a_{n+i}}{c'_d},\qquad c'_d\neq 0,$$

with a periodic forcing w'. If $d=0$, this says directly that a_n is eventually periodic. If $d\ge 1$, the state $(n \bmod q, a_n,\dots,a_{n+d-1})$ belongs to a finite set, because the coefficients a_n are bounded integers. The recurrence determines its next state uniquely. Two equal states therefore have identical subsequent trajectories; an infinite trajectory in this finite set is eventually periodic. Again a_n is eventually periodic, a contradiction. All coefficients in the proposed relation must vanish. □

This theorem supplies complete two-sided hierarchies for several familiar coefficient sequences:

1. The prime indicator $a_n=\mathbf{1}_{n \text{ prime}}$. There are infinitely many ones, while factorial blocks give arbitrarily long intervals of zeros at arbitrarily large indices. Such a sequence cannot be eventually periodic.

2. The Möbius function $a_n=\mu(n)$ and its square $a_n=\mu(n)^2$, with value zero assigned at $n=0$. For every desired block length choose distinct primes p_i and impose $N+i\equiv 0 \pmod{p_i^2}$ by CRT. This gives arbitrarily late zero blocks, while primes supply infinitely many nonzero terms.

3. The Liouville function $a_n=(-1)^{\Omega(n)}$ for $n\ge 1$, again assigning any integer at zero. If it had eventual period q, its values on all sufficiently large multiples qn would be constant. Complete multiplicativity would then make a_n eventually constant. The identity $a_{2n}=-a_n$ contradicts this.

4. For each fixed integer $k\ge 2$, the indicator of perfect kth powers; or the indicator of the set $\{2^m : m\ge 0\}$. Each has infinitely many ones and gaps of zeros tending to infinity.

For example the prime factorial function

$$\mathcal{P}(z)=\sum_{p \text{ prime}}\frac{z^p}{p!}$$

has $\{1,e^2,\mathcal{P}^{[j]}(2):j\in\mathbb{Z}\}$ linearly independent over $\mathbb{Q}$. The same assertion holds separately for each other function just listed. These separate families must not be united without checking eventual relations between their coefficient sequences. Nor does the theorem automatically adjoin such arbitrary sequences to the positive arithmetic-profile family: their combined coefficients need not be sublinear.

20.1 A simultaneous signed multiplicative family

The sequence criterion also allows several of the preceding arithmetic functions to be combined, provided their eventual relations are checked. For $c \geq 1$ define

$$B_c(n) = c^{\omega(n)}, \quad L_c(n) = \lambda(n)c^{\omega(n)}, \quad S_c(n) = \mu(n)^2 c^{\omega(n)}, \quad M_c(n) = \mu(n)c^{\omega(n)}.$$

Write $W_c(n) = c^{\Omega(n_{>c})}$ for $c \geq 2$, where $n_{>c}$ is the product of all prime-power factors of n with prime greater than c. The growth bound for this function is also proved in the rough-prime extension above. For any coefficient function f here, write $A_f(z) = \sum_{n\geq 1} f(n)z^n/n!$.

Theorem 20.4 (A joint signed arithmetic block)**.** *At every permitted $x = \pm 2/b$, the family*

$$\begin{gathered} 1,\ e^x,\ \mathcal{P}(x), \\ A_{d_r}(x)\ (r \geq 2), \qquad A_{B_c}(x)\ (c \geq 2), \qquad A_{W_c}(x)\ (c \geq 2), \\ A_{L_c}(x),\ A_{S_c}(x),\ A_{M_c}(x) \qquad (c \geq 1) \end{gathered}$$

is linearly independent over $\mathbb{Q}$. Here $\lambda(n) = (-1)^{\Omega(n)}$ and $\mathcal{P}$ is the prime factorial function defined above.

Proof. All the coefficient sequences are $o(n)$. For d_r and c^ω this follows from the polynomial-valuation growth bound; multiplying by μ, μ^2, or λ does not increase absolute values. If p_c is the least prime greater than c, then $W_c(n) \leq n^{\log c/\log p_c} = o(n)$. By Corollary 20.2 it suffices to exclude a finite nontrivial eventual relation between these coefficient sequences, including the constant sequence $B_1 = 1$ for e^x.

Choose a prime p larger than every c occurring in the proposed relation, and evaluate at $n = p^a$ as $a \to \infty$. The W_c terms are c^a, the d_r terms are the polynomials $\binom{a+r-1}{r-1}$, the B_c terms are constants c, and the L_c terms are $c(-1)^a$. The prime, S_c, and M_c terms vanish for $a \geq 2$. Successively dividing by the largest exponential c^a shows that all the W_c coefficients vanish. The remaining polynomial plus an alternating constant can vanish for all large a only if its polynomial is constant and its alternating coefficient is zero. The distinct positive degrees of the d_r polynomials then force all their coefficients to vanish.

Now choose k distinct primes. Give them all exponent two, and then change one exponent to three. The numbers can be chosen arbitrarily large. In both cases $\omega(n) = k$, while $\lambda(n)$ has opposite signs and all S_c, M_c, and prime-indicator terms are zero. Adding and subtracting the two eventual relations gives

$$\sum_c b_c c^k = 0, \qquad \sum_c \ell_c c^k = 0 \qquad (k \geq 1).$$

Finite Vandermonde matrices on the distinct positive bases c are invertible, so all B_c and L_c coefficients vanish, including b_1.

For a product of $k \geq 2$ distinct primes, the remaining relation is

$$\sum_c s_c c^k + (-1)^k \sum_c m_c c^k = 0.$$

Even k and a Vandermonde matrix on the bases c^2 give $s_c + m_c = 0$; odd $k \geq 3$ give $s_c - m_c = 0$. Thus all S_c and M_c coefficients vanish. Evaluation at arbitrarily large primes finally kills the prime indicator coefficient. This proves the asserted eventual independence and hence the value independence. □

In particular the constants

$$\sum_{n\geq 1} \frac{2^n \mu(n)}{n!}, \quad \sum_{n\geq 1} \frac{2^n \lambda(n)}{n!}, \quad \sum_{n\geq 1} \frac{2^{n+\omega(n)}\mu(n)^2}{n!}, \quad \sum_{p \text{ prime}} \frac{2^p}{p!}$$

belong to one explicit joint independent family together with all the d_r, B_c, and W_c values in the theorem. This is a base-value statement. Combining the entire derivative hierarchies of different bounded sequences would require additional eventual-shift independence, which has not been assumed or asserted here.

The theorem proves arithmetic properties of the displayed factorial values. It does not prove digit aperiodicity of an unrelated named constant, and it does not imply transcendence of these values.

21 All cyclotomic orders and two quadratic coefficient fields

Fix a positive permitted radius $x = 2/b$, with an integer $b \geq 1$. The residue-class theorem has an infinite angular consequence: one can simultaneously use every cyclotomic order, provided that the angular values are expressed in their canonical rational coordinates. At orders four and six this also gives linear independence of the actual complex values over a quadratic number field. These statements do not require a new denominator estimate.

21.1 Canonical coordinates at every root of unity

Let $\Phi_d(T)$ be the dth cyclotomic polynomial, let $\zeta_d = \exp(2\pi i/d)$, and put $\varphi_d = \deg \Phi_d$. For an entire function H with real Taylor coefficients define the polynomial of degree less than φ_d

$$\mathcal{C}_d[H](T) = \sum_{\substack{1\leq k\leq d\\(k,d)=1}} H(\zeta_d^k x)\frac{\Phi_d(T)}{(T-\zeta_d^k)\Phi_d'(\zeta_d^k)} = \sum_{\ell=0}^{\varphi_d-1} C_{d,\ell}[H](x)T^\ell. \tag{73}$$

For $d = 1$, this means $C_{1,0}[H](x) = H(x)$. Conjugating the interpolation formula pairs k with $-k$, so every $C_{d,\ell}[H](x)$ is real. These are explicitly defined finite linear combinations of actual angular values, with algebraic coefficients.

Theorem 21.1 (All cyclotomic coordinates simultaneously)**.** *For the canonical base family $\mathcal{B}$ of Theorem 14.1, the real numbers*

$$\{1\} \ \cup\ \{C_{d,\ell}[e^z](x) : d \geq 1,\ 0 \leq \ell < \varphi_d\} \ \cup\ \{C_{d,\ell}[A_f^{[j]}](x) : f \in \mathcal{B},\ j \in \mathbb{Z},\ d \geq 1,\ 0 \leq \ell < \varphi_d\}$$

are linearly independent over $\mathbb{Q}$. In particular, every displayed coordinate is irrational, and every nonzero finite rational combination of the coordinates is irrational.

Proof. Consider a finite proposed relation and take a positive integer q divisible by every cyclotomic order that occurs. For each f, j put $U_a = A_{f,q,a}^{[j]}(x)$, $0 \leq a < q$, and define

$$P_{f,j}(T) = \sum_{a=0}^{q-1} U_a T^{\langle a-j\rangle_q},$$

where $\langle\cdot\rangle_q$ denotes the representative in $\{0,\dots,q-1\}$. The derivative and antiderivative series give

$$P_{f,j}(\zeta_d^k) = A_f^{[j]}(\zeta_d^k x) \qquad (d \mid q).$$

Indeed a derivative changes the exponent of z from n to $n-j$; the same exponent formula applies to a normalized antiderivative. Thus the remainder of $P_{f,j}$ modulo Φ_d is exactly the interpolant in (73).

The map

$$\mathbb{Q}[T]/(T^q-1) \longrightarrow \bigoplus_{d\mid q} \mathbb{Q}[T]/(\Phi_d), \qquad P \longmapsto (P \bmod \Phi_d)_{d\mid q},$$

is an isomorphism over $\mathbb{Q}$. To see this without a dimension assumption, the cyclotomic factors are pairwise coprime, so a polynomial in the kernel is divisible by their product T^q-1; the dimensions on both sides are q, since $\sum_{d\mid q}\varphi_d = q$. Taking remainders therefore gives an invertible rational matrix on the q coefficient coordinates. The initial permutation $a \mapsto \langle a-j\rangle_q$ is invertible as well.

For the exponential use $P_0(T) = \sum_a E_{q,a}(x)T^a$ and the same remainder map. Hence, block by block, all coordinates of orders dividing q are an invertible rational transform of the family in Corollary 14.2. That family, together with 1, is linearly independent. Its transformed family is independent, and so is the subfamily in the proposed relation. The order q can be chosen for every finite relation; this proves the assertion for all d simultaneously. □

The theorem is not a claim that arbitrary algebraic-coefficient combinations of the original real constants are independent. The interpolants are special: reduction modulo Φ_d identifies their coefficients as a rational, invertible transform of residue-class coordinates. Nor does a decomposition $H(\zeta_d x) = \sum_\ell C_{d,\ell}[H](x)\zeta_d^\ell$, by itself, prove transcendence of $H(\zeta_d x)$.

21.2 Six directions and twelve explicit real transforms

Write $\omega = \exp(i\pi/3)$. For any H as above, the six real transforms

$$H(x),\quad H(-x),\quad \Re H(\omega x),\quad \frac{\Im H(\omega x)}{\sqrt{3}},\quad \Re H(\omega^2 x),\quad \frac{\Im H(\omega^2 x)}{\sqrt{3}} \tag{74}$$

are an invertible rational transform of the cyclotomic coordinates of orders $1,2,3,6$. For example, at order six the two coordinates satisfy

$$H(\omega x) = C_{6,0} + \omega C_{6,1},\qquad \Re H(\omega x) = C_{6,0} + \tfrac{1}{2}C_{6,1},\qquad \frac{\Im H(\omega x)}{\sqrt{3}} = \tfrac{1}{2}C_{6,1}.$$

At order three the sign of $C_{3,1}/2$ in the real part is negative. Consequently all six transforms of e^z, and all six transforms of every $A_f^{[j]}$, are jointly independent together with 1. For the exponential these are

$$e^x,\ e^{-x},\quad e^{x/2}\cos\frac{\sqrt{3}x}{2},\quad \frac{e^{x/2}}{\sqrt{3}}\sin\frac{\sqrt{3}x}{2},\quad e^{-x/2}\cos\frac{\sqrt{3}x}{2},\quad \frac{e^{-x/2}}{\sqrt{3}}\sin\frac{\sqrt{3}x}{2}.$$

One can retain these six transforms, add the order-four pair $\Re H(ix), \Im H(ix)$, and add four further transforms. Put $\eta = \exp(i\pi/6)$ and define

$$D_0[H] = \frac{\Re H(\eta x) + \Re H(\eta^5 x)}{2},\qquad D_1[H] = \frac{\Re H(\eta x) - \Re H(\eta^5 x)}{\sqrt{3}},$$
$$D_2[H] = \frac{\Im H(\eta x) - \Im H(\eta^5 x)}{\sqrt{3}},\qquad D_3[H] = \frac{\Im H(\eta x) + \Im H(\eta^5 x)}{2}.$$

Since $\Phi_{12}(T) = T^4 - T^2 + 1$, writing its interpolant as $C_0 + C_1T + C_2T^2 + C_3T^3$ gives

$$D_0 = C_0 + \tfrac{1}{2}C_2,\qquad D_1 = C_1,\qquad D_2 = C_2,\qquad D_3 = \tfrac{1}{2}C_1 + C_3.$$

This rational transformation is invertible. Therefore the twelve transforms just listed, for every arithmetic function and every integer derivative order, are jointly independent with the twelve exponential transforms and 1. The last four exponential transforms have the familiar forms

$$\cosh\frac{\sqrt{3}x}{2}\cos\frac{x}{2},\qquad \frac{2}{\sqrt{3}}\sinh\frac{\sqrt{3}x}{2}\cos\frac{x}{2},$$

$$\frac{2}{\sqrt{3}}\sinh\frac{\sqrt{3}x}{2}\sin\frac{x}{2}, \qquad \cosh\frac{\sqrt{3}x}{2}\sin\frac{x}{2}.$$

The factors $1/\sqrt{3}$ are part of the stated coordinates and must not be silently removed when asserting joint independence. The general theorem supplies higher cyclotomic orders by the same explicit interpolation procedure, without a fixed upper bound on d.

21.3 Gaussian and Eisenstein linear independence

Lemma 21.2 (Imaginary quadratic scalar extension)**.** *If a family of real numbers is linearly independent over $\mathbb{Q}$, it is linearly independent over every imaginary quadratic field $K = \mathbb{Q}(\sqrt{-D})$, $D \in \mathbb{N}$.*

Proof. Write each coefficient as $a + i\sqrt{D}\,b$ with $a, b \in \mathbb{Q}$. The real and imaginary parts of a finite relation are two rational relations among the same real numbers. Both are trivial. □

Corollary 21.3 (Actual angular values over quadratic fields)**.** *For $q = 4$, put $K = \mathbb{Q}(i)$ and $\zeta = i$; for $q = 6$, put $K = \mathbb{Q}(\sqrt{-3})$ and $\zeta = \omega$. In either case the complex numbers*

$$\{1\} \ \cup\ \{e^{\zeta^k x} : 0 \le k < q\} \ \cup\ \{A_f^{[j]}(\zeta^k x) : f \in \mathcal{B},\ j \in \mathbb{Z}, 0 \le k < q\}$$

are linearly independent over K. In particular, each displayed arithmetic value lies outside K, and every nonzero finite K-linear combination of the exponential and arithmetic values lies outside K.

Proof. The residue-class family in Corollary 14.2 is real. Lemma 21.2 extends its independence to K. For each fixed f, j, the evaluation map is the discrete Fourier map

$$A_f^{[j]}(\zeta^k x) = \zeta^{-kj}\sum_{a=0}^{q-1}\zeta^{ka}A_{f,q,a}^{[j]}(x).$$

The Fourier matrix is invertible over K, with inverse obtained by conjugating its entries and dividing by q. Its diagonal phase factors ζ^{-kj} are nonzero elements of K. The exponential block has the same Fourier matrix without those phase factors. Apply these invertible transformations to each finite collection of blocks. Independence together with 1 also proves the final assertion. □

This quadratic-field statement is stronger than individual irrationality, but it is still a linear-independence result, not an algebraic-independence or transcendence theorem. For a general cyclotomic field its real subfield is nontrivial, so the scalar-extension lemma does not apply. The canonical real-coordinate theorem remains valid at every order.

Finally, all arguments in this section concern a common absolute radius x. For a relation combining unrelated rational radii, rationality of the sum does not imply rationality of each summand; one cannot apply Lemma 10.1 separately at each radius. No independence across unrelated radii is claimed here.

22 Logarithmic kernels and positive integrals in every dimension

The rational-weight theorem has an exact integral calculus. It includes logarithmic kernels after the indicated differential factors are supplied, as well as positive integrals over cubes of arbitrarily large dimension. The results in this section are consequences of the earlier theorem;

different integral representations are not counted as independent constants unless their rational multipliers are independent.

Throughout, $\mathcal{B}$ is the canonical arithmetic family in Theorem 14.1, $x = \pm 2/b$ with $b \geq 1$, and $A_f(z) = \sum_{n\geq 1} f(n) z^n/n!$. Write $\theta = z\, d/dz$. For integers $a, r \geq 1$ and $P \in \mathbb{Q}[X]$, define

$$\mathscr{L}_{a,r,P}[A_f](x) = \frac{1}{(r-1)!} \int_0^1 t^{a-1} (-\log t)^{r-1} (P(\theta) A_f)(xt)\, dt.$$

Proposition 22.1 (An exact test for logarithmic integral expressions)**.** *Consider finitely many polynomials $P_{f,a,r} \in \mathbb{Q}[X]$ and put*

$$R_f(X) = \sum_{a,r} \frac{P_{f,a,r}(X)}{(X+a)^r}.$$

Suppose each reduced rational function R_f has at most simple poles. If some R_f is nonzero, then for all $u, v \in \mathbb{Q}$ the number

$$u + ve^x + \sum_{f,a,r} \mathscr{L}_{a,r,P_{f,a,r}}[A_f](x)$$

is irrational. The same conclusion holds with separate rational multipliers on each residue class, adjoining the corresponding periodic exponential functions.

The simple-pole condition can be checked by finite rational arithmetic: for every f, a and every $j \geq 2$, it is precisely

$$\sum_{r\geq j} \frac{P_{f,a,r}^{(r-j)}(-a)}{(r-j)!} = 0. \tag{75}$$

Only the finitely many indices present in the expression need be tested.

Proof. For $n \geq 1$, substitution $t = e^{-u}$ and repeated integration by parts give

$$\frac{1}{(r-1)!} \int_0^1 t^{n+a-1} (-\log t)^{r-1}\, dt = \frac{1}{(n+a)^r}.$$

Moreover $P(\theta) A_f(z) = \sum_{n\geq 1} P(n) f(n) z^n/n!$. Termwise integration is absolutely justified: the arithmetic coefficients have polynomial growth, the factorial series converges uniformly on the compact set of arguments, and its terms vanish at least linearly at zero. Consequently

$$\mathscr{L}_{a,r,P}[A_f](x) = \sum_{n\geq 1} \frac{P(n) f(n) x^n}{n!(n+a)^r}.$$

The entire expression therefore has multiplier $R_f(n)$ in channel f. Apply Corollary 14.4. Taylor expansion of each $P_{f,a,r}$ at $-a$ gives exactly (75) as the coefficient of $(X+a)^{-j}$, proving the finite test. Residue restrictions commute with all these operations and use the residue version of the same corollary. □

In particular, for every $a, r \geq 1$,

$$\frac{1}{(r-1)!} \int_0^1 t^{a-1} (-\log t)^{r-1} ((\theta + a)^{r-1} A_f)(xt)\, dt = M_{f,a}(x) \notin \mathbb{Q}. \tag{76}$$

This includes arbitrary logarithmic powers, with an explicit differential operator. For fixed f, a, changing r in (76) gives the same number. Without that operator, the repeated-pole multiplier $(n+a)^{-r}$ is outside the asserted theorem when $r > 1$. Failure of (75) therefore means that this particular criterion does not apply; it does not imply rationality.

Theorem 22.2 (Product kernels and a jointly independent tower)**.** *Let $a_1, a_2, \dots$ be any sequence of pairwise distinct positive integers. For $m \geq 1$ define*

$$I_{f,m}(x) = \int_{[0,1]^m} \left(\prod_{j=1}^{m} t_j^{a_j - 1} \right) A_f \left(x \prod_{j=1}^{m} t_j \right) dt_1 \cdots dt_m.$$

All numbers $I_{f,m}(x)$, for $f \in \mathcal{B}$ and $m \geq 1$, are jointly linearly independent over $\mathbb{Q}$ together with $1, e^x$ and all polynomial moments $\sum_{n \geq 1} n^d f(n) x^n / n!$. They may also be restricted jointly to every residue class of a fixed modulus. Each is irrational.

Proof. Fubini's theorem and absolute convergence give

$$I_{f,m}(x) = \sum_{n \geq 1} \frac{f(n) x^n}{n! \prod_{j=1}^{m} (n + a_j)}.$$

The multiplier $R_m(X) = \prod_{j=1}^{m} (X + a_j)^{-1}$ has only simple negative-integer poles. The family consisting of all R_m and all polynomials is linearly independent over $\mathbb{Q}$: in a finite relation, the pole at $-a_M$ occurs only in the term with largest index M; its residue is nonzero, forcing that coefficient to vanish. Continue downward, and then use polynomial independence. Corollary 14.4 now rules out every proposed relation of the displayed numbers, separately in each canonical arithmetic channel and residue class. □

For example, taking $a_j = j$ gives the particularly simple identity

$$I_{f,m}(x) = \sum_{n \geq 1} \frac{f(n) x^n}{(n + m)!}.$$

More generally,

$$I_{f,m}(x) = \sum_{j=1}^{m} \frac{M_{f,a_j}(x)}{\prod_{\ell \neq j} (a_\ell - a_j)}.$$

These are exact alternative representations, not additional independent coordinates on top of all the $M_{f,a}$. Multiplication by $\prod_j a_j$ gives the expectation of $A_f(x \prod_j U_j)$ when the U_j are independent and have densities $a_j t^{a_j - 1}$ on $(0, 1)$. Thus the corresponding expectation constants inherit the same independence statement.

Arbitrary logarithmic powers can also be inserted simultaneously in this cube integral. For positive integers $r_1, \dots, r_m$, replace the weight by

$$\prod_{j=1}^{m} \frac{t_j^{a_j - 1} (-\log t_j)^{r_j - 1}}{(r_j - 1)!}$$

and replace A_f by $\prod_{j=1}^{m} (\theta + a_j)^{r_j - 1} A_f$. The resulting integral is exactly $I_{f,m}(x)$, as is seen term by term. This supplies a positive-dimensional integral identity even when the logarithmic kernel is of arbitrarily high order.

Theorem 22.3 (Positive bi-Vandermonde cube integrals)**.** *Fix a positive integer a. For $m \geq 1$, set*

$$V_m(t) = \prod_{1 \leq i < j \leq m} (t_j - t_i),$$
$$W_m(t) = \prod_{1 \leq i < j \leq m} (t_j^m - t_i^m),$$

$$K_{f,a,m}(x)=\int_{[0,1]^m}\left(\prod_{j=1}^{m}t_j^{a-1}\right)V_m(t)W_m(t)A_f\left(x\prod_{j=1}^{m}t_j\right)\,dt.$$

Write $(y)_d=y(y+1)\cdots(y+d-1)$ *and*

$$C_m=m!\,m^{m(m-1)/2}\prod_{j=1}^{m-1}(j!)^2.$$

Then

$$K_{f,a,m}(x)=C_m\sum_{n\ge1}\frac{f(n)x^n}{n!(n+a)_{m^2}}. \tag{77}$$

For this fixed a*, all* $K_{f,a,m}(x)$*, as* $m\ge1$ *and* $f\in\mathcal{B}$ *vary, are jointly linearly independent over* $\mathbb{Q}$ *together with* $1,e^x$ *and all polynomial moments. In particular they are irrational. The kernel multiplying* A_f *is nonnegative on the cube and positive away from equal coordinates and the boundary.*

Proof. Expansion of the two Vandermonde determinants and Fubini's theorem give, for each integer $n\ge0$,

$$\int_{[0,1]^m}\left(\prod_j t_j^{n+a-1}\right)V_m(t)W_m(t)\,dt=m!\det\left[\frac{1}{n+a+mi+j}\right]_{0\le i,j<m}.$$

Indeed, expand both determinants over permutations; integrating the product of monomials gives this determinant identity after fixing one permutation and reindexing the other. The Cauchy determinant formula then yields

$$m!\det\left[\frac{1}{n+a+mi+j}\right]=\frac{m!\prod_{0\le i<j<m}m(j-i)^2}{\prod_{i,j=0}^{m-1}(n+a+mi+j)}=\frac{C_m}{(n+a)_{m^2}}.$$

Here the integers $mi+j$ run through $0,\dots,m^2-1$ exactly once. The Cauchy identity itself follows by multiplying the determinant by its common denominator: the numerator is alternating separately in the row and column parameters, its degree gives the two Vandermonde factors, and a residue at one pole gives constant one by induction. Termwise integration now proves (77).

For a finite linear relation, the multiplier with largest m has a simple pole at $-a-m^2+1$ that none of the smaller multipliers has. Its coefficient therefore vanishes, and descending induction proves independence of these rational multipliers together with polynomials. Apply Corollary 14.4. Finally, $(t_j-t_i)(t_j^m-t_i^m)\ge0$ on $[0,1]^2$, with strict inequality for distinct positive coordinates, proving the kernel assertion. □

At $a=1$, formula (77) simplifies to

$$K_{f,1,m}(x)=C_m\sum_{n\ge1}\frac{f(n)x^n}{(n+m^2)!}.$$

For example, with $m=2$ one obtains the explicit double integral

$$\int_0^1\int_0^1(v-u)^2(u+v)A_f(xuv)\,du\,dv=4\sum_{n\ge1}\frac{f(n)x^n}{(n+4)!}\notin\mathbb{Q}.$$

The weight has total mass $C_m/(a)_{m^2}$, a positive rational number. Dividing by this mass therefore gives another probabilistic interpretation without changing irrationality. As throughout this section, the integral representations inherit the proved hierarchy; they are not claims to new historical resolutions of named problems.

23 Polynomial integral operators and exact projective classification

This section extracts further consequences of the established independent family. The claims are closure and classification results, rather than new irrationality criteria for unrelated constants. Throughout, $q \geq 1$ is fixed, $\mathcal{B}$ is a canonical base family satisfying Theorem 14.1, and $x = \pm 2/b \neq 0$ is a permitted rational point. Write $A^{[j]}_{f,q,a}$ for the two-sided jets in Corollary 14.2.

23.1 A normal form for arbitrarily nested polynomial kernels

Let D denote differentiation, let

$$(JG)(z) = \int_0^z G(t)\,dt,$$

and allow multiplication by any polynomial in $\mathbb{Q}[z]$. Start with the functions 1, $E_{q,a}$, and $A_{f,q,a}$, and apply finitely many of these operations, rational linear combinations, and compositions. In particular, this permits repeated polynomial-kernel integrals

$$G(z) \longmapsto \int_0^z K(z,t)G(t)\,dt, \qquad K \in \mathbb{Q}[z,t],$$

interspersed with differentiation.

Theorem 23.1 (Polynomial-operator normal form)**.** *Every resulting function has a unique finite representation*

$$P_{\mathrm{c}}(z) + \sum_{a=0}^{q-1} P_a(z)E_{q,a}(z) + \sum_{f,a,j} P_{f,a,j}(z)A^{[j]}_{f,q,a}(z), \qquad P_{\mathrm{c}}, P_a, P_{f,a,j} \in \mathbb{Q}[z]. \tag{78}$$

At a permitted point x, its value is rational if and only if

$$P_a(x) = 0 \quad \textit{for every } a, \qquad P_{f,a,j}(x) = 0 \quad \textit{for every } f,a,j.$$

In that case the value is exactly $P_{\mathrm{c}}(x)$.

Proof. The displayed module is closed under polynomial multiplication and D. Indeed, $DE_{q,a} = E_{q,a-1}$, with residues read modulo q, and $DA^{[j]}_{f,q,a} = A^{[j+1]}_{f,q,a}$ for every integer j. For integration, repeated integration by parts gives, for a polynomial P,

$$\int_0^z P(t)A^{[j]}_{f,q,a}(t)\,dt = \sum_{\ell=0}^{\deg P} (-1)^\ell \left[P^{(\ell)}(t)A^{[j-\ell-1]}_{f,q,a}(t)\right]_{t=0}^{t=z}. \tag{79}$$

All lower-endpoint terms are rational: the value of a nonpositive jet at zero is zero, whereas for $k \geq 1$,

$$A^{[k]}_{f,q,a}(0) = f(k)\mathbf{1}_{k\equiv a\ (q)} \in \mathbb{Z}.$$

Also $JE_{q,a} = E_{q,a+1} - \mathbf{1}_{a+1\equiv 0\ (q)}$; the same integration-by-parts argument treats polynomial multiples of these functions. Thus the module is closed under J as well. Expanding a polynomial kernel in its two variables reduces it to these operations. This proves existence and gives a finite exact algorithm.

For uniqueness, any nonzero rational-function relation among the displayed generators could be evaluated at one of the infinitely many points $2/b$, avoiding every pole and the zeros of a nonzero coefficient. It would contradict Corollary 14.2. In fact the generators are linearly independent over $\mathbb{Q}(z)$, which is stronger than the required uniqueness over $\mathbb{Q}[z]$. At a fixed permitted x, their rational linear independence gives exactly the stated rationality test. □

For example, for $q = 1$ and $A = A_f$, an actual nested expression reduces to

$$\int_0^x t^2 \left(\int_0^t (s+1)A^{(3)}(s)\, ds \right) dt$$
$$= (x^3 + x^2)A'(x) - (4x^2 + 2x)A(x) + (8x + 2)A^{[-1]}(x) - 8A^{[-2]}(x) - \frac{x^3}{3}(f(2) - f(1)).$$

The coefficient of $A^{[-2]}(x)$ is -8, so this number is irrational at every permitted nonzero x. At $x = 2$ and $f = \sigma_5$, the formula is

$$12A_5'(2) - 20A_5(2) + 18A_5^{[-1]}(2) - 8A_5^{[-2]}(2) - \frac{256}{3}.$$

More generally, normal-form coefficient vectors decide all rational linear relations among any finite collection of nested expressions. Identities such as $DJG = G$ are automatically respected; the theorem does not incorrectly regard different operator descriptions as independent.

23.2 All polynomial modes, beta averages, and selected multiple integrals

For $P \in \mathbb{Q}[t]$ define

$$\mathcal{L}_{f,a,P}(x) = \int_0^1 P(t) A_{f,q,a}(xt)\, dt.$$

If $P(t) = \sum_{k=0}^{d} p_k t^k$, then

$$\mathcal{L}_{f,a,P}(x) = \sum_{\substack{n \geq 1 \\ n \equiv a \ (q)}} \frac{f(n)x^n}{n!} \sum_{k=0}^{d} \frac{p_k}{n+k+1}. \tag{80}$$

The map $P \mapsto \sum_k p_k/(X + k + 1)$ is injective, as is seen from the residues at its distinct poles. Corollary 14.4 therefore gives the following exact criterion: within each fixed (f, a) block, polynomial kernels produce independent values precisely when the kernels themselves are linearly independent over $\mathbb{Q}$. Independent kernel families in all blocks may be adjoined jointly to 1, the exponential residue values, and every polynomial arithmetic moment.

One familiar choice is the complete family of shifted Legendre polynomials

$$L_m(t) = \frac{1}{m!} \frac{d^m}{dt^m} \big[t^m (t-1)^m \big], \qquad m \geq 0.$$

Their degrees are m with nonzero rational leading coefficient, so they form a basis of $\mathbb{Q}[t]$. Thus all the Legendre coefficients

$$\Lambda_{f,a,m}(x) = \int_0^1 L_m(t) A_{f,q,a}(xt)\, dt \qquad (f \in \mathcal{B},\ 0 \leq a < q,\ m \geq 0)$$

are jointly independent together with those moments and exponential values. At a positive permitted point they are also strictly positive: integrating by parts m times gives

$$\Lambda_{f,a,m}(x) = \frac{x^m}{m!} \int_0^1 t^m (1-t)^m A_{f,q,a}^{(m)}(xt)\, dt > 0.$$

There are infinitely many positive coefficients in every residue class, so the last derivative is positive for $0 < t < 1$. The same kernel-rank criterion applies to any other rational polynomial basis, including orthogonal bases for polynomial weights, without a separate arithmetic argument.

For integers $u, v \geq 1$, the normalized beta average is

$$\begin{aligned}\mathcal{B}_{u,v}A_{f,q,a}(x) &= \frac{(u+v-1)!}{(u-1)!(v-1)!}\int_0^1 t^{u-1}(1-t)^{v-1}A_{f,q,a}(xt)\,dt \\ &= \frac{(u+v-1)!}{(u-1)!}\sum_{\substack{n\geq 1\\ n\equiv a\ (q)}}\frac{f(n)x^n}{n!(n+u)(n+u+1)\cdots(n+u+v-1)}.\end{aligned}$$

For each fixed u, all $v \geq 1$ give an independent family, jointly with the polynomial moments and exponential coordinates: the kernels $t^{u-1}(1-t)^{v-1}$ have distinct degrees. These are also the expectations of $A_{f,q,a}(xT_0)$ under integer-parameter Dirichlet distributions on any simplex, with u the parameter of T_0 and v the sum of the other parameters. This interpretation does not create additional independent values when the two parameters u, v agree.

There is a useful multiple-integral version with a different geometry. For distinct positive integers $u_1, \ldots, u_s$, termwise integration gives

$$\int_{[0,1]^s}\left(\prod_{j=1}^s t_j^{u_j-1}\right)A_{f,q,a}(xt_1\cdots t_s)\,dt_1\cdots dt_s = \sum_{\substack{n\geq 1\\ n\equiv a\ (q)}}\frac{f(n)x^n}{n!\prod_{j=1}^s(n+u_j)}.$$

Its rational weight is nonzero with simple negative-integer poles, so the value is irrational. Partial fractions decide independence of several such integrals; different subsets of poles need not give independent weights. Distinctness of the u_j is essential for this particular deduction from Corollary 14.4.

23.3 Every quotient of rational affine forms

Let $(\beta_i)_{i\in I}$ denote any one of the independent coordinate families proved above, with $\beta_0 = 1$. Coefficient vectors below always have finite support in this fixed family, and $L_v = \sum_i v_i\beta_i$ for $v \in \mathbb{Q}^{(I)}$.

Theorem 23.2 (Exact projective and rank tests)**.** *Every nonzero vector v has $L_v \neq 0$. For $v \neq 0$,*

$$\frac{L_u}{L_v} \in \mathbb{Q} \quad\Longleftrightarrow\quad u \in \mathbb{Q}v.$$

More generally, for $v_0 \neq 0$, the numbers

$$1, \quad \frac{L_{v_1}}{L_{v_0}}, \ldots, \frac{L_{v_m}}{L_{v_0}}$$

are linearly independent over $\mathbb{Q}$ if and only if $v_0, v_1, \ldots, v_m$ are linearly independent over $\mathbb{Q}$.

Proof. The map $v \mapsto L_v$ is injective by the underlying independence. The first equivalence follows by applying this injection to $u - rv$, where $r \in \mathbb{Q}$. For the last assertion, multiply a proposed rational relation by the nonzero common denominator L_{v_0} and use the same injection. Both implications follow. □

Consequently, whenever L_u, L_v have independent coefficient vectors, every rational Möbius transform

$$\frac{aL_u + bL_v}{cL_u + dL_v}, \qquad a, b, c, d \in \mathbb{Q}, \quad ad - bc \neq 0,$$

is well defined and irrational. This includes arbitrary rational affine shifts in the original coordinates, not merely quotients of positive sums. It does not assert a result for general nonlinear rational functions.

For $q = 1$, division of the two-sided hierarchy by e^x proves joint independence of

$$\{1, e^{-x}\} \ \cup \ \{e^{-x} A_f^{[j]}(x) : f \in \mathcal{B}, \ j \in \mathbb{Z}\}.$$

At a positive permitted x, fix one f_* and divide instead by the positive number $A_{f_*}(x)$. Then

$$1, \quad \frac{1}{A_{f_*}(x)}, \quad \frac{e^x}{A_{f_*}(x)}, \quad \frac{A_f^{[j]}(x)}{A_{f_*}(x)} \quad ((f, j) \neq (f_*, 0))$$

are jointly independent. In particular $A_f'(x)/A_f(x)$, every ratio of two distinct jets of the same base, and every ratio of two distinct Legendre modes are irrational. Higher derivatives of $\log A_f$ involve products of coordinates and are not included by this linear argument.

23.4 Truncated and arithmetic-biased Poisson laws

Let N be Poisson with positive permitted mean x, fix $M \geq 1$, and let S be a nonempty set of residues modulo q. Condition on

$$C = \{N \geq M, \ N \bmod q \in S\}.$$

Its unnormalized mass is

$$D_C = \sum_{a \in S} E_{q,a}(x) - \sum_{\substack{0 \leq n < M \\ n \bmod q \in S}} \frac{x^n}{n!} > 0.$$

The second sum is rational. Removing finitely many terms from any arithmetic coordinate likewise subtracts a rational number. Theorem 23.2, applied to this common denominator, therefore proves joint independence of 1 and all the numbers

$$\mathbb{E}\Big[N^d f(N) \mathbf{1}_{N \equiv a \ (q)} \mid C\Big], \qquad f \in \mathcal{B}, \ a \in S, \ d \geq 0,$$
$$\mathbb{E}\Big[\frac{f(N)}{N+u} \mathbf{1}_{N \equiv a \ (q)} \mid C\Big], \qquad f \in \mathcal{B}, \ a \in S, \ u \geq 1.$$

The moment and simple-pole coordinates are independent by Corollary 14.4; each retains its own coefficient after the finite truncation, whereas D_C has a nonzero exponential coordinate. This verifies the required rank directly. Hence arbitrary finite cutoffs and residue conditioning preserve these joint irrationality statements.

There is a further family using an arithmetic weight in the probability law itself. Let f have a nontrivial admissible profile and let $m \geq 1$. On the same infinite support C, define

$$\Pr_m(N = n) = \frac{f(n)^m x^n / n!}{\displaystyle\sum_{\substack{k \geq M \\ k \bmod q \in S}} f(k)^m x^k / k!}.$$

All its moments exist. Then

$$\{1\} \ \cup \ \{\mathbb{E}_m[f(N)^j] : j \in \mathbb{Z}, \ j \geq -m, \ j \neq 0\} \quad \text{is linearly independent over } \mathbb{Q}. \tag{81}$$

Indeed, its numerator at exponent j is the truncated factorial sum for f^{m+j}. The case $m+j = 0$ is an exponential residue sum minus a rational finite sum. All positive powers have admissible profiles $R, R^2, R^3, \ldots$, which are pairwise distinct: a nontrivial positive rational profile cannot satisfy $R^a = R^b$ for $a \neq b$. The numerator vectors, together with the denominator vector for f^m, are consequently independent. The projective rank test proves (81).

For example, for the law on positive integers proportional to $\sigma_5(n)^m 2^n/n!$, all positive integer moments of $\sigma_5(N)$ and the reciprocal moments of orders $1,\dots,m$ are irrational and jointly independent together with 1. The same conclusion applies to the totient, the radical, and every other nontrivial admissible base. This is a statement about linear relations among expectations. Variances, cumulants, entropy, and products of the expectations require additional arguments and are not inferred here.

24 A classical companion: all inverse exponential moments

There is a separate, classical transcendence theorem for the pure exponential channel. It reaches every nonzero algebraic argument and arbitrary repeated poles, including a pole at zero. Its essential input is Beukers's refinement of the Siegel–Shidlovskii theorem [13, Corollary 1.4]; it is not a new result of the fiber-cancellation method. No joint independence between the constants of this section and the nontrivial arithmetic channels is asserted.

For $s \ge 1$ define the entire inverse-moment functions

$$\mathcal{E}_s(z) = \sum_{n\ge 1} \frac{z^n}{n! n^s}, \qquad \mathcal{E}_0(z) = e^z - 1.$$

They satisfy $\theta\mathcal{E}_s = \mathcal{E}_{s-1}$.

Theorem 24.1 (Classical inverse-moment independence)**.** *At every nonzero algebraic α, the infinite family*

$$1, \quad e^\alpha, \quad \mathcal{E}_1(\alpha), \mathcal{E}_2(\alpha), \dots$$

is linearly independent over $\overline{\mathbb{Q}}$. In particular every $\mathcal{E}_s(\alpha)$ is transcendental. Every nonconstant $\overline{\mathbb{Q}}$-linear combination of these entries is transcendental; here nonconstant means that some coefficient other than that of 1 is nonzero.

Proof. We check explicitly the hypotheses of the cited theorem for any finite initial segment. The functions $1, e^z, \mathcal{E}_1, \dots, \mathcal{E}_S$ are E-functions. Indeed their factorial-series coefficients are rational, bounded in absolute value, and the denominators through index N divide $\operatorname{lcm}(1,\dots,N)^S = \exp(O_S(N))$. For completeness, the elementary bound used here follows by letting $\vartheta(y) = \sum_{p\le y} \log p$: primes in $(m, 2m]$ divide $\binom{2m}{m}$, so dyadic summation gives $\vartheta(y) = O(y)$. Consequently

$$\log \operatorname{lcm}(1,\dots,N) = \sum_{j\ge 1} \vartheta(N^{1/j}) = O(N) + O(\sqrt{N}\log N) = O(N).$$

Finally they satisfy the homogeneous first-order system

$$1' = 0, \qquad (e^z)' = e^z, \qquad \mathcal{E}_1' = (e^z - 1)/z, \qquad \mathcal{E}_s' = \mathcal{E}_{s-1}/z \quad (s \ge 2).$$

Its matrix has coefficients in $\mathbb{Q}[z, 1/z]$, so every nonzero algebraic argument is an ordinary point of the system.

It remains to prove functional linear independence. As $t \to +\infty$,

$$\mathcal{E}_s(-t) = -\frac{(\log t)^s}{s!} + O_s((\log t)^{s-1}) \qquad (s \ge 1). \tag{82}$$

For $s = 1$ this follows from $\mathcal{E}_1(-t) = \int_0^t (e^{-u} - 1)\, du/u$ by splitting at $u = 1$. The identity $d\mathcal{E}_s(-t)/dt = \mathcal{E}_{s-1}(-t)/t$ then proves the formula inductively. Suppose a linear relation over $\mathbb{C}(z)$ exists and clear its denominators, obtaining polynomial coefficients. Among the coefficients of $1, \mathcal{E}_1, \dots, \mathcal{E}_S$, take their largest degree D, and among those of degree D take

the largest index s, with 1 assigned index zero. Along $z=-t$, division by $t^D(\log t)^s$ leaves a nonzero limit by (82). Every smaller polynomial degree, every smaller logarithmic degree at the same polynomial degree, and the polynomial multiple of e^{-t} vanish in this normalization. This is impossible. Thus all these polynomial coefficients vanish, and the coefficient of e^z vanishes as well.

Beukers's linear-independence theorem for E-function systems now transfers this functional independence to $\overline{\mathbb{Q}}$-linear independence at $\alpha\neq 0$. If a nonconstant algebraic linear combination had an algebraic value, moving that value to the coefficient of 1 would contradict this independence. □

The theorem also makes quotients explicit. If V,W are two distinct entries other than 1 in the displayed independent family, then V/W is transcendental; otherwise $V-(V/W)W=0$ would be an algebraic linear relation. The same argument gives transcendence of $\mathcal{E}_s(\alpha)/e^{\alpha}$. These are statements over the algebraic numbers, stronger than irrationality, but they are classical consequences of the cited theorem rather than new priority claims.

Corollary 24.2 (All integer-pole rational weights in the pure channel)**.** *Let $R\in\overline{\mathbb{Q}}(X)$ have poles only at $0,-1,-2,\dots$, of arbitrary finite orders, and let $\alpha\neq 0$ be algebraic. The value*

$$V_R(\alpha)=\sum_{n\geq 1}\frac{R(n)\alpha^n}{n!}$$

has an effective finite expression

$$V_R(\alpha)=c_0+c_e e^{\alpha}+\sum_{s=1}^{S}c_s\mathcal{E}_s(\alpha),\qquad c_0,c_e,c_s\in\overline{\mathbb{Q}}.$$

It is algebraic if and only if $c_e=c_1=\cdots=c_S=0$; otherwise it is transcendental. When algebraic, its value is the explicitly computed number c_0.

Proof. Partial fractions reduce the calculation to polynomial weights and $(n+a)^{-r}$, where $a\geq 0$ and $r\geq 1$. A pole at zero gives $\mathcal{E}_r(\alpha)$ directly. For $a\geq 1$, write the falling factorial polynomial as

$$X(X-1)\cdots(X-a+1)=\sum_{j=1}^{a}c_{a,j}X^j.$$

For every integer ℓ, put $\mathcal{F}_\ell(z)=\sum_{m\geq 1}z^m/(m!m^\ell)$. Thus $\mathcal{F}_\ell=\mathcal{E}_\ell$ for $\ell\geq 1$, $\mathcal{F}_0=e^z-1$, and $\mathcal{F}_{-d}=\theta^d(e^z-1)$ for $d\geq 1$. Reindexing by $m=n+a$ yields the exact identity

$$\sum_{n\geq 1}\frac{\alpha^n}{n!(n+a)^r}=\alpha^{-a}\sum_{j=1}^{a}c_{a,j}\mathcal{F}_{r-j}(\alpha)-\frac{1}{a^r}. \tag{83}$$

In extending the reindexed sum to $m\geq 1$, the falling factorial vanishes at $1,\dots,a-1$; the remaining boundary term $m=a$ is exactly $1/a^r$. Every $\mathcal{F}_{-d}(\alpha)$ is an algebraic multiple of e^{α}: the relevant polynomials are generated by $Q_0(z)=1$ and $Q_{d+1}(z)=z(Q_d(z)+Q_d'(z))$. Polynomial weights are treated by the same Euler operator, with the constant weight contributing $e^{\alpha}-1$. These finite operations give the claimed coefficients. The final classification is exactly Theorem 24.1. □

In particular, for every $a\geq 1$, $r\geq 2$ and nonzero algebraic α,

$$\sum_{n\geq 1}\frac{\alpha^n}{n!(n+a)^r}\quad\text{is transcendental.} \tag{84}$$

Indeed its coefficient of $\mathcal{E}_{r-1}(\alpha)$ in (83) is $\alpha^{-a}c_{a,1} = \alpha^{-a}(-1)^{a-1}(a-1)! \neq 0$, and no larger inverse moment occurs. Simple-pole values have genuine exceptions, which the criterion detects; for example

$$\sum_{n\geq 1} \frac{1}{n!(n+2)} = \frac{1}{2}.$$

This example also explains why the nontrivial arithmetic-profile hypothesis cannot simply be discarded from the earlier rational-weight theorem.

25 Exact finite verification and reproducibility

The proof for every K is the argument above, not a numerical experiment. The checks below test indexing, arithmetic normalizations, nonzero output, and the progression calculation.

25.1 An entirely explicit one-exponent example

For $K = 1$, the general construction gives

$$L = 2, \quad b = 2, \quad F = D = 2, \quad B = 5, \qquad (p_0, t_0) = (5, 0), \quad (p_1, t_1) = (7, 2).$$

The integer weights $(c_0, c_1) = (21, -20)$ satisfy

$$21\frac{5}{\sigma_1(5)} - 20\frac{7}{\sigma_1(7)} = 21\frac{5}{6} - 20\frac{7}{8} = 0.$$

The lower shifts both equal 5, and the terminal shifts are 10 and 12. Their surviving coefficients are $175/2$ and $-245/2$. Thus the zero-limit contradiction is already visible in a two-vertex example. This example recovers an individual result; it is not the new joint conclusion.

25.2 A smaller two-exponent certificate

For a compact actual computation, we discard zero-coefficient constraints and also retain the lower-degree residual, using a smaller grid with the required shift and coprimality properties. At $K = 2$, the own-degree expansions have the terms

$$T_1(n) = \frac{\sigma_1(n+1)}{n+1} + \frac{\sigma_1(n+2)}{(n+2)^2} + O(n^{-3/2}),$$

and

$$T_2(n) = \frac{\sigma_2(n+1)}{n+1} + \frac{\sigma_2(n+2)}{(n+2)^2} + \frac{\sigma_2(n+2)}{(n+2)^3} + \frac{\sigma_2(n+3)}{(n+3)^3} + O(n^{-3/2}).$$

For the check, impose the three array types $(k, i, j) = (1, 0, 0)$, $(2, 0, 0)$, and $(2, 1, 1)$ on a 4×4 grid. Unlike the stronger general construction, this smaller certificate leaves the lower-degree residual as well. It is checked as a mixed survivor; the isolated terminal shift belongs only to $k = 2$. The progression comparison still works term by term using (15). This is an audit of the mechanism, not a claim that the smaller matrix satisfies every equation of Lemma 4.2.

Take radix 7, $I_{\max} = 24$,

$$D = \prod_{\substack{\ell \leq 24 \\ \ell \text{ prime}}} \ell = 223092870, \qquad B = 1 + 2D \cdot 24 = 10708457761.$$

The pairwise-coprimality argument only requires all primes up to $\max(I_{\max}, K+1)$ to divide D, so a primorial is sufficient. Using exact rational elimination gives the following observed outputs.

Quantity	Verified value
Constraint matrix	12×16
Exact rank / nullity	$12 / 4$
Nonzero entries of the selected integer vector	12
Maximum decimal digits in an integer weight	281
Isolated terminal vertex	$(0, 0)$
Isolated terminal shift	32125373283
Separating prime	38148880801
Exact arithmetic checks	Passed

The program also reconstructs each multiplier from its prime factorization, computes σ_1 and σ_2, checks every matrix equation, constructs the simultaneous CRT residue, verifies all multiplicativity coprimalities, and computes a nonzero difference of refined progression means divided by $\zeta(3)$ using rational Euler factors. The infinite mean identity itself is proved in Lemma 5.1.

The complete verifier is embedded below using `filecontents*`. Compiling this single source writes `check_joint_grid.py`; the PDF does not require that script to be executed. Reproduce with

```
pdflatex -interaction=nonstopmode Simultaneous_Fiber_Cancellation.tex
pdflatex -interaction=nonstopmode Simultaneous_Fiber_Cancellation.tex
python3 check_joint_grid.py
```

The verifier uses only the Python standard library and writes the full weights, factorizations, CRT data, and rational mean certificate to `joint_grid_K2.json`. It does not approximate the constants α_k or infer linear independence from floating-point data.

25.3 Exact checks for mixed-prime interpolation

The expanded source also embeds `check_prime_responses.py`. It uses only integers and Python's standard-library `Fraction` class. The following results were obtained from executing it:

Finite check	Result
Divisor/Jordan response ranks, $H = 1, \ldots, 6$	$2, 4, 6, 8, 10, 12$
Falling-factorial inverse transforms	Degrees 0–12
Exact polynomial evaluation checks	546
Single-prime progression-mean checks	588
Six-channel single-prime rank	5
Six-channel mixed-prime isolation rank	6
Nonempty-subset on/off contrast checks	1152

The six channels are $\sigma_1, \varphi, \psi, J_3, \varphi^3, \psi^2$, whose normalized response functions are independent of any polynomial weights needed to align their growth degrees. The single-prime rank defect is genuine: symbolic polynomial arithmetic verifies $\delta_\psi = -\delta_\varphi + 2\delta_{J_3}$. For each target, the checker expands the interpolation expression into 32 nonempty-subset terms and verifies that every other channel is zero while the target is nonzero. It also checks the canonical identity $(1 - X)(1 + X) = 1 - X^2$, so $\varphi\psi$ and J_2 are not mistakenly counted as distinct functions.

These are finite auxiliary checks, not a computational proof of Theorem 8.1. The general proof is Lemma 7.1 and Theorem 7.2. After compiling the standalone source, reproduce the additional checks with

```
python3 check_prime_responses.py
```

The program writes `prime_response_checks.json`, containing the exact matrices and interpolation certificates. These verification scripts are embedded in this one source; no external data or package is required to execute them.

25.4 Exact denominator checks and the concrete value

The additional embedded program `check_dyadic_height.py` checks 129 factorial valuations, 8385 factorial-quotient valuations, 100001 binary-height bounds, and 60 extremal binary cases. It verifies 6888 scaled partial-sum terms and 2624 scaled rational-total terms for both signs and several denominators, including even denominators. All checks passed. It also computes the exact rational enclosure of C_* stated in Section 10; the lower and upper bounds have the same first 50 decimal places. Execute it after compiling with

```
python3 check_dyadic_height.py
```

The output `dyadic_height_checks.json` stores the exact fractions and check counts. These finite computations supplement, and do not replace, Lemma 10.1 and Theorem 10.3.

25.5 Exact checks for the integrated extension

The fourth embedded standard-library program, `check_extensions_v4.py`, checks the new structures using rational arithmetic. Its independent formulas compare divisor, radical, lattice, and lcm coefficients with their prime-power profiles; test the local-response formulas and mixed-prime isolation; and verify the residue and offset-grid transformations.

Finite check	Passed instances
Arithmetic coefficients versus prime-power profiles	960
Least-common-multiple identities	80
Local means and valuation responses	720
Radical valuation-depth checks	10
Mixed-prime profile isolation	144
Modulus-four inverse transformations	625
Periodic-grid pairs	268
Negative-offset-grid pairs	386
Shifted-factorial coefficient expansions	54
Periodic polynomial rearrangements	155

In particular, the radical profile has a zero depth-one response and a nonzero depth-two response, demonstrating why the valuation-vector extension is needed. The oscillatory transformation has determinant -8. The offset checks include the repaired congruence $n_e \equiv R + 1 \pmod{p_e}$, including negative shifts. Run the additional verifier after compiling the standalone source:

```
python3 check_extensions_v4.py
```

It writes `extensions_v4_checks.json`. These finite checks supplement the uniform proofs; they are not proof-assistant certification.

25.6 Exact checks for the expanded edition

Four further standard-library Python verifiers are embedded as `filecontents*` blocks. Compiling this single source writes all eight verifier scripts beside the build. Run the four original commands above and the following commands:

```
python3 check_profile_universe_v5.py
python3 check_projective_functionals.py
python3 check_rational_kernels_v5.py
python3 check_scope_v5.py
```

The new checks compare weighted factorizations, gcd and unit-gcd sums, and subgroup profiles with exact formulas and direct finite counts; verify nested-integral and Legendre identities coefficient by coefficient; check product-kernel, bi-Vandermonde, and inverse-moment reductions; and check thirty cyclotomic CRT matrices and thirty finite-ring counts. A 42×19 exact sample matrix for the signed arithmetic family has rank 19. The general signed-family proof uses eventual relations and Vandermonde arguments, not this finite rank test. All numerical quantities in these scripts are integers or exact rational numbers.

The new checks validate identities and finite constructions. The universal growth, progression-mean, and independence claims are proved in the text; they are not inferred from finite samples. None of the scripts is a Lean formalization or a substitute for external mathematical review.

26 Attribution, novelty, and remaining verification

Component	Provenance and role
Original joint target	Deajim–Siksek (2011); corroborated by Pratt's introduction. The constants and all-rank quantifier are retained.
Individual all-exponent architecture	Tokengrinder (2026): factorial expansion, coprime grids, integer limits, and isolated-shift means. These are attributed and reproved here.
Added joint step	Rational fiber constraints include every arithmetic normalization simultaneously; a wider radix makes all terminal shifts injective.
Added separation step	Mixed differences and interpolation separate local responses; exact-valuation refinements extend this to full prime-power profile vectors.
Denominator extension	Binary digit sums bound the common denominator by $O(N)$; one extra cancellation order covers reduced numerators ± 2.
New application range	Polynomial valuation profiles of every logarithmic order; arithmetic convolutions and rough multiplicity; signed sublinear sequences; all cyclotomic coordinates; integral operators and quotient classification.
Logical status	Theorems 1.1, 11.4, 12.2, 13.3, 14.1, 16.5, 19.4, and 20.1 have complete unconditional arguments in this manuscript. No unproved estimate or conjecture is an input.
Review status	Separate AI proof reviews of the original and integrated arguments, and eight exact finite arithmetic verifiers; no external human referee or Lean formalization of this new joint proof.
Historical status	No identical unconditional joint resolution was located in the searches performed. Exhaustive absence of prior work and a historically unprecedented foundational principle are not certified.

The bounded literature check on 28 October 2026 used the exact title and author names of [1], searches for unconditional joint independence of divisor-sum factorial series, Pratt's primary introduction, the original and extension directories of [4], and a scholarly citation search. The full 2011 article was not accessible in the retrieved sources; its mathematical target is attested by the primary abstract and by [2]. The extension directory was read at commit `46b9fd26361faf669bcd7e4495e63678bfa6a735`. The source and limits of the priority check are stated so that a mathematical proof is not confused with a historical certification. The additional search examined Jordan and Dedekind factorial series, their powers, and weighted

divisor-sum moments. Pratt's introduction provides prior motivation for the moment hierarchy; it does not state a named conjecture identical to our full polynomial-Euler theorem. The degree-one overlap with Erdős–Straus [3] is credited explicitly. No assertion is made that every specialization was previously open.

Accordingly, this manuscript does not claim that every requested success condition has already been independently established. It provides a complete proof for an unchanged, pre-existing, author-identified target and a concrete additional method. Confirmation of worldwide priority and external acceptance remain separate tasks. No solution of the other proposed Erdős problems, and no transcendence conclusion for an individual α_k, is asserted. Neither Robin's inequality nor irrationality of Euler's constant follows from the results proved here.

The additional bounded literature check on 29 October 2026 examined factorial series with products, periodic coefficients, and shifted denominators, and the primary sources for the classical counting functions used here. Hančl–Tijdeman [7] already study higher differences and affine-product denominators; these general ideas are not claimed as new. The specific offset-grid argument and the joint profile-and-residue theorem are the statements supplied here. The search neither located an identical full theorem nor established historical open status for every individual application. Subpolynomial numerators at $x = 1$ have elementary classical factorial-tail treatments and are not presented as novelty flagships.

The expanded edition separates three kinds of results: new arguments developed in this manuscript, explicit applications of those arguments, and classical results reproduced with attribution. The sublinear criterion and finite-state recurrence argument are elementary and are not presented as historically new discoveries. Classical subgroup counts and Menon-type identities are credited to their literature; their factorial-value applications are verified by the stated profile conditions. The entire inverse-exponential-moment section is explicitly based on Beukers's classical theorem, including the transcendence statements there. No claim is made that adding these families resolves the unrelated irrationality questions for Euler's constant, odd zeta values, or rational Lambert values. The enlarged arithmetic statements have received separate internal proof reviews, but remain unrefereed mathematical claims.

References

[1] A. Deajim and S. Siksek, *On the $\mathbb{Q}$-linear independence of the sums $\sum_{n=1}^{\infty} \sigma_k(n)/n!$*, Journal of Number Theory **131** (2011), no. 4, 745–749. doi:10.1016/j.jnt.2010.11.009. Primary abstract: `https://wrap.warwick.ac.uk/id/eprint/41557/`.

[2] K. Pratt, *The irrationality of a divisor function series of Erdős and Kac*, arXiv:2209.11124v1, 22 September 2022, Introduction (the Deajim–Siksek criterion, weighted moment functions, and Theorem 1). Published in Acta Arithmetica **211** (2023), 193–228, doi:10.4064/aa220927-1-9. `https://arxiv.org/html/2209.11124v1`.

[3] P. Erdős and E. G. Straus, *On the irrationality of certain series*, Pacific Journal of Mathematics **55** (1974), 85–92, Theorem 3.7. doi:10.2140/pjm.1974.55.85. `https://msp.org/pjm/1974/55-1/pjm-v55-n1-p08-p.pdf`.

[4] Tokengrinder (GitHub account `tokengr1nder`), *Erdős 252: irrationality of the factorial divisor-sum series*, public Lean development, September 2026. `https://github.com/tokengr1nder/Erdos252`. Original reviewed source commit: `dc071aafce41bbae41caf4c015499db6dafafd11`. The present proof reconstructs the mathematics and does not claim to inherit formal verification from that source.

[5] Tokengrinder, *Genuine extensions of the E252 argument*, `Generalizations/README.md` in [4], inspected at commit `46b9fd26361faf669bcd7e4495e63678bfa6a735`. The section

on polynomial divisor sums explicitly distinguishes joint independence from the proved polynomial-numerator extensions.

[6] KEIO, *Verifiable Mathematical Reasoning Corpus*, snapshot of 28 October 2026, commit `510a32c9a3fd2aa0b5c78d3f6812ded723b9ac8f`. https://github.com/QEEDDD/Verifiable-Mathematical-Reasoning-Corpus. Selected relevant drafts: [013], *Profinite Recognition of Mahler Equations and D-Finite Witt Escape*; [078], *Mixed Mahler Congruences*. These are methodological context, not assumed theorems.

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A Complete exact verifier

The following code is included in the single source. Its role is the finite audit described in Section 25, not a substitute for the general proof. Every primality check uses exact trial division.

```
#!/usr/bin/env python3
"""Exact finite audit of the K=2 joint-divisor factorial-tail grid.

Uses only the Python standard library.  This verifies one auxiliary grid;
it is not a numerical proof of infinite-series linear independence.
"""
from fractions import Fraction as F
from itertools import product
from math import gcd, lcm, prod
from pathlib import Path
import json


def factor(n):
    out = {}
    d = 2
    while d * d <= n:
        while n % d == 0:
            out[d] = out.get(d, 0) + 1
            n //= d
        d = 3 if d == 2 else d + 2
    if n > 1:
        out[n] = out.get(n, 0) + 1
    return out


def sigma(fac, k):
    return prod(sum(p ** (k*j) for j in range(v+1)) for p,v in fac.items())


def nullspace(rows):
    a = [[F(x) for x in row] for row in rows]
    nr, nc = len(a), len(a[0])
    pivots = []
    r = 0
    for c in range(nc):
        pivot = next((j for j in range(r,nr) if a[j][c]), None)
        if pivot is None:
            continue
        a[r],a[pivot] = a[pivot],a[r]
        val = a[r][c]
        a[r] = [x/val for x in a[r]]
        for j in range(nr):
            if j != r and a[j][c]:
                val = a[j][c]
                a[j] = [x-val*y for x,y in zip(a[j],a[r])]
        pivots.append(c)
        r += 1
        if r == nr:
            break
    free = [c for c in range(nc) if c not in pivots]
    vectors = []
    for c in free:
        v = [F(0)]*nc
        v[c] = F(1)
        for row,pivot in enumerate(pivots):
            v[pivot] = -a[row][c]
        vectors.append(v)
    return pivots, vectors


def is_prime(n):
    # Exact trial division; no probable-prime assertion is used.
    return n >= 2 and factor(n) == {n: 1}
```

```
def as_json_fraction(x):
    x = F(x)
    return {"numerator":str(x.numerator), "denominator":str(x.denominator)}


def main():
    # General-construction K=1 instance: p=(5,7), t=(0,2), w=(21,-20).
    assert F(21*5,6)-F(20*7,8)==0
    assert [h*pp-tt for h in (1,2) for pp,tt in ((5,0),(7,2))]==[5,5,10,12]
    assert 100%25==0 and 100%49==2
    K,L,B = 2,4,7
    E = list(product(range(L), repeat=K))
    I = (L-1)*(1+B)
    D = prod(p for p in range(2,max(I,K+1)+1) if is_prime(p))
    P = 1+K*D*I
    p = {e:P+D*(e[0]+B*e[1]) for e in E}
    t = {e:D*(e[0]+2*B*e[1]) for e in E}
    fac = {e:factor(p[e]) for e in E}
    sig = {(k,e):sigma(fac[e],k) for k in (1,2) for e in E}
    assert all(prod(q**v for q,v in fac[e].items()) == p[e] for e in E)
    assert all(is_prime(q) for e in E for q in fac[e])
    assert all(gcd(p[e],p[f])==1 for ix,e in enumerate(E) for f in E[ix+1:])
    assert all(gcd(p[e],h)==1 for e in E for h in range(1,K+2))

    # k=2,i=1,h=1 has S(1,0)=0 and is omitted legitimately.
    types = [(1,0,1),(2,0,1),(2,1,2)]
    rows=[]
    labels=[]
    for k,i,h in types:
        for fixed in range(L):
            labels.append([k,i,h,fixed])
            rows.append([F(p[e]**(i+1),sig[k,e]) if e[2-h]==fixed else F(0) for e in E])
    pivots,ns = nullspace(rows)
    assert ns
    v=ns[0]
    denominator=lcm(*(x.denominator for x in v))
    w=[int(x*denominator) for x in v]
    common=gcd(*w)
    w=[x//common for x in w]
    assert any(w)
    assert all(sum(x*y for x,y in zip(row,w))==0 for row in rows)
    shifts={(e,h):h*p[e]-t[e] for e in E for h in range(1,K+2)}
    assert all(r>0 for r in shifts.values())
    assert max(r for (e,h),r in shifts.items() if h<=K)<(K+1)*P
    assert min(shifts[e,K+1] for e in E)==(K+1)*P
    assert len({shifts[e,K+1] for e in E})==len(E)
    active=[e for e,x in zip(E,w) if x]
    star_e=active[0]
    star=shifts[star_e,K+1]
    assert [(e,h) for (e,h),r in shifts.items() if r==star]==[(star_e,K+1)]
    for h in range(1,K+1):
        for fixed in range(L):
            assert len({shifts[e,h] for e in E if e[2-h]==fixed})==1

    # Exact simultaneous residue class and the required multiplicativity.
    Q=prod(p[e]**2 for e in E)
    A=sum(t[e]*(Q//p[e]**2)*pow(Q//p[e]**2,-1,p[e]**2) for e in E)%Q
    assert all((A-t[e])%p[e]**2==0 for e in E)
    N=A+Q
    for e in E:
        n=(N-t[e])//p[e]
        assert n>0 and n%p[e]==0
        for h in range(1,K+2):
            assert p[e]*(n+h)==N+shifts[e,h]
            assert gcd(p[e],n+h)==1

    # Compute mu_{Q,2}(A+star)/zeta(3) exactly through Euler factors.
    qfac={q:2*v for e in E for q,v in fac[e].items()}
    assert prod(q**v for q,v in qfac.items())==Q
    ratio=F(1)
    for q,vQ in qfac.items():
```

```
        a=A+star
        va=0
        while a%q==0:
            va+=1
            a//=q
        if va<vQ:
            local=sum(F(1,q**(K*j)) for j in range(va+1))
        else:
            local=sum(F(1,q**(K*j)) for j in range(vQ))
            local+=F(1,q**(K*vQ))/(1-F(1,q**(K+1)))
        ratio*=local*(1-F(1,q**(K+1)))
    assert ratio>0
    ell=max(shifts.values())+1
    while not is_prime(ell):
        ell+=1
    assert ell<2**64 and gcd(ell,Q)==1
    v0=(-A*pow(Q,-1,ell))%ell
    v1=(-(A+star)*pow(Q,-1,ell))%ell
    assert all((Q*v0+A+r)%ell!=0 for r in shifts.values())
    assert all(((Q*v1+A+r)%ell==0)==(r==star) for r in shifts.values())
    top_coeff=F(w[E.index(star_e)]*p[star_e]**(K+1),sig[K,star_e])
    diff_ratio=top_coeff*ratio/F(ell**K)
    assert diff_ratio!=0
    out={
        "scope":"One exact K=2 auxiliary grid; not a proof by finite computation",
        "parameters":{"K":K,"L":L,"B":B,"I":I,"D":D,"P":P},
        "rows":len(rows),"columns":len(E),"rank":len(pivots),"nullity":len(ns),
        "constraint_labels_k_i_h_fixed":labels,
        "vertices":[{"e":e,"p":p[e],"t":t[e],"factorization":fac[e],
            "sigma1":sig[1,e],"sigma2":sig[2,e],"weight":str(w[j])}
            for j,e in enumerate(E)],
        "isolated_endpoint":{"vertex":star_e,"shift":star,"top_coefficient":as_json_fraction(top_coeff
    )},
        "crt":{"modulus":str(Q),"residue":str(A)},
        "fresh_prime":{"ell":ell,"v_miss":v0,"v_hit":v1,
            "mean_difference_divided_by_zeta3":as_json_fraction(diff_ratio)},
        "all_checks":"passed"
    }
    target=Path(__file__).with_name("joint_grid_K2.json")
    target.write_text(json.dumps(out,indent=2)+"\n")
    print(json.dumps({k:out[k] for k in ("parameters","rows","columns","rank","nullity","all_checks")},
     indent=2))
    print("weight_support_size",len(active))
    print("max_weight_digits",max(len(str(abs(x))) for x in w))
    print("isolated_vertex",star_e,"isolated_shift",star)
    print("fresh_prime",ell)
    print("json",target)


if __name__=="__main__":
    main()
```